

Uniform Stability of the Spherical Laughlin Gap

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Abstract

We prove that the fermionic Laughlin V_1 Hamiltonian at filling $1/3$ on expanding round spheres retains a unique ground state and a uniform spectral gap under sufficiently weak bounded real scalar one-body potentials projected to the lowest Landau level. With coefficient one for each pair projector and Laughlin flux $q = 3(N - 1)$, the gap and perturbation threshold are uniform for all sufficiently large particle numbers and all potential profiles of supremum norm at most one. The result uses the uniform unperturbed Fock-space gap and gives existential constants.

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1 Introduction

Laughlin’s wave function describes a strongly correlated state at fractional filling of the lowest Landau level [7]. Haldane’s spherical geometry and pseudopotentials give a particularly simple parent Hamiltonian: each electron pair is penalized by the orthogonal projection onto relative angular momentum one [5]. The cubic Laughlin polynomial is its unique zero state at filling $1/3$ with the spherical shift. A spectral gap separates this state from excitations. Stability asks whether that separation survives a weak external potential whose total many-particle norm grows with the area of the system.

We prove uniform stability for the full spherical interaction under bounded scalar potentials projected to the lowest Landau level. The potential may vary with system size and need not have any symmetry. The proof uses the uniform unperturbed Fock-space gap of [16], and derives the additional local estimates needed to pass from a gap at zero disorder to a gap on a fixed disorder interval.

1.1 Model and main result

For $N \geq 2$, set $q = 3(N - 1)$ and let S_q be the round sphere of radius $\sqrt{q/2}$. Magnetic length is one, so the sphere carries q flux quanta and has area $2\pi q$. Let U_q be its one-particle lowest

Landau level, equivalently the spin- $q/2$ representation of $SU(2)$. On the entire antisymmetric space $\bigwedge^N U_q$, put

$$H_{N,q} = \sum_{1 \leq i < j \leq N} P_{ij}^{(1)}, \quad (1.1)$$

where $P_{ij}^{(1)}$ projects the indicated pair onto total spin $q - 1$. Each unordered pair has coefficient one. The normalized Laughlin vector spans the line generated by

$$\Psi_{L,N} = \prod_{i < j} (z_{i0} z_{j1} - z_{i1} z_{j0})^3. \quad (1.2)$$

If Π_q denotes the lowest-level projection and φ is a bounded real scalar function on S_q , its Toeplitz operator and many-particle perturbation are

$$T_q(\varphi) = (\Pi_q M_\varphi \Pi_q)|_{U_q}, \quad V_{N,q}(\varphi) = \sum_{i=1}^N T_q(\varphi)^{(i)}.$$

Let $E_0(N, \lambda, \varphi) \leq E_1(N, \lambda, \varphi)$ be the two lowest eigenvalues, counted with multiplicity, of

$$H_{N,q}(\lambda, \varphi) = H_{N,q} + \lambda V_{N,q}(\varphi).$$

Theorem 1.1 (Uniform scalar-disorder stability). *There are constants $\lambda_* > 0$, $\Delta_* > 0$, and $N_* < \infty$ such that, for every $N \geq N_*$, every real bounded measurable φ on $S_{3(N-1)}$ with $\|\varphi\|_\infty \leq 1$, and every real $|\lambda| \leq \lambda_*$,*

$$E_1(N, \lambda, \varphi) - E_0(N, \lambda, \varphi) \geq \Delta_*. \quad (1.3)$$

In particular the perturbed ground state is unique.

The constants are uniform over the potential as well as particle number. Thus the theorem applies, in particular, to smooth potentials satisfying $\max_{0 \leq k \leq 2} \|\nabla^k \varphi\|_\infty \leq 1$ in the physical round metric. Its stronger amplitude-only formulation comes from the coherent-orbital representation of Toeplitz multiplication. The ground vector is allowed to move, and its energy is subtracted in the stability argument. Adding a constant to the potential changes only that energy. We give existential constants; the proof does not optimize the disorder threshold.

1.2 Historical context and the additional difficulty

Exact parent Hamiltonians have been central to the mathematical study of fractional quantum Hall states since the work of Trugman and Kivelson [20]. Haldane pseudopotentials also arise from suitable short-range limits of repulsive interactions, as proved by Seiringer and Yngvason [19]. The spectral gap is a separate quantitative question: identifying a zero polynomial does not control the energy of vectors perpendicular to it. Rougerie's survey [17] discusses the gap problem and its role in the response of the Laughlin phase to perturbations.

Rigorous uniform gaps were established for truncated pseudopotentials in thin-cylinder and thin-torus geometries by Nachtergaele, Warzel, and Young [15] and Warzel and Young [21]. The unperturbed input used here concerns the full interaction on expanding round spheres [16]. We state its exact normalization in Theorem 2.2. In addition, we use and reproduce a strengthening of its four-body comparison: certain positive blocks that were discarded in obtaining the gap can be retained. This stronger estimate supplies the energy-loss construction below.

Another line of work controls density within the Laughlin-correlated class. Rougerie and Yngvason [18], followed by Lieb, Rougerie, and Yngvason [9], established incompressibility estimates

for Laughlin functions multiplied by analytic symmetric factors. The sharp local density bound in [9] is averaged over mesoscopic disks. Our zero-mode estimate controls general local observables, with total cost proportional to the number of missing electrons; it is then combined with an estimate for states outside the zero space. Algebraic descriptions and recursions for pseudopotential zero modes were developed in [4, 11]. We use the symmetric-polynomial description and add quantitative bounds for its orthogonal branching.

General gap-stability theorems explain the role of local information. Bravyi, Hastings, and Michalakis [3] treated stability under sufficiently weak, suitably decaying local perturbations of commuting-projector models with topological-order conditions. Michalakis and Zwolak [12] proved stability for frustration-free lattice Hamiltonians under local topological-order and local-gap hypotheses. Subsequent infinite-volume formulations also retain local conditions in addition to a bulk gap [14]. Those hypotheses contain more information than a uniform global gap. For the spherical lowest Landau level, the local control needed here is established through flux pinning and a particle-loss evolution. The resulting relative-bound argument has the same purpose as local block diagonalization in the lattice theory, with a different source of control for the diagonal terms.

Our final continuation of the ground line uses the quasiadiabatic construction of Hastings and Wen [6] and the exact spectral-flow framework of Bachmann, Michalakis, Nachtergaele, and Sims [2]. It is applied inside a gap bootstrap, where its gap hypothesis is available. Locality is obtained using redundant localized frames. This connects to the continuum fermionic frame method of Bachmann and De Nittis [1]; we prove the finite-sphere estimates, including comparisons at neighboring fluxes, that the present argument requires. The propagation estimates follow the Lieb–Robinson method [8] and its fermionic form [13].

1.3 The proof and its reusable estimates

The main difficulty is extensive perturbation size: $\|V_{N,q}(\varphi)\|$ can be of order N . A norm bound on the whole perturbation therefore gives a disorder interval that shrinks with particle number. We instead compare suitably centered local terms with the interaction energy. Two estimates make that comparison possible.

Local observables in zero spaces. At fixed flux $q = 3(N - 1)$, write $Z_{n,q} = \ker H_{n,q}$. For an interaction W that is a bounded-density sum of uniformly localized neutral terms, we prove

$$|\langle \psi, W\psi \rangle - \langle \Omega, W\Omega \rangle| \leq C_W(N - n), \quad \psi \in Z_{n,q}, \quad \|\psi\| = 1, \quad (1.4)$$

where Ω is the normalized filled Laughlin vector. This is Theorem 6.6; the constant is uniform in flux. Neutral means that a term preserves particle number; localization is defined precisely in Section 3. The linear dependence on $N - n$ is essential.

To prove (1.4), we read a zero mode from one pole. Either the last orbital is empty, which removes one unit of flux and records a flag, or contraction in that orbital removes an electron and three units of flux. The branches are orthogonal and give coordinates for the entire zero space. The number of flags is the *quasi-hole degree* $h = q + 3 - 3n$. At the filled flux it equals $3(N - n)$, three times the electron deficit; a one-hole sector below means $h = 1$, not one missing electron. Pinning zeros at the pole realizes the required change of flux. The uniform Fock gap persists along that pinning deformation and gives local unitary transport. A history of branch choices then bounds a local observable by weighted flag values along the history. A first, slowly decaying weight supplies enough information to close a second estimate with an integrable weight. Rotation averaging yields (1.4).

Local descent of interaction energy. Extend the pair Hamiltonian to all particle sectors by $H_q = \bigoplus_{n=0}^{q+1} H_{n,q}$, with $H_{0,q} = H_{1,q} = 0$, and let $\mathcal{N}$ act as multiplication by n on the n -particle sector. On this Fock space we construct local particle-loss operators J_y such that (Theorem 7.1)

$$\int J_y^* J_y dy = H_q, \quad \int J_y^* H_q J_y dy \leq H_q^2 - c H_q. \quad (1.5)$$

Each jump removes between two and a fixed number of particles. The associated completely positive evolution decreases interaction energy exponentially, and its expected particle loss is bounded by the initial energy. The construction first removes a pair and then partially empties a fixed neighborhood. A rotation average shows that the remaining pair correlations are paid for by the retained four-body blocks. This also proves a uniform energy cost for particle excess above Laughlin filling.

From descent to perturbation theory. For a small fixed $\mu > 0$, the operator $G = H_q + \mu(N - \mathcal{N})$ controls both interaction energy and particle-number deficit or excess. If each neutral local term A_x annihilates Ω , following its expectation through the loss evolution gives

$$\pm \sum_x A_x \leq CG.$$

The proof closes a quadratic inequality for the best relative-bound constant, using (1.4) at the limiting zero state (Proposition 8.1). This implication from local energy descent and linear electron-deficit control is useful independently of the particular flux construction. Finally spectral transport makes the derivative of the pulled-back perturbed Hamiltonian termwise centered in this sense. On the N -particle sector $G = H_q$, so integration gives a uniform lower bound above the transported ground state.

Section 2 fixes the algebra and zero-mode conventions. Sections 3–6 establish the local zero-mode estimate. Section 7 constructs the loss evolution, and Section 8 proves the relative bound and completes Theorem 1.1. Appendix A contains the retained-block comparison and its finite arithmetic data.

2 Lowest Landau level and zero modes

We fix the normalization of the interaction before discussing locality. The exterior-algebra formulation is useful because both the pinned Hamiltonians and the particle-loss evolution change particle number. Throughout the paper, constants are independent of flux and particle number unless an explicit dependence is indicated. A statement for large flux means that its threshold is independent of the particle sector.

2.1 Coherent orbitals and pair annihilators

Let $q \geq 1$ be an integer. The sphere S_q has radius $\sqrt{q/2}$, area $2\pi q$, and magnetic field one. The lowest Landau level U_q is the spin- $q/2$ representation of $SU(2)$. We use homogeneous polynomials of degree q in spinor coordinates (z_0, z_1) , with invariant inner product normalized so that

$$e_j = \binom{q}{j}^{1/2} z_0^{q-j} z_1^j, \quad 0 \leq j \leq q,$$

is an orthonormal basis. This normalization identifies U_q unitarily with the physical lowest Landau level; it does not rescale any operator. Relative to a chosen north pole, $j = 0$ is the north orbital and $j = q$ the south orbital.

Write k_x for the unit coherent vector at $x \in S_q$, defined up to a phase by the Riesz vector for evaluation at x . Its components in the axis basis have squared moduli

$$|\langle e_j, k_x \rangle|^2 = \binom{q}{j} (1 - m/q)^{q-j} (m/q)^j, \quad m = q \sin^2(\theta/2). \quad (2.1)$$

Schur's Lemma and the trace give the resolution

$$\kappa_q \int_{S_q} |k_x\rangle \langle k_x| dA(x) = I_{U_q}, \quad \kappa_q = \frac{q+1}{2\pi q}. \quad (2.2)$$

In particular the density κ_q is uniformly bounded. For a bounded measurable real function φ , the Toeplitz operator is

$$T_q(\varphi) = \kappa_q \int_{S_q} \varphi(x) |k_x\rangle \langle k_x| dA(x). \quad (2.3)$$

Indeed the quadratic form on the right is the integral of φ times the product of the two evaluated lowest-level sections, which is the quadratic form of compression of multiplication by φ .

The fermionic Fock space and number operator are

$$\mathcal{F}_q = \bigoplus_{n=0}^{q+1} \bigwedge^n U_q, \quad \mathcal{N}|_{\bigwedge^n U_q} = nI.$$

Increasing wedges of orthonormal vectors have norm one. For $a \in \bigwedge^k U_q$, let $B(a)$ be the adjoint of left wedge multiplication by a . Thus B is conjugate-linear, $B(a \wedge b) = B(b)B(a)$, and $c_j = B(e_j)$ satisfy the canonical anticommutation relations. For a one-particle vector f we also write $c(f) = B(f)$. For an operator on $\bigwedge^k U_q$, its lift to Fock space is the linear extension of

$$\mathcal{L}_k(|a\rangle \langle b|) = B(a)^* B(b). \quad (2.4)$$

The lift preserves positivity and rotations. It vanishes on sectors with fewer than k particles and equals its argument on the k -particle sector.

Let $V_q \subset \bigwedge^2 U_q$ be the spin- $(q-1)$ summand. Its orthonormal axis basis v_p , $0 \leq p \leq 2q-2$, can be written as

$$v_{b-1} = Z_{q,b}^{-1/2} \sum_{\substack{0 \leq i < j \leq q \\ i+j=b}} (j-i) \binom{q}{i}^{1/2} \binom{q}{j}^{1/2} e_i \wedge e_j, \quad Z_{q,b} = \frac{b(2q-b)}{2(2q-1)} \binom{2q}{b}, \quad (2.5)$$

where $1 \leq b \leq 2q-1$. The formula follows by lowering the highest vector $e_0 \wedge e_1$. For its normalization, the sum over ordered i, j with $i+j=b$ is a hypergeometric second moment: its total mass is $\binom{2q}{b}$ and the mean of $(j-i)^2$ is $b(2q-b)/(2q-1)$. Dividing by two gives $Z_{q,b}$. We set $B_p = B(v_p)$. The interaction is

$$H_q = \mathcal{L}_2(\Pi_{V_q}) = \sum_{p=0}^{2q-2} B_p^* B_p, \quad H_{n,q} = H_q|_{\bigwedge^n U_q}. \quad (2.6)$$

On n particles this is exactly $\sum_{i<j} P_{ij}^{(1)}$, with coefficient one for each unordered pair. One way to check the coefficient is to identify a unit wedge with its normalized antisymmetric tensor: contraction in two prescribed tensor positions is $B(a)/\sqrt{\binom{n}{2}}$, and summing the $\binom{n}{2}$ pair projections gives (2.6).

We shall also use the wedge maps

$$\begin{aligned} W_3 : V_q \otimes U_q &\longrightarrow \bigwedge^3 U_q, & W_3(a \otimes u) &= a \wedge u, \\ W_4 : V_q \otimes V_q &\longrightarrow \bigwedge^4 U_q, & W_4(a \otimes b) &= a \wedge b. \end{aligned} \tag{2.7}$$

In particular $V_q \otimes V_q$ uses ordered pairs. This convention fixes the coefficient of the four-body term in the normal-ordered square used in Appendix A.

2.2 The zero spaces and the unperturbed input

The polynomial and second-quantized descriptions of these kernels are closely related; see [4, 11] for algebraic constructions of pseudopotential zero modes. Put $Z_{n,q} = \ker H_{n,q}$ and let $P_{n,q}$ be its orthogonal projection. When all sectors are being considered, write $Z_q = \ker H_q$ and P_q for the Fock-space zero projection.

Lemma 2.1 (Polynomial description). *For $n \geq 1$, a vector lies in $Z_{n,q}$ if and only if its polynomial has the form*

$$\prod_{i<j} (z_{i0}z_{j1} - z_{i1}z_{j0})^3 S(z_1, \dots, z_n), \tag{2.8}$$

where S is symmetric and homogeneous of degree $h = q + 3 - 3n$ in each spinor. Consequently $Z_{n,q} = \{0\}$ when $h < 0$, and, when $h \geq 0$,

$$\dim Z_{n,q} = \binom{n+h}{n}.$$

At $q = 3(N-1)$ the space $Z_{N,q}$ is the Laughlin line, and every nonempty zero sector has particle number at most N . The vacuum sector $Z_{0,q}$ has dimension one.

Proof. Positivity of the pair terms says that a zero vector has no pair relative-angular-momentum-one component. An antisymmetric polynomial of the same homogeneous degree in each of two spinors vanishes when the spinors coincide. Separate homogeneity extends that vanishing to proportional spinors, so the irreducible bracket $[z_1, z_2]$ divides it. Dividing by the bracket and then setting $z_1 = z_2 = z$ gives an equivariant map to the degree- $(2q-2)$ polynomials in z , the spin- $(q-1)$ representation. This map is nonzero on $e_0 \wedge e_1$ and hence is onto. Its kernel is exactly divisibility by the bracket cubed: the divided polynomial is symmetric and separately homogeneous, and if it vanishes on the diagonal it is divisible by the bracket; the resulting quotient is antisymmetric and therefore divisible by the bracket once more. By the two-spin decomposition, the map detects precisely the spin- $(q-1)$ summand. Thus absence of that component is equivalent to cubic divisibility.

It follows that every pair bracket cubed divides the polynomial. The distinct pair brackets are nonassociate irreducible polynomials; unique factorization makes their product divide it. Dividing by the cubic product changes antisymmetry to symmetry and leaves degree $q - 3(n-1)$ in each variable. This proves both directions of (2.8). A symmetric polynomial with individual degree h has the symmetrized monomial basis indexed by $0 \leq j_1 \leq \dots \leq j_n \leq h$, giving the binomial dimension. The remaining claims follow at once, with the vacuum treated separately. $\square$

The quantitative unperturbed result used in this paper is the following Fock-space gap theorem. We recall it with the normalization just fixed; its proof is in the companion manuscript [16, Theorem 1.1 and Corollary 1.2].

Theorem 2.2 (Uniform unperturbed gap). *Let $\gamma_* = 4616733319001/10^{14}$. For every $0 < \gamma < \gamma_*$ there is q_γ such that*

$$H_q^2 \geq \gamma H_q \quad (q \geq q_\gamma) \quad (2.9)$$

on all of $\mathcal{F}_q$. In particular, for all sufficiently large N and $q = 3(N - 1)$,

$$H_{N,q} \geq \frac{1}{25}(I - P_{N,q}). \quad (2.10)$$

The all-sector statement gives separation of the spectrum from zero. We use it for zero-space transport, including particle numbers below Laughlin filling. The extra four-body comparison needed for energy descent is proved separately in Appendix A; it is stronger than (2.9).

2.3 Geometric scales

An axis on S_q gives latitude coordinate $m = q \sin^2(\theta/2)$ and longitude ϕ . In these coordinates $dA = dm d\phi$. We use

$$u = 1 + m, \quad v = 1 + q - m, \quad w = \sqrt{\frac{uv}{q+1}}. \quad (2.11)$$

Here v is an end-distance weight; the potential is always denoted by φ . The width w converts index distance to physical distance, including in the polar caps. Section 3 proves the geometric estimates that make these scales useful for local operators.

3 A local calculus in the lowest Landau level

The orbital basis diagonalizes rotations about an axis, but its middle orbitals extend around an entire latitude. We first replace that basis by a redundant family localized in both latitude and longitude. Its analysis map embeds the physical fermions into a larger collection of fermionic modes with a genuine notion of spatial support. The use of localized frames to obtain continuum fermionic locality is developed in [1]; we give the finite-sphere construction and estimates needed here. We then prove the propagation and functional-calculus estimates needed to manipulate local operators. All constants in this section are independent of the flux and the particle number.

3.1 Localized frames and auxiliary fermions

Fix an axis and write

$$m = q \sin^2(\theta/2), \quad u = 1 + m, \quad v = 1 + q - m, \quad w_q(m) = \left(\frac{uv}{q+1}\right)^{1/2}. \quad (3.1)$$

The area form is $dm d\phi$. Away from the two polar balls of fixed radius, changing m by $w_q(m)$ or changing ϕ by $1/w_q(m)$ moves a bounded physical distance. The following elementary comparisons will also be used at the poles, with $u, v \geq 1$:

$$\frac{u(x)}{u(y)} + \frac{v(x)}{v(y)} + \frac{w_q(x)}{w_q(y)} \leq C(1 + d(x, y))^2. \quad (3.2)$$

Indeed m is one half the squared chordal distance from the north pole; the triangle inequality gives the assertion for u , and the south pole gives it for v . The assertion for w_q follows from their product.

Here is an explicit frame construction. Choose a nonnegative smooth function η , supported in $(-2, 2)$ and positive on $[-1, 1]$, and a smooth function $0 \leq \chi \leq 1$ on $[0, 1]$ equal to one on $[0, 1/2]$ and zero on $[2/3, 1]$. For positive integers k use the nonzero functions

$$\chi(j/q)\eta(\sqrt{1+j}-k), \quad (1-\chi(j/q))\eta(\sqrt{1+q-j}-k), \quad 0 \leq j \leq q.$$

Divide each by the square root of the sum of their squares. The resulting windows ξ_a satisfy

$$\sum_a \xi_a(j)^2 = 1, \quad \text{supp } \xi_a \subset I_a, \quad |I_a| \leq Cw_a, \quad |\Delta^\ell \xi_a(j)| \leq C_l w_a^{-l}, \quad (3.3)$$

where I_a is an interval of integers, m_a is the corresponding window center, and $w_a = w_q(m_a)$. At the first few windows the constants absorb the bounded widths. Every index belongs to a bounded number of intervals. To check the difference bound, extend each expression smoothly in j . On a northern window $k \asymp \sqrt{1+j} \asymp w_q(j)$; each derivative of $\sqrt{1+j}$ costs at least $w_q(j)^{-1}$, and derivatives of $\chi(j/q)$ cost $q^{-1} \leq Cw_q(j)^{-1}$. The denominator is bounded above and below, and obeys the same estimates. The southern windows have the identical verification.

For a window bearing the index k , choose the fixed integer $L_a = 16(k+1)$, which is larger than the diameter of the full bump support before restriction to $[0, q]$. Thus $L_a \asymp w_a$ and, for a northern window, its Fourier length is independent of q . For $0 \leq \ell < L_a$ set

$$z_{a\ell} = L_a^{-1/2} \sum_{j=0}^q \xi_a(j) e^{2\pi i j \ell / L_a} e_j, \quad x_{a\ell} = (m_a, 2\pi \ell / L_a), \quad (3.4)$$

where e_j is the normalized orbital of Section 2. Repeated points near the poles are retained as distinct slots. Their multiplicities are bounded. Write Λ_q for this set of slots. Fourier orthogonality and (3.3) give the exact identity

$$\sum_{a,\ell} |z_{a\ell}\rangle \langle z_{a\ell}| = I_{U_q}. \quad (3.5)$$

Thus $S_q \psi = (\langle z_x, \psi \rangle)_{x \in \Lambda_q}$ is an isometry from U_q to $\ell^2(\Lambda_q)$. The slots have bounded density:

$$\#\{x \in \Lambda_q : d(x, y) \leq r\} \leq C(1+r)^2. \quad (3.6)$$

The window centers have bounded density in radial physical distance, and their circles carry $O(w_a)$ slots with spacing bounded below except inside the bounded polar balls. This proves (3.6). Figure 1 shows the two scales in this construction.

We need two quantitative consequences of this construction. Let k_y be a normalized coherent orbital centered at y .

Lemma 3.1 (Uniform frame localization). *For every D there is C_D such that*

$$|\langle z_x, k_y \rangle| \leq C_D (1 + d(x, y))^{-D}. \quad (3.7)$$

For two frames constructed as above, possibly with different axes, their atoms satisfy the same bound with k_y replaced by an atom centered at y . For each fixed integer J , (3.7) also holds, with constants depending on J , for every rotated polar orbital $e_j(y)$ with $0 \leq j \leq J$.

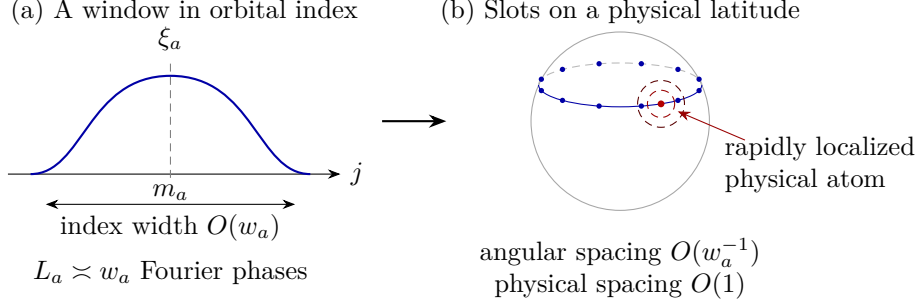


Figure 1: The index window has width comparable to w_a ; its Fourier phases place a comparable number of slots around the latitude. Together with unit-order radial spacing, this gives bounded slot density in physical distance. The drawing is schematic: dashed circles indicate rapidly decaying tails, not compact spatial support.

Proof. Choose the axial gauge and azimuth convention in which the coefficient of e_j in k_y is $e^{ij\phi_y} p_{q,m}(j)^{1/2}$. Its modulus is

$$p_{q,m}(j)^{1/2}, \quad p_{q,m}(j) = \binom{q}{j} (m/q)^j (1 - m/q)^{q-j}.$$

They obey, after extension by zero outside $[0, q]$,

$$|\Delta^l p_{q,m}(j)^{1/2}| \leq C_{D,l} w_q(m)^{-1/2-l} \left(1 + \frac{|j-m|}{w_q(m)}\right)^{-D}. \quad (3.8)$$

Here and below finite differences may shift the argument by a fixed number of indices, which only changes the constant. We include a verification to specify uniformity near the ends. First put $p = m/q$ and $\sigma^2 = qp(1-p)$, so $w_q(m)^2 = 1 + q\sigma^2/(q+1) \asymp 1 + \sigma^2$. Fourier inversion of a binomial law of variance σ^2 gives

$$\sup_j p_{q,m}(j) \leq \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 - 4p(1-p) \sin^2(t/2))^{q/2} dt \leq \frac{C}{1+\sigma}.$$

For $0 < p < 1$, tilt the law by $e^{\lambda j}$. Its new parameter is $p_\lambda = pe^\lambda/(1 - p + pe^\lambda)$, and its variance differs from σ^2 by a factor between $e^{-|\lambda|}$ and $e^{|\lambda|}$. Therefore the centered log moment generating function satisfies $\log \mathbb{E} e^{\lambda(X-m)} \leq \sigma^2 \lambda^2$ for $|\lambda| \leq 1/2$. At $z = j - m$, choose $\lambda = z/(2(\sigma^2 + |z|))$. The exact tilting identity and the preceding point-mass bound give

$$p_{q,m}(j) \leq \frac{C}{w_q(m)} \exp\left(-\frac{c|j-m|^2}{w_q(m)^2 + |j-m|}\right). \quad (3.9)$$

The cases $p = 0, 1$ are direct. This proves every polynomial tail bound with the required height, uniformly also when $w_q(m)$ is bounded.

For the derivative gain first assume $w_q(m)^2 \geq 16(l+1)$, and let $h = \min(m, q-m) \asymp w_q(m)^2$. Interpolate the square root of the binomial probability by gamma functions, writing the resulting positive function as $F(x)$. Put $\psi = (\log \Gamma)'$ and $\psi_1 = (\log \Gamma)''$. On $|x-m| \leq h/2$, the explicit formulas

$$\begin{aligned} (\log F)'(x) &= \frac{1}{2} \left(\log \frac{p}{1-p} - \psi(x+1) + \psi(q-x+1) \right), \\ (\log F)''(x) &= -\frac{1}{2} (\psi_1(x+1) + \psi_1(q-x+1)) \end{aligned}$$

and the positive-real polygamma series give $|(\log F)'(m)| \leq Cw^{-2}$, $(\log F)''(x) \leq -cw^{-2}$, and $|(\log F)^{(k)}(x)| \leq C_k w^{-k}$ for $k \geq 2$. Stirling's inequalities give $F(m) \leq Cw^{-1/2}$. Differentiating $F = e^{\log F}$ consequently proves

$$|F^{(l)}(x)| \leq C_{D,l} w^{-1/2-l} (1 + |x - m|/w)^{-D}, \quad |x - m| \leq h/2.$$

If $|j - m| \leq h/4$, integrate this derivative over the unit cube that represents $\Delta^l F(j)$; the entire stencil stays in the displayed interval. Outside that region, each nonzero stencil entry has $|j + a - m| \geq cw^2$, $0 \leq a \leq l$. Equation (3.9) then supplies an exponential in that distance, which pays both w^{-l} and every polynomial tail. This also handles stencils crossing an endpoint. Finally, if $w^2 < 16(l+1)$, bound the finite difference by its $l+1$ values and use (3.9); the factor w^{-l} is now a constant depending only on l . This completes the proof of (3.8), including the bounded-width regime.

Insert (3.8) in (3.4). For windows whose radial centers are a bounded distance apart, their widths are comparable; summation by parts l times gives the factor $(1 + w_a |\phi_x - \phi_y|)^{-l}$, with angular difference modulo 2π . The absolute sum before summation by parts is bounded. For separated radial windows, the tail in (3.8) first supplies arbitrary powers of radial distance. The ratios of the widths cost only polynomial powers by (3.2). Spending additional tail orders therefore gives the same angular estimate. Radial distance plus the shorter latitude arc bounds the spherical distance from above, proving (3.7).

The coherent resolution of the identity has density $(q+1)/(2\pi q)$ with respect to area. Inserting it between two frame atoms reduces the frame-to-frame bound to

$$\int_{S_q} (1 + d(x, z))^{-D-3} (1 + d(z, y))^{-D-3} dz \leq C_D (1 + d(x, y))^{-D}.$$

The inequality follows by dividing the integral into the sets where $d(x, z) \geq d(x, y)/2$ and where $d(z, y) \geq d(x, y)/2$. Finally

$$|\langle e_j(y), k_z \rangle|^2 = \binom{q}{j} \sin^{2j}(\vartheta/2) \cos^{2(q-j)}(\vartheta/2),$$

where ϑ is the angle between y and z . For bounded j , this is bounded by $C_j (1 + d(y, z))^{2j} e^{-cd(y, z)^2}$; increasing the constant covers bounded q . Inserting the coherent resolution once more proves the last assertion. $\square$

Lemma 3.2 (Common frames for nearby fluxes). *For each fixed d_0 , frames at fluxes q and t with $0 \leq q - t \leq d_0$ may be embedded in the same slots so that index identification $i_{q,t} : e_j^{(t)} \mapsto e_j^{(q)}$, $j \leq t$, satisfies the exact identity $S_t = S_q i_{q,t}$. All preceding estimates remain uniform. In the northern third the windows can be chosen from one system independent of the flux; there one may use the common metric with radial variable $\sqrt{1+m}$.*

Proof. Use the windows and Fourier lengths at flux q for both spaces, and restrict their index sequences to $0 \leq j \leq t$ for the smaller space. Equation (3.5) still holds on those indices. Only a bounded number of end indices have been removed. The windows meeting them have bounded width, or have rapidly small tails in the estimates in which an endpoint extension occurs. The slot positions for the two sphere metrics differ by a bounded distance; the constants may depend on d_0 . For all sufficiently large q , every window centered in the northern third has its full support below $q/2$, because its width is $O(\sqrt{q})$. On these supports the construction uses only $\eta(\sqrt{1+j} - k)$ and

their fixed normalization, and their Fourier length is the fixed integer $16(k+1)$. These complete slot vectors are therefore independent of q in index coordinates. The buffer between $q/3$ and $q/2$ permits changing the frame outside this region without changing its northern slots. Bounded fluxes may use pairwise padded frames; no long sequence of common slots is needed there. $\square$

Whenever operators at these two fluxes are subtracted, we first use $i_{q,t}$ and extend the smaller operator by the identity on the added orbital modes. Thus, under the graded decomposition $\mathcal{F}_q = \mathcal{F}_t \hat{\otimes} \wedge \text{span}\{e_{t+1}, \dots, e_q\}$, the padded operator is $A \hat{\otimes} I$. Extra modes are put in their vacuum only when comparing states or taking the physical compression.

The fermionic Fock space over $\ell^2(\Lambda_q)$ is a graded tensor product of two-dimensional slot spaces. The exterior isometry $\Gamma(S_q)$ identifies $\mathcal{F}_q$ with the subspace in which the orthogonal complement of $S_q U_q$ is vacant. A *physical* operator is an operator in the CAR algebra generated by $S_q U_q$, extended as the identity on the complementary modes. It preserves that vacant-complement subspace. For $X \subset \Lambda_q$, let $\mathfrak{A}(X)$ be the CAR algebra of its slots. Put $\mathcal{N}_{\text{slot}} = \sum_{x \in \Lambda_q} c_x^* c_x$. An operator has number grade $k \in \mathbb{Z}$ if $[\mathcal{N}_{\text{slot}}, A] = kA$, and parity $\epsilon_A \in \{0, 1\}$ if $(-1)^{\mathcal{N}_{\text{slot}}} A (-1)^{\mathcal{N}_{\text{slot}}} = (-1)^{\epsilon_A} A$. Operators of even parity on disjoint sets commute; homogeneous operators obey the graded relation

$$[A, B]_{\text{gr}} = AB - (-1)^{\epsilon_A \epsilon_B} BA = 0.$$

All Hamiltonians below have even parity. Local expansions can be projected onto any prescribed number grade by averaging the gauge action of the total slot number $\mathcal{N}_{\text{slot}}$. This action preserves every slot algebra and does not increase approximation errors. On physical operators its grading agrees with that of the physical number operator $\mathcal{N}$.

3.2 Local norms and summation

For a center x , an operator A , and $l > 0$, define

$$\|A\|_{x,l} = \|A\| + \sup_{r \geq 1} (1+r)^l \inf_{A_r \in \mathfrak{A}(B_r(x))} \|A - A_r\|. \quad (3.10)$$

We call A rapidly local at x if this is finite for every l , with bounds uniform in the volume under consideration. An interaction is a family A_x integrated against a measure whose mass in every unit ball is bounded. A bound of *size* $a(x)$ means $\|A_x\|_{x,l} \leq C_l a(x)$ for every l . A positive function a is called tempered if, for fixed C, κ ,

$$a(x) \leq C a(y) (1 + d(x, y))^\kappa. \quad (3.11)$$

The constants in estimates involving a may depend on C, κ . Powers and products of u, v, w are examples.

Taking near-best approximations at radii 2^k and subtracting successive ones gives a ball decomposition

$$A = \sum_{k \geq 0} A^{(k)}, \quad A^{(k)} \in \mathfrak{A}(B_{2^{k+1}}(x)), \quad \|A^{(k)}\| \leq C_l \|A\|_{x,l} 2^{-kl}. \quad (3.12)$$

Conversely such a decomposition implies the approximation bound, with any slightly smaller exponent. We use this elementary equivalence to pass between operators and interactions on balls.

An observable supported initially on a large ball needs a separate convention. A family has *core radius* r and *amplitude* a if

$$\|A\| \leq Ca, \quad \inf_{A_R \in \mathfrak{A}(B_R(x))} \|A - A_R\| \leq C_D a (1 + R)^{-D} \quad (R \geq C_* r), \quad (3.13)$$

where the geometric constant C_* is independent of D . The amplitude may contain a specified power of r . This convention does not assert that every centered norm (3.10) is bounded by Ca without a further radius factor.

Lemma 3.3 (Shells outside a fixed core). *If (3.13) holds with $r \geq 1$, there is a decomposition $A = \sum_{\nu \geq 0} A_\nu$ with $A_\nu \in \mathfrak{A}(B_{C_* 2^{\nu+1} r}(x))$ and $\|A_\nu\| \leq C_D a 2^{-\nu D}$ for every D . Consequently, for each fixed $B \geq 0$,*

$$\sum_{\nu \geq 0} (C_* 2^{\nu+1} r)^B \|A_\nu\| \leq C_B a r^B. \quad (3.14)$$

Proof. Take approximants at radii $C_* 2^\nu r$ and telescope, starting with the first approximant as the core. The core norm is Ca ; the other differences are bounded by the two adjoining errors in (3.13). Sum with any $D > B + 1$ to obtain (3.14). $\square$

Lemma 3.4 (Products, commutators, and weighted sums). *The following bounds are uniform over the slot systems above.*

- (i) *Products of operators rapidly local at the same center, and adjoints, preserve rapid locality. If A, B are centered at x, y , respectively, with bounded local norms, then $\|[A, B]_{\text{gr}}\| \leq C_D (1 + d(x, y))^{-D}$ for every D .*
- (ii) *If A is rapidly local at x and a rapidly local interaction K_y has tempered size $a(y)$, then $[\int K_y dy, A]_{\text{gr}}$ is rapidly local at x , of size $a(x)$ times the local bounds of A .*
- (iii) *Suppose A is even, $\|A\|_{x,l} \leq 1$, and J_y has bounded rapid local norms. With $D = 1 + d(x, y)$ and $C_{xy} = [A, J_y]$,*

$$\|C_{xy}\| \leq CD^{-l}, \quad \|C_{xy}\|_{x,l} \leq C, \quad \|C_{xy}^* C_{xy}\|_{x,l} \leq CD^{-l}. \quad (3.15)$$

For an operator initially supported in $B_r(x)$, the corresponding core estimates cost a fixed polynomial in r . At distances larger than a fixed multiple of r , any prescribed decay order is available.

Proof. The product estimate follows from

$$\|AB - A_r B_r\| \leq \|A - A_r\| \|B\| + \|A_r\| \|B - B_r\|.$$

For the commutator use approximations of radius less than $d(x, y)/3$; their graded commutator vanishes. For the weighted assertion expand both factors as in (3.12). Terms with disjoint balls vanish, while intersecting balls of radii R, S require $d(x, y) \leq R + S$. The number or measure of their possible centers is $O((1 + R + S)^2)$, and (3.11) costs at most $C(1 + R + S)^\kappa$. To obtain an output moment of order D , bound these costs by $C(1 + R + S)^{D+\kappa+2}$ and choose input decay order larger than $D + \kappa + 6$. The two geometric sums then converge. This argument also permits assigning each output term to the center of either input; that convention will be used for differences of interactions.

For (3.15), the first estimate follows by the disjoint-ball argument at exponent l . For approximation radii $r \leq 3D$, use zero as an approximant to C_{xy} , giving $r^l \|C_{xy}\| \leq C$. For $r > 3D$, approximate A in $B_{r/3}(x)$ and J_y in $B_{r/3}(y)$; their commutator is supported in $B_r(x)$ and has error Cr^{-l} . Thus the second estimate holds. Apply the product estimate with the small norm $\|C_{xy}\| \leq CD^{-l}$ on one factor to obtain the third estimate. Finally a ball $B_r(x)$ contains $O((1 + r)^2)$ slots, and tempered ratios within it cost $O((1 + r)^\kappa)$. The same sums give the stated polynomial core costs. For tails beyond Cr , the distance from the original ball is comparable to the tail radius, so increasing a tail order does not increase the core-volume exponent. $\square$

Corollary 3.5 (The physical interactions). *The unperturbed Hamiltonian H_q and the Fock lift of any bounded real scalar symbol φ have physical, neutral, Hermitian rapid interactions of uniform sizes C_D and $C_D\|\varphi\|_\infty$, respectively.*

Proof. Coefficient localization gives operator-norm localization directly. If E_R is the one-particle projection onto slots in $B_R(x)$, the CAR identity $\|c(f)\| = \|f\|_2$ gives

$$\|c(S_q f) - c(E_R S_q f)\| = \|(I - E_R)S_q f\|_2.$$

For $f = k_x$ or a bounded polar mode at x , Lemma 3.1 and the slot count (3.6) make the right side at most $C_D(1 + R)^{-D}$ for every D ; the approximating annihilator is supported in $B_R(x)$.

The coherent resolution gives

$$d\Gamma(T_q(\varphi)) = \frac{q+1}{2\pi q} \int_{S_q} \varphi(x) c^*(k_x) c(k_x) dA(x).$$

For H_q , the highest pair vector is $v_0 = e_0 \wedge e_1$. Schur's Lemma on the spin- $(q-1)$ pair space and its dimension $2q-1$ give, with Haar mass one,

$$H_q = (2q-1) \int_{SU(2)} B(gv_0)^* B(gv_0) dg. \quad (3.16)$$

The product $B(gv_0) = c(ge_1)c(ge_0)$ is rapidly local at the rotated pole by Lemmas 3.1 and 3.4. Its phase along the stabilizer disappears in its square, so the center measure on the sphere has density $(2q-1)/(2\pi q)$. This and $(q+1)/(2\pi q)$ are uniformly bounded. The same two lemmas localize $c^*(k_x)c(k_x)$. No derivative of φ enters either the representation or its bound. $\square$

3.3 Propagation and spectral filters

We next establish that a time filter preserves this local calculus. The relevant propagation bound is needed only with arbitrary polynomial decay, which permits a direct truncation argument. The underlying commutator iteration is the Lieb–Robinson method [8], in its graded fermionic form [13]; its use with time filters is the locality mechanism behind quasi-adiabatic continuation and spectral flow [6, 2].

Lemma 3.6 (Propagation of local observables). *Let K be self-adjoint and even, with a rapidly local interaction of uniform size. Write $\tau_t^K(A) = e^{itK} A e^{-itK}$. For each D there are D' and M_D , independent of volume, such that*

$$\|\tau_t^K(A)\|_{x,D} \leq C_D(1 + |t|)^{M_D} \|A\|_{x,D'}. \quad (3.17)$$

The assertion holds also for time-dependent generators with the same uniform bounds. In particular evolution over a bounded parameter interval preserves rapid locality. If A is supported in $B_r(x)$, the error in approximating $\tau_t^K(A)$ on $B_R(x)$, for $R \geq C(r + (1 + |t|)^4)$, is bounded by $C_D\|A\|(1+r)^2 R^{-D}$.

Proof. Use (3.12) to write K as a sum of terms on balls. Taking Hermitian and even parts makes each term Hermitian and even without changing the total or its estimates. First retain only terms of diameter at most ℓ . For this truncated interaction define

$$M_{ab} = 2 \sum_{X \ni a,b} \|K_X\|.$$

Uniform summability, including the number of sites in X , gives $\sup_a \sum_b M_{ab} e^{d(a,b)/\ell} \leq C$; indeed each retained distance is at most ℓ , and the ball-volume costs are polynomial. The iterated commutator inequality, with internal evolution removed unitarily at each step, therefore gives

$$\|[\tau_t^{K \leq \ell}(A), B]_{\text{gr}}\| \leq C \|A\| \|B\| |X| |Y| e^{C|t| - d(X,Y)/\ell} \quad (3.18)$$

for $A \in \mathfrak{A}(X)$ and $B \in \mathfrak{A}(Y)$. For completeness, the n th iterated integral contributes $|t|^n/n!$ times a chain of n matrix factors M . Multiplication by $e^{d(X,Y)/\ell}$ is bounded by the product of the exponential weights along the chain. Summing the intermediate sites gives C^n , and summing n gives (3.18). Evenness of the generator makes the same calculation valid for odd A, B with the graded commutator.

Restore the removed terms using Duhamel's formula. For each removed ball term use the smaller of the trivial commutator bound and (3.18). The centers within distance $C\ell(1 + |t|)$ of X , together with the radius of that term, occupy at most a polynomial volume. Outside that region the exponential bound is summable. Consequently, for every s , the restoration error is at most

$$C_s \|A\| |X| (1 + |t|)(1 + r + \ell(1 + |t|))^4 \ell^{-s}, \quad (3.19)$$

when $X \subset B_r(x)$. To see the power allowance explicitly, a removed term of radius a contributes its norm times at most $C(1 + r + \ell(1 + |t|) + a)^2$ possible centers and at most $C(1 + a)^2$ sites. Its arbitrarily high radius moment makes the sum over $a > \ell/2$ bounded by the right side of (3.19), after increasing the input moment by $s + 6$.

Now let $R \geq C(r + (1 + |t|)^4)$ and choose $\ell = R^{1/3}$. The truncated evolution differs from the evolution generated by its terms contained in $B_R(x)$ by $C\|A\||X|\text{poly}(R)e^{C|t| - cR^{2/3}}$. This is again Duhamel's formula and (3.18), summed over the boundary terms. Both restoration errors are bounded by $C_D\|A\||X|R^{-D}$, choosing s sufficiently large in (3.19). This supplies an approximation in $\mathfrak{A}(B_R(x))$. At smaller radii use the norm bound $\|\tau_t(A)\| = \|A\|$. Since $|X| \leq C(1 + r)^2$, this proves the assertion for supported A and then, by (3.12), (3.17). The proof uses only uniform bounds on the interaction at each time, so also proves the time-dependent assertion. $\square$

Lemma 3.7 (Time filters). *Suppose $F \in L^1(\mathbb{R})$ and $\int (1 + |t|)^j |F(t)| dt < \infty$ for every j . Under the hypotheses of Lemma 3.6,*

$$\mathcal{T}_{F,K}(A) = \int_{\mathbb{R}} F(t) \tau_t^K(A) dt \quad (3.20)$$

is rapidly local whenever A is, with uniform bounds. The same is true for a finite measure consisting of such a function and a mass at zero. Number grade and physicality are preserved when K is neutral and physical. Suitable symmetry of F preserves adjoints.

Proof. Integrate (3.17). The approximation norm is a seminorm up to its operator-norm summand, so the integral of local approximants gives the required estimate. The remaining assertions follow directly by commuting the evolution with gauge transformations, the physical algebra, and adjoints. $\square$

The frequency functions used below are smooth, vanish on an interval about zero, and have tails proportional to $1/\omega$ on one or both half-lines. Their inverse Fourier transforms satisfy the hypotheses of Lemma 3.7. Here is a convenient verification. Every positive-order derivative of such a multiplier is integrable, so integration by parts gives arbitrary powers of decay for $|t| \geq 1$. For $0 < |t| < 1$, split the frequency integral at $|\omega| = |t|^{-1}$. The inner portion is $O(1 + |\log |t||)$, and one integration by parts on the outer portion is $O(1)$. Thus the possible singularity at zero is integrable. A smooth compactly supported multiplier has a Schwartz kernel. These facts cover both spectral transport and the filters that retain only small energy differences.

Lemma 3.8 (Weighted differences). *Let K, K' satisfy Lemma 3.6 uniformly, and assume $K - K'$ has a rapidly local interaction of tempered size $a(z)$. For a rapidly local A centered at x ,*

$$\|\tau_t^K(A) - \tau_t^{K'}(A)\|_{x,D} \leq C_D a(x)(1 + |t|)^{M_D} \|A\|_{x,D'}. \quad (3.21)$$

The same conclusion holds after an all-moments time filter, and for differences of bounded-parameter evolutions. If the input itself changes by local size $b(x)$, its separate contribution is $Cb(x)$. All estimates retain the product of a local input size and $a(x)$. If $A \in \mathfrak{A}(B_r(x))$, put $s = \kappa + 2$, with κ from (3.11). The difference has operator norm at most

$$Ca(x)\|A\|(1+r)^s|t|(1+|t|)^{4s}. \quad (3.22)$$

For $R \geq C_(1+r)$, with a fixed geometric constant C_* , it has an approximation supported in $B_{r+R}(x)$ with error at most*

$$C_D a(x)\|A\|(1+r)^s(1+|t|)^{4D+4s+1}(1+R)^{-D}. \quad (3.23)$$

Thus the power of the initial radius is independent of the tail order. If the differing interaction vanishes throughout a ball of radius d about x , the bounds for a fixed rapidly local input gain arbitrary inverse powers of $1+d$.

Proof. Use the exact formula

$$\tau_t^K(A) - \tau_t^{K'}(A) = i \int_0^t \tau_s^K([K - K', \tau_{t-s}^{K'}(A)]) ds.$$

Apply Lemma 3.6 to the inside observable, Lemma 3.4(ii) to the commutator, and Lemma 3.6 once more to the outside evolution. The tempered inequality (3.11) is precisely what allows the insertion at z to be charged to $a(x)$. All time costs are polynomial and therefore integrable against the specified filters. The time-dependent version follows from its identical Duhamel formula. Linearity handles a changed input.

We give the additional radius bookkeeping for (3.22)–(3.23). For a ball decomposition about x , use the weighted sum $\sum_R (1+R)^p \|A_R\|$. Lemma 3.6 bounds its evolved value by $C_p(1+|t|)^{4p}$ times its initial value. Indeed, for a radius- r input, begin a dyadic decomposition at $Q = C(1+r+(1+|t|)^4)$. The core contributes $C\|A\|Q^p$; the subsequent shells contribute at most $C_M\|A\|(1+r)^2 \sum_{R \geq Q} R^{p-M}$. Choosing $M > p+2$ bounds this by $C_p\|A\|Q^p$. The weighted insertion estimate in the proof of Lemma 3.4 costs $s = \kappa + 2$ powers of the input radius. Duhamel at $p = 0$ proves (3.22).

For the tail first take $|t| \leq cR^{1/4}$ in the Duhamel integrand, with $c > 0$ a sufficiently small fixed constant; bounded R are covered by (3.22). Fix also a small $\varepsilon > 0$. Choose ε, c and then C_* so that the propagation lemma applies both from radius r to $r + \varepsilon R$ and from radius $r + 3\varepsilon R$ to $r + R$. This requires only fixed constants: if C is the constant in that lemma, take $\varepsilon < (100C)^{-1}$, $c^4 < \varepsilon/(10C)$, and $C_* > 100C/\varepsilon$. Cut the inner evolved observable at radius $r + \varepsilon R$. Its remaining shells have weighted sum of order s at most $C_M\|A\|(1+r)^2 R^{s-M}$, by the short-time bound in Lemma 3.6; their commutator with the insertion has the same bound times $Ca(x)$. In the retained part, discard insertion terms of radius greater than εR . Their arbitrary radius moments give an error $C_M a(x)\|A\|R^{-M}$. The remaining commutator is supported in $B_{r+3\varepsilon R}(x)$ and has norm at most $Ca(x)\|A\|R^s$. Evolving it in the outside factor of Duhamel and using the short-time propagation estimate gives an approximation on $B_{r+R}(x)$ with error $C_M a(x)\|A\|R^{s+2-M}$. Choose M in each of these three bounds larger than the requested order plus $s+4$, and integrate over an

interval of length at most $R^{1/4}$. This gives the stated tail bound for short times. For $|t| > cR^{1/4}$, use (3.22) and $(1 + |t|)^{4D}(1 + R)^{-D} \geq c_D$.

Finally suppose an insertion ball of radius a can occur only if its center z satisfies $d(x, z) + a \geq d$. If that ball meets an input shell of radius R , then $d \leq R + 2a$. Inserting $(1 + d)^{-M}(1 + R + 2a)^M$ into the shell summation proves the last assertion, after increasing the input moments. This includes changes supported outside a common comparison region. $\square$

To specify physical compression, let $S : U_q \rightarrow K_S$ and $T : U_q \rightarrow K_T$ be two slot analysis isometries. Write $J_S = \Gamma(S)$ for the vacuum-complement embedding and ι_T for the embedding of the physical CAR algebra into the output slots, acting as the identity on complementary modes. The map that we estimate is

$$\mathcal{C}_{T \leftarrow S}(A) = \iota_T(J_S^* A J_S). \quad (3.24)$$

When $T = S$ this is compression followed by identity extension; it is a unital contractive completely positive map. On physical input operators it is simply change of frame.

Lemma 3.9 (Compression, changes of frame, and smooth functions). *The map (3.24), for the frames of Lemma 3.1, preserves rapid locality. An operator supported in $B_r(x)$ remains localized about that ball, with polynomial core costs and arbitrary inverse-power tails outside $B_{Cr}(x)$. If $A = A^*$ is bounded and rapidly local, and f is smooth on a neighborhood of its spectrum, then $f(A)$ is rapidly local. The constants depend on finitely many local bounds and derivatives for each requested decay order.*

Proof. Put two slot spaces side by side. If S, T are their analysis isometries, the partial isometry TS^* has rapidly decaying matrix entries by Lemma 3.1. On their direct sum, the anti-self-adjoint one-particle operator

$$\begin{pmatrix} 0 & -ST^* \\ TS^* & 0 \end{pmatrix}$$

rotates the two physical subspaces into each other through angle $\pi/2$ and acts as zero on their complements. Its second quantization is a bounded-strength, rapidly local quadratic interaction: absolute row sums and every distance moment are bounded by (3.6). Lemma 3.6 applies over this fixed angle. Conjugate $A \otimes I$ by this rotation and take the vacuum expectation in the entire first slot copy, retaining the entire second copy. On a CAR monomial, the physical components have moved from SU_q to TU_q , while the complementary components in the first copy are evaluated in their vacuum. The resulting map is exactly (3.24), including for odd monomials; linearity gives the identity for every operator. Vacuum expectation is contractive and preserves support in the doubled slot geometry. This proves the assertion, and $S = T$ gives re-embedded physical compression. In particular no projection onto the output complementary vacuum is left in the resulting operator.

For the functional-calculus assertion extend f smoothly to a compactly supported function on $\mathbb{R}$. If A_r approximates A locally, its Hermitian part does so with the same error. The Fourier formula and Duhamel's identity give

$$\|f(A) - f(A_r)\| \leq \left(\int_{\mathbb{R}} |t| |\hat{f}(t)| dt \right) \|A - A_r\|.$$

Since $f(A_r)$ belongs to the same local algebra, the approximation bounds follow. A constant value of f contributes only a scalar identity, which is local in every ball. $\square$

3.4 Pair rows and diagonal symbols

The coherent orbit representations already prove locality of the unperturbed Hamiltonian and bounded scalar perturbations. Pinning zeros will deform the pair coefficients and remove full rotation invariance. To control those deformations and compare neighboring fluxes, we now prove a local calculus directly for angular-momentum rows and diagonal orbital symbols.

Write $Q_b = B_{b-1}$ for the normalized pair annihilator with index sum b in (2.5), $1 \leq b \leq 2q - 1$. Define its ordered coefficients by

$$a_b(i) = \frac{(b-2i)\sqrt{\binom{q}{i}\binom{q}{b-i}}}{\sqrt{S_b}}, \quad S_b = \binom{2q}{b} \frac{b(2q-b)}{2q-1}. \quad (3.25)$$

In this formula the sum for S_b is over ordered i . With a fixed choice of annihilator sign, $Q_b = 2^{-1/2} \sum_{i+j=b} a_b(i) c_j c_i = \sqrt{2} \sum_{i < j, i+j=b} a_b(i) c_j c_i$. Changing this common sign leaves every square unchanged.

Lemma 3.10 (Pair profiles and local row squares). *With $m = b/2$ and $w = w_q(m)$, the zero-extended coefficients satisfy*

$$|\Delta_i^r \Delta_b^s a_b(i)| \leq C_{D,r,s} w^{-1/2-r-s} \left(1 + \frac{|i-m|}{w}\right)^{-D}. \quad (3.26)$$

There are pair annihilators $L_{\nu,\phi}$, rapidly local with uniform bounds about centers $x_{\nu,\phi}$, and a measure $d\mu$ of bounded mass per unit ball, such that

$$H_q = \sum_{\nu} \int_0^{2\pi} L_{\nu,\phi}^* L_{\nu,\phi} d\mu_{\nu}(\phi). \quad (3.27)$$

The construction is linear in the row coefficients. In particular symbol bounds (3.26) with an additional tempered factor give local bounds with that factor; the same holds for differences of row families in common frames.

Proof. Vandermonde's identity identifies $\binom{q}{i}\binom{q}{b-i}/\binom{2q}{b}$ as a hypergeometric law of mean $b/2$ and variance $b(2q-b)/(4(2q-1))$. Multiplying its second centered moment by four proves (3.25). The moment formula itself also follows directly by applying $i\binom{q}{i} = q\binom{q-1}{i-1}$ twice and Vandermonde's identity.

Here are symbol estimates that include the dependence on the moving row center. Set

$$\ell_q(x) = \log \Gamma(q+1) - \log \Gamma(x+1) - \log \Gamma(q-x+1), \quad t = i - m,$$

and

$$E(m, t) = \frac{1}{2} \{\ell_q(m+t) + \ell_q(m-t)\} - \ell_q(m), \quad T(m) = S_{2m} e^{-2\ell_q(m)},$$

where the gamma formula defines the smooth extension. Then $a_b(i) = -2te^{E(m,t)}/\sqrt{T(m)}$. The gamma duplication formula gives the exact expression

$$T(m) = \frac{4m(q-m)}{2q-1} \sqrt{\pi} \frac{R(m)R(q-m)}{R(q)}, \quad R(x) = \frac{\Gamma(x+1)}{\Gamma(x+1/2)}. \quad (3.28)$$

For $m, q-m \geq 1/2$, Stirling's inequalities yield $T(m) \asymp w_q(m)^3$. Its logarithmic derivatives of order $r \geq 1$ are bounded by $C_r(u^{-r} + v^{-r})$. These derivative assertions can be obtained directly by differentiating Euler's gamma product: for $k \geq 1$,

$$\frac{d^{k+1}}{dx^{k+1}} \log \Gamma(x) = (-1)^{k+1} k! \sum_{n=0}^{\infty} (x+n)^{-k-1}.$$

In the region $|t| \leq c \min(u, v)$, the second derivative ℓ_q'' is negative and comparable in magnitude to $u^{-1} + v^{-1} \asymp w^{-2}$. Thus $E(m, t) \leq -ct^2/w^2$. Differentiating E in i and b gives polynomials in t/w , with one factor w^{-1} per derivative; (3.28) has the same property. The discrete product rule, or integration of these derivative bounds over unit cubes, proves (3.26) in this region. There is also a global Gaussian bound, including the ends: convexity of the trigamma function $\psi_1 = (\log \Gamma)''$ and $\psi_1(x) \geq 1/x$ give

$$E_{tt}(m, t) = -\frac{1}{2} \{ \psi_1(m+t+1) + \psi_1(m-t+1) + \psi_1(q-m+t+1) + \psi_1(q-m-t+1) \} \leq -\frac{1}{u} - \frac{1}{v} \leq -\frac{c}{w^2}.$$

Since $E(m, 0) = E_t(m, 0) = 0$, we have $E(m, t) \leq -ct^2/w^2$ on the entire allowed interval. Outside the central region $|t|/w \geq cw$ when w is large, so increasing the tail order pays all derivative and endpoint-extension factors. The rows with bounded w require only bounded finite differences. This proves the full uniform estimate without differentiating a lattice normalization sum.

Choose smooth windows $\zeta_\nu(b)$ with $\sum_\nu \zeta_\nu(b)^2 = 1$, supported on intervals of length $O(W_\nu)$ around $b = 2M_\nu$, where $W_\nu = w_q(M_\nu)$. The construction (3.3), with $m = b/2$, supplies them. Define

$$L_{\nu, \phi} = W_\nu^{-1/2} \sum_b \zeta_\nu(b) e^{-ib\phi} Q_b, \quad d\mu_\nu(\phi) = W_\nu \frac{d\phi}{2\pi}. \quad (3.29)$$

Fourier orthogonality proves (3.27) exactly. The radial centers have bounded density in physical distance, while their circle lengths are comparable to W_ν outside the polar balls. Hence μ has bounded mass on unit balls.

After removing its linear phase $e^{-i(i+j)\phi}$, the two-index coefficient of $L_{\nu, \phi}$ has height $O(W_\nu^{-1})$, and every mixed finite difference of orders r, s costs W_ν^{-r-s} . Its tails are arbitrary powers of $1 + |i - M_\nu|/W_\nu + |j - M_\nu|/W_\nu$. To express it in slot fermions, multiply in each index by the window coefficients from (3.4) and sum. On comparable windows, repeated summation by parts in both indices gives, for arbitrary D , coefficients bounded by

$$C_D (1 + d(x, x_{\nu, \phi}))^{-D} (1 + d(y, x_{\nu, \phi}))^{-D}$$

for the slot monomial $c_x c_y$. The height is bounded because the two summation lengths are $O(W_\nu)$, the coefficient height is $O(W_\nu^{-1})$, and the two Fourier normalizations contribute $O(W_\nu^{-1})$. Noncomparable windows have radial separation; their scale ratios cost polynomial powers by (3.2), paid by increasing the radial tail orders. This proves the displayed bound for all slots. Its absolute sum, including every distance moment, is bounded by (3.6). Since $\|c_x c_y\| \leq 1$, truncating that sum proves rapid locality in operator norm. Products give locality of $L_{\nu, \phi}^* L_{\nu, \phi}$. Each operation used was linear in the row coefficients before the final square; keeping any extra tempered factor proves the last assertion. $\square$

For a smooth real function a on $[0, q]$, fix a constant C_0 larger than the window-support constants and define

$$\mathfrak{s}_K(a; m) = \max_{0 \leq k \leq K} \sup_{\substack{j \in [0, q] \\ |j-m| \leq C_0 w_q(m)}} w_q(m)^k |a^{(k)}(j)|. \quad (3.30)$$

The order-zero term is part of this seminorm. Scaled derivatives, rather than merely bounded unscaled derivatives, are necessary: orbital oscillations on the scale of one index translate an operator around a macroscopic latitude.

Lemma 3.11 (Diagonal symbols). *Suppose $A(m) > 0$ is tempered with exponent κ and $\mathfrak{s}_K(a; m) \leq C_K A(m)$ for every integer $K \geq 0$. Then $D_q(a) = \sum_j a(j) c_j^* c_j$ has a physical, neutral, Hermitian interaction of size $A(m)$. For each output approximation order D , only the seminorms through $K_D = \lceil 4D + 4\kappa + 40 \rceil$ are needed. The construction and the estimate are linear in the symbol, so the same assertion holds for differences of symbols.*

Proof. Insert the analysis map on both sides of the diagonal one-particle matrix. The slot matrix elements are

$$(L_{a_1} L_{a_2})^{-1/2} \sum_j a(j) \xi_{a_1}(j) \xi_{a_2}(j) e^{ij(\phi_2 - \phi_1)}.$$

The two radial windows must overlap, so their widths are comparable and their radial physical separation is bounded. Taking K finite differences of the summand and summing by parts gives the bound $C_K A(m_1) (1 + d(x_1, x_2))^{-K}$. The normalization cancels the length of the summation interval. At bounded-width end windows the same statement follows from finite sums; all involved slots are inside a bounded polar ball.

Assign the quadratic monomial to x_1 , and pair it with its adjoint to obtain Hermitian terms. Vacuum-compress each assigned term using (3.24). Compression fixes the physical total $D_q(a)$, so the resulting decomposition is still exact and now each term is physical. A monomial joining points at distance R is supported in $B_{1+R}(x_1)$; the compression lemma bounds its approximation norm of order D by $C_D (1 + R)^{D+2}$. The slot count (3.6), the tempered ratios, and $K > D + \kappa + 6$ therefore make the assigned absolute sum converge with the required D th moment. The stated value of K_D more than covers these losses and the fixed finite differences of the windows. The grade and reality are preserved by compression. Every step is linear before pairing adjoints, which also proves the difference assertion. $\square$

We record explicitly how the symbol criterion applies after subtracting an affine function. Let $\rho(j, m)$ denote meridional distance between the latitudes of indices j, m , and fix a center x at index m . Suppose there are $\beta > 0$, $\kappa \geq 0$, and constants C_k such that, for every $k \geq 2$ and $j \in [0, q]$,

$$w_q(j)^k |a^{(k)}(j)| \leq C_k \beta (1 + \rho(j, m))^\kappa. \quad (3.31)$$

Define $\tilde{a}_x(j) = a(j) - a(m) - a'(m)(j - m)$. Taylor's formula and (3.2) imply

$$\mathfrak{s}_K(\tilde{a}_x; m_y) \leq C_K \beta (1 + d(x, y))^{\kappa+8}. \quad (3.32)$$

Here is the bound for the two orders not already present in (3.31). The index excursion between m and j is at most a constant times $w_q(j)(1 + \rho(j, m))^2$, and the ratio of $w_q(j)$ to the minimum of w_q on that meridian segment is polynomially bounded by $(1 + \rho(j, m))^2$. Integrating a'' once or twice therefore bounds $w_q(j) |\tilde{a}'_x(j)|$ and $|\tilde{a}_x(j)|$ by the right side of (3.32). On each window the widths are comparable and meridional distance changes by a bounded amount, proving the displayed seminorm bound.

Lemma 3.11 and the weighted commutator estimate now show that $[D_q(\tilde{a}_x), O_x]$ has rapid local size $C\beta$ whenever O_x has bounded rapid local norms at x . In applications one can take β comparable to $w_q(m)^2 |a''(m)|$ and verify (3.31) directly. For an annular sum commuting with $\mathcal{N}$ and $\sum_j j c_j^* c_j$, replacing a by $\tilde{a}_x$ leaves the commutator unchanged: the removed operator is a linear combination of those two conserved quantities.

Commutators, evolution, and spectral functions of physical operators remain physical, and neutral evolution preserves number grade. The strictly supported approximants used to measure locality need not preserve the vacant-complement subspace; physical vacuum compression is

available when a physical representative is required. The frame may therefore be fixed while proving an operator estimate, changed when rotating an observable, or chosen compatibly at nearby fluxes. Lemmas 3.7 and 3.8 ensure that the same local estimates remain available after the spectral transports used next.

4 Pinning zeros and transporting weighted observables

We compare zero modes at fluxes differing by one or three. Multiplication by a zero at the south pole gives the algebraic comparison, but is not unitary. We construct a unitary comparison whose effect on a local observable is small away from that pole. For a convex orbital weight we also retain a sign in this comparison. That sign will be used in the first charge estimate in Section 6.

Fix $d \in \{1, 3\}$ and put $t = q - d$. The identification of orbital labels $0, \dots, t$ embeds U_t isometrically into U_q ; write $\iota_{q,d}$ for its exterior-algebra extension. Its image has the last d orbitals empty. Set

$$g_{q,d}(j) = \frac{1}{2} \log \frac{\binom{t}{j}}{\binom{q}{j}} = \frac{1}{2} \sum_{r=0}^{d-1} \log \frac{q-j-r}{q-r}, \quad 0 \leq j \leq t. \quad (4.1)$$

Multiplication of a polynomial by the d south-pole zero factors, one factor for each particle, is $\iota_{q,d} \exp D_t(g_{q,d})$, up to a scalar on each number sector. In particular, the subspace of $Z_{n,q}$ having these zeros is

$$\iota_{q,d} \exp D_t(g_{q,d}) Z_{n,t}. \quad (4.2)$$

Throughout this section, $m, u = 1 + m, v = 1 + q - m$, and $w = (uv/(q+1))^{1/2}$ use the larger flux q . Replacing q by t in these size functions changes them by bounded factors. For every integer $\ell \geq 1$, the smooth extension of Equation (4.1) satisfies, for $0 \leq m \leq t$,

$$g_{q,d}^{(\ell)}(m) = -\frac{(\ell-1)!}{2} \sum_{r=0}^{d-1} (q-m-r)^{-\ell}, \quad |g_{q,d}^{(\ell)}(m)| \leq C_\ell v^{-\ell}, \quad -g_{q,d}''(m) \asymp v^{-2}. \quad (4.3)$$

Here and below constants can depend on d and on a stated differentiation or localization order, but not on q, n , or m .

4.1 A uniformly gapped path

For $1 \leq b \leq 2t-1$, let $B_{t,b}$ be the normalized pair annihilator whose coefficient at $i < j, i+j=b$, is proportional to $(j-i)\sqrt{\binom{t}{i}\binom{t}{j}}$. For $0 \leq s \leq 1$ define

$$\begin{aligned} B_{q,d,b}(s) &= \sum_{\substack{0 \leq i < j \leq t \\ i+j=b}} a_{q,d,b,s}(i,j) c_j c_i, \\ a_{q,d,b,s}(i,j) &= \frac{a_{t,b}(i,j) e^{-s(g_{q,d}(i)+g_{q,d}(j))}}{\left(\sum_{k < l, k+l=b} |a_{t,b}(k,l)|^2 e^{-2s(g_{q,d}(k)+g_{q,d}(l))} \right)^{1/2}}. \end{aligned} \quad (4.4)$$

and

$$H_{q,d}(s) = \sum_b B_{q,d,b}(s)^* B_{q,d,b}(s), \quad P_{q,d}(s) = \text{the orthogonal projection onto } \ker H_{q,d}(s).$$

All these operators act on $\mathcal{F}_t$. Direct conjugation of each annihilator gives

$$\ker H_{q,d}(s) = e^{sD_t(g_{q,d})} \ker H_t. \quad (4.5)$$

The normalization in Equation (4.4) only multiplies each row by a positive scalar and therefore does not affect this identity.

Lemma 4.1 (Gap along the pinning path). *There exist q_{pin} and $\gamma_{\text{pin}} > 0$ such that*

$$H_{q,d}(s) \geq \gamma_{\text{pin}}(1 - P_{q,d}(s)) \quad (q \geq q_{\text{pin}}, d \in \{1, 3\}, 0 \leq s \leq 1)$$

on the entire Fock space $\mathcal{F}_{q-d}$.

Proof. Fix an allowed unperturbed Fock gap $\gamma > 0$. We compare columns of annihilators, rather than Hamiltonians. This avoids a factor equal to the number of particles.

Choose a smooth function $\ell_B(x)$ on $x \geq 1$, constant for $x \leq B$, equal to $\log x$ for $x \geq 2B$, and with $|\ell_B^{(k)}(x)| \leq C_k(B+x)^{-k}$ for $k \geq 1$. Replace each logarithm $\log(q-j-r)$ in Equation (4.1) by $\ell_B(q-j-r)$, retaining its constant denominator term, and denote the result by $\bar{g}$. Let $\bar{B}_b(s)$ be the corresponding normalized rows. At $m = b/2$, the pair-row moment bounds of Lemma 3.10 imply

$$\|\bar{B}_b(s) - B_{t,b}\| \leq C\sqrt{w} \frac{w^2}{(v+B)^2}. \quad (4.6)$$

Here is the normalization detail in this application. Subtract the affine Taylor polynomial of $\bar{g}$ at m from each of its two arguments. The subtracted contribution is constant on $i+j=b$ and cancels in row normalization. Write $z = |i-m|/w$. Taylor's formula, the derivative bounds on ℓ_B , and $w^2 \leq v$ give a bound $Cw^2(v+B)^{-2}(1+z)^K$ for the remaining exponent; exponentiating costs at most another fixed power of $1+z$. To check this also outside $|i-m| \leq (v+B)/2$, observe that this exterior region has $z \geq c\sqrt{v}$ whenever $v \geq B$, whereas logarithmic differences are bounded by $C \log(1+v/B)$. If $v < B$, all the logarithms have bounded scaled derivatives on the whole row. Thus the elementary inequality $|e^x - 1| \leq |x|e^{|x|}$ and the row moment estimates give both an ℓ^2 coefficient error $Cw^2/(v+B)^2$ and an ℓ^1 error $C\sqrt{w}w^2/(v+B)^2$. The normalization factor differs from one by the first of these errors. Since $w^2/(v+B)^2 \leq C/B$, it is bounded away from zero when B is large. Finally $\|\sum e_{ij}c_jc_i\| \leq \sum |e_{ij}|$ proves Equation (4.6).

Let $\mathcal{C}_0\psi = (B_{t,b}\psi)_b$ and $\bar{\mathcal{C}}_s\psi = (\bar{B}_b(s)\psi)_b$. These are operators from $\mathcal{F}_t$ to a direct sum of copies of $\mathcal{F}_t$. Equation (4.6) gives

$$\|\bar{\mathcal{C}}_s - \mathcal{C}_0\|^2 \leq C \sum_b \frac{w_b^5}{(v_b+B)^4} \leq C \int_0^\infty \frac{(1+x)^{5/2}}{(B+1+x)^4} dx \leq CB^{-1/2}. \quad (4.7)$$

The middle inequality uses $w_b^2 \leq v_b$ and the spacing $1/2$ of v_b . Consequently the operator-column error is at most $CB^{-1/4}$. Choose B , independent of q and s , so that this is at most $\sqrt{\gamma}/2$. Equation (4.5), also valid for $\bar{g}$, shows that the two columns have the same nullity in each number sector. The min-max principle for singular values and the supplied Fock gap therefore give

$$\|\bar{\mathcal{C}}_s\psi\| \geq \frac{\sqrt{\gamma}}{2} \text{dist}(\psi, \ker \bar{\mathcal{C}}_s).$$

It remains to remove the rounding. Put $\delta = g_{q,d} - \bar{g}$ and $S_s = e^{sD_t(\delta)}$. At most $2B+d$ orbital entries of δ are nonzero, and each is bounded in modulus by $C_d \log(2B)$. Thus

$$\|S_s\| \|S_s^{-1}\| \leq \exp\left(\sum_j |\delta(j)|\right) =: K_B < \infty$$

uniformly on the entire Fock space. If $\mathcal{C}_s$ is the column of the unrounded normalized rows, row normalization gives

$$\mathcal{C}_s = R_s(\mathbb{K} \otimes S_s)\bar{\mathcal{C}}_s S_s^{-1}, \quad \|R_s\| + \|R_s^{-1}\| \leq C_B,$$

where R_s is diagonal in the row index. Since $\ker \mathcal{C}_s = S_s \ker \bar{\mathcal{C}}_s$, the preceding distance estimate implies

$$\|\mathcal{C}_s \psi\| \geq \frac{\sqrt{\gamma}}{2C_B K_B} \text{dist}(\psi, \ker \mathcal{C}_s).$$

Squaring proves the assertion. The flux threshold only has to ensure $t \geq q_\gamma$ and $q \geq 4B + d$. $\square$

4.2 Local transport and comparison of consecutive fluxes

For the remainder of this section all compared path fluxes are assumed to be at least q_{pin} . Taking $q \geq q_{\text{pin}} + 6$ ensures this for the comparisons of size at most three and the growth steps used below. Larger fixed comparison ranges require only a corresponding fixed increase in this margin.

We compare different orbital dimensions by a fixed convention. If $t \leq t'$, identify orbitals with the same labels and extend an even operator A on $\mathcal{F}_t$ to $A \otimes I$ on $\mathcal{F}_{t'}$, using the graded tensor decomposition over the additional modes. Its original matrix elements are recovered by putting those modes in the vacuum. We then use the common slot frames of Lemma 3.2. All operator differences below use this padding convention.

The local rate of the transport will be

$$\alpha_q(m) = \frac{w^2}{v^2}. \quad (4.8)$$

In the north this rate is of order $(1 + m)/q^2$; changing the flux by a bounded amount will gain one more inverse power of v .

Proposition 4.2 (Transport of the pinned spaces). *There is a number-preserving and axially invariant unitary path $U_{q,d}(s)$ on $\mathcal{F}_t$, starting at the identity, such that*

$$U_{q,d}(s) \ker H_t = \ker H_{q,d}(s).$$

Writing $\partial_s U_{q,d}(s) = Y_{q,d}(s) U_{q,d}(s)$, the anti-Hermitian generator $Y_{q,d}(s)$ is a rapid interaction of size $C\alpha_q(m)$. In common frames, a bounded change in q , with d fixed, changes the generator interaction by size $C\alpha_q(m)/v$.

Proof. We first prove uniform row regularity and its flux difference, then estimate the affine-subtracted commutator that enters the spectral filter.

Row regularity. The unnormalized coefficients in Equation (4.4) are

$$(j - i) \left[\binom{t}{i}^{1-s} \binom{q}{i}^s \binom{t}{j}^{1-s} \binom{q}{j}^s \right]^{1/2}.$$

They obey the pair-row bounds in Lemma 3.10 uniformly for $0 \leq s \leq 1$. We give the normalization argument because the normalization is now a sum with moving endpoints. Write $z = i - m$. Relative to the undeformed row, after canceling its midpoint value, the additional factor is exactly

$$E_s(m, z) = \prod_{r=0}^{d-1} \left(1 - \frac{z^2}{(q - m - r)^2} \right)^{-s/2}. \quad (4.9)$$

When v is bounded, all allowed z and w are bounded and the estimates follow from finite differences. When v is large and $|z| \leq cv$, the logarithm of this factor and its derivatives in scale- w variables are bounded by $C(w^2/v^2)$ times fixed polynomials in $|z|/w$. The positive exponential factor can be absorbed into half the Gaussian decay of the undeformed row, since its exponent is $O(z^2/v^2)$ while that decay has exponent $-cz^2/w^2$ and $w^2 \leq v$. For $|z| > cv$, the factor in Equation (4.9) is at most $Cv^{d/2}$, whereas $|z|/w \geq c\sqrt{v}$. The Gaussian tail therefore pays this factor and every required inverse power of w .

To control normalization differences, first retain a smooth central cutoff supported in $|z| \leq c \min(u, v)$ and equal to one on a smaller such interval. Use the smooth log-gamma expression for the undeformed row before normalization, and sum its square times E_s^2 and this cutoff over all fixed integers i . This is a smooth function of real m , bounded below by cw^3 after extracting the common midpoint binomial factor: the $O(w)$ indices with $|i - m| \asymp w$ contribute $\asymp w^2$ each. Its derivative of order k is at most $C_k w^{3-k}$, by the Gaussian moment bounds. The discarded tail is $O_D(w^{3-D})$ for every D , also after any fixed finite difference: bound that difference by its finitely many values at adjacent centers, where the scales are comparable. The chain rule for the retained positive normalization, followed by this tail estimate, proves the full finite-difference estimates for the exactly normalized rows. For bounded w the normalization is bounded above and below directly. Thus no derivative of a moving-endpoint sum is being assumed. Lemma 3.10 now proves that $H_{q,d}(s)$ is a rapid interaction of bounded strength.

Flux differences. Use common windows for fluxes a bounded distance apart, as in Lemma 3.2. With orbital labels fixed from the north, their Hamiltonian interactions differ by size C/v . Indeed

$$\log \binom{Q+1}{j} - \log \binom{Q}{j} = \log(Q+1) - \log(Q+1-j).$$

For the path profiles, after canceling their common midpoint values, the corresponding ratio is exactly

$$\left(1 - \frac{z^2}{(t+1-m)^2}\right)^{-(1-s)/2} \left(1 - \frac{z^2}{(q+1-m)^2}\right)^{-s/2}.$$

Its difference from one has size $Cw^2/v^2 \leq C/v$ in the row-symbol bounds. The preceding central-cutoff normalization argument, applied to the difference of the two profiles, retains this extra factor in every finite-difference bound. In using that argument, the factor w^2/v^2 is retained throughout $|z| \leq cv$, including the part discarded by the smaller central cutoff. Only on $|z| > cv$ is it recovered from tail powers, using $v^2/w^2 \leq C(|z|/w)^2$. This distinction is necessary when v is much larger than u . If an individual pair coefficient is present only at the larger flux, its index excursion satisfies $|i - m| \geq v - O(1)$. For large v its Gaussian tail pays both the inverse of w^2/v^2 and every required symbol seminorm; for bounded v it is a bounded-scale estimate. The entirely new rows lie in a bounded south zone, where $1/v$ is bounded below. This proves the claimed common-frame interaction difference, including the changed endpoint supports. Repeating the one-flux identity handles any fixed bounded change in flux.

Affine subtraction. For a smooth real function a on the orbital interval, set $X_a = [H_{q,d}(s), D_t(a)]$. Decompose the Hamiltonian into annuli by a sum-of-squares partition $\sum_k \vartheta_k(b)^2 = 1$, of row-index width comparable to w_k , and then use its Fourier wavepackets. If m_k is the annulus center, put

$$a_k^\circ(j) = a(j) - a(m_k) - a'(m_k)(j - m_k).$$

The affine contribution commutes with the annular sum, because every row has a fixed pair index sum. More explicitly, if C_x is its Fourier wavepacket and $C_x(a)$ is the same wavepacket with row

coefficients multiplied by $a_k^\circ(i) + a_k^\circ(j)$, then

$$X_a = \int X_{a,x} d\mu(x), \quad X_{a,x} = C_x^* C_x(a) - C_x(a)^* C_x. \quad (4.10)$$

The center measure has bounded mass in unit balls. The wavepacket normalization is $w_k^{-1/2}$ and its circle measure is $w_k d\phi/(2\pi)$, so Fourier orthogonality proves Equation (4.10) exactly. The weighted row bounds give size $Cw^2|a''(m)|$ whenever Equation (3.31) holds with $\beta = w^2|a''(m)|$. For $a = g_{q,d}$ this size is $C\alpha_q(m)$: for every $k \geq 2$,

$$w(j)^k |g_{q,d}^{(k)}(j)| \leq C_k \frac{w(j)^2}{v(j)^2} \left(\frac{w(j)}{v(j)} \right)^{k-2} \leq C_k \alpha_q(j),$$

and α_q has tempered ratio exponent four, independently of k . Upon changing q by a bounded amount, the source interaction changes by size $C\alpha_q/v$: differences of g'' gain v^{-1} , and the row wavepackets change by C/v . These assertions include every fixed rapid localization order, since the estimates just used also apply after the finite differences used in Fourier localization.

Spectral filter. We use the spectral-filter construction of quasi-adiabatic continuation [6, 2], with locality supplied by Section 3. Choose a smooth odd real function η that is zero on $[-\gamma_{\text{pin}}/2, \gamma_{\text{pin}}/2]$ and equals $\text{sgn}(\omega)$ for $|\omega| \geq \gamma_{\text{pin}}$. Let $r(\omega) = \eta(\omega)/\omega$, with value zero near zero, and use the spectral calculus of $\text{ad}_H(A) = [H, A]$ to set

$$Y_{q,d}(s) = r(\text{ad}_{H_{q,d}}(s))X_g.$$

The function r and each of its derivatives belong to $L^2(\mathbb{R})$. Its inverse Fourier transform therefore has every absolute polynomial moment: use Cauchy–Schwarz on $|t| \leq 1$, and the L^2 norm of a sufficiently high derivative of r on $|t| > 1$. Lemma 3.7 and Equation (4.10) give rapid locality with size α_q .

The spectral multiplier of $D_t(g)$ is $\eta(\omega)$, so $Y^* = -Y$. If $P = P_{q,d}(s)$ and $Q = 1 - P$, the gap gives

$$YP = QD_t(g)P, \quad PY = -PD_t(g)Q, \quad PYP = 0.$$

Differentiating Equation (4.5) gives $\partial_s P = QD_t(g)P + PD_t(g)Q = [Y, P]$. The unitary solution therefore transports P . Both $H_{q,d}(s)$ and $D_t(g)$ commute with number and with $\sum_j j c_j^* c_j$, proving the two symmetries.

Finally compare two fluxes in common frames. The change of X_g has size α_q/v . The Hamiltonian change has size $1/v$. Inserting the latter into the time evolution in the filter by Duhamel's formula costs α_q/v , by Lemma 3.8. All filters are the same for every flux and every path parameter. This proves the difference assertion. $\square$

The isometry used subsequently is $\mathcal{U}_{q,d} = \iota_{q,d} U_{q,d}(1)$ restricted to $Z_{n,q-d}$. By Equation (4.2), its image is exactly the subspace with d south-pole zeros.

Corollary 4.3 (A fixed observable in the northern region). *Let O be a neutral operator of norm at most one supported on $B_r(x)$, where $r \geq 2$ and x has latitude index m . There are fixed constants b, C_* , and $c = 8$, independent of every tail order below, such that, if $q > Cr^b(1+m)$, the operator*

$$\Delta_{q,d}(O) = U_{q,d}(1)^* O U_{q,d}(1) - O$$

satisfies

$$\|\Delta_{q,d}(O)\| \leq Cr^c \frac{1+m}{q^2}, \quad (4.11)$$

$$\inf_{A \in \mathfrak{A}(B_{r+R}(x))} \|\Delta_{q,d}(O) - A\| \leq C_D r^c \frac{1+m}{q^2} (1+R)^{-D} \quad (R \geq C_*(1+r)). \quad (4.12)$$

In particular, it admits a decomposition

$$\Delta_{q,d}(O) = \sum_{\nu \geq 0} D_\nu, \quad D_\nu \in \mathfrak{A}(B_{C_* 2^\nu r}(x)), \quad \|D_\nu\| \leq C_D r^c \frac{1+m}{q^2} 2^{-\nu D}. \quad (4.13)$$

For $|q - q'| \leq 3$, with the same d and identical northern slots, $\Delta_{q,d}(O) - \Delta_{q',d}(O)$ obeys all these estimates with q^{-2} replaced by q^{-3} . In particular, for every fixed B , the shell decomposition can be chosen so that

$$\sum_{\nu \geq 0} (C_* 2^\nu r)^B \|D_\nu\| \leq C_B r^{B+c} \frac{1+m}{q^2}, \quad (4.14)$$

and the same moment estimate holds for the flux difference with q^{-3} .

Proof. At the center the cutoff gives $v \asymp q$, so $\alpha_q(m) \leq C(1+m)/q^2$ and $\alpha_q(m)/v(m) \leq C(1+m)/q^3$. Use the supported-observable bounds in Lemma 3.8, first for the self-adjoint path generators $iY_{q,d}(s)$ and 0. Since $\alpha_q = u/((q+1)v)$, its tempered ratio exponent can be taken to be four: each of u and v^{-1} costs at most two powers of distance. Equations (3.22)–(3.23) give a core factor r^6 and every tail order beyond radius Cr . Apply the same bounds directly to $iY_{q,d}(s)$ and $iY_{q',d}(s)$ for the comparison. Their difference has size $\alpha_q/v = u/((q+1)v^2)$, whose ratio exponent is at most six, and therefore costs r^8 . Thus $c = 8$ suffices for both assertions. Choose b large enough that the initial ball uses identical northern slots. The common-frame generator comparison already includes the padded south modes, so this same bound covers the endpoint changes. Take the ball approximations by the normalized partial trace onto the corresponding slot algebra. This contraction has error at most twice the best approximation error and preserves number grade. Differences at radii $C_* 2^\nu r$ give one decomposition satisfying Equation (4.13) for all tail orders, after increasing C_* by one fixed factor. Choose the tail order greater than $B+1$ and sum the resulting geometric series to obtain Equation (4.14). $\square$

4.3 Convex weighted observables

For fixed $0 < p < 2$ and $L \geq 2$, put

$$f_{L,p}(j) = \left(\frac{L}{L+j} \right)^p, \quad \beta_q(m) = w^2 f_{L,p}''(m), \quad A_q(m) = \alpha_q(m) \beta_q(m).$$

Constants in the next statements can depend on p , but are uniform in $L \geq 2$. Direct calculation gives

$$A_q(m) = \frac{u^2}{(q+1)^2} f_{L,p}''(m) \asymp \frac{f_{L,p}(m)}{(q+1)^2} \left(\frac{u}{L+u} \right)^2. \quad (4.15)$$

We use $d\mu$ for the center measure above; its integral of a radial tempered size is bounded by a constant times $\int_0^t (\cdot) dm$. These weights satisfy the precise symbol hypothesis needed for affine subtraction. For every $k \geq 2$,

$$w(j)^k |f_{L,p}^{(k)}(j)| \leq C_{k,p} \beta_q(j) \left(\frac{w(j)}{L+j} \right)^{k-2} \leq C_{k,p} \beta_q(j).$$

The ratios of $\beta_q(j)$ have a fixed polynomial bound in meridional distance, independent of k and $L \geq 2$. Thus Equation (3.31) applies with $\beta = \beta_q(m)$ at each chosen center.

Proposition 4.4 (Change of a weighted number observable). *The operator*

$$K_{q,d}(f) = U_{q,d}(1)^* D_t(f) U_{q,d}(1) - D_t(f)$$

has a rapid interaction representation of size $CA_q(m)$. Its interaction difference under a bounded change in q , with d fixed, has size $CA_q(m)/v$ in common frames. The pieces may be assigned to their original annular centers throughout the construction.

Proof. Retain the annular decomposition of Y used in its construction. Each annular total commutes with number and axial angular momentum: this is true of its source in Equation (4.10) and remains true under the time filter. We may consequently replace $f(j)$, in its commutator with this annular total, by $f(j) - f(m_k) - f'(m_k)(j - m_k)$. Lemma 3.11 bounds the resulting one-body interaction near the annulus by $C\beta_q$; its bounds away from the center grow only polynomially with distance. The weighted commutator calculus thus bounds $[D_t(f), Y(s)]$ by $C\alpha_q\beta_q$. Integrating

$$K_{q,d}(f) = \int_0^1 U_{q,d}(s)^* [D_t(f), Y(s)] U_{q,d}(s) ds$$

proves the first assertion. For the difference, use the same padded $D(f)$ on both systems: its additional end-orbital number operators commute with the smaller system's generator. Thus no uncompensated end value of f occurs. The source and generator differences supply the factor v^{-1} proved above; the difference of the transporting evolutions supplies the same factor by Duhamel's formula. The weights A_q and v^{-1} have uniformly polynomial ratios, so Lemma 3.8 applies also to their product. All decompositions start with an annular source and expand its evolved operator in shells about that source. This gives the last assertion. $\square$

The preceding norm estimate costs $C \int A_q(m) dm$. Summation over successive fluxes requires the stronger one-sided estimate below. The weight f is convex whereas the pinning logarithm g is concave. Their affine-subtracted quadratic parts therefore have opposite signs. After filtering, these leading parts form a negative square; the cubic Taylor remainder costs the additional factor $1/w$.

Proposition 4.5 (One-sided change in a zero space). *Writing P_0 for the zero projection of H_t , one has*

$$P_0 K_{q,d}(f) P_0 \leq C \left(\int_0^t \frac{A_q(m)}{w(m)} dm \right) P_0. \quad (4.16)$$

Proof. Fix s and abbreviate $H = H_{q,d}(s)$ and $P = P_{q,d}(s)$. The derivative in the transported zero space is the compression of

$$D_t(g)(1 - P)D_t(f) + D_t(f)(1 - P)D_t(g). \quad (4.17)$$

We estimate this operator before integrating in s .

Choose a smooth χ_+ equal to zero for $\omega \leq \gamma_{\text{pin}}/2$ and to one for $\omega \geq \gamma_{\text{pin}}$. Apply the filter $r_+(\omega) = \chi_+(\omega)/\omega$ separately to the pieces in Equation (4.10), and set

$$G_x = r_+(\text{ad}_H)X_{g,x}, \quad F_x = r_+(\text{ad}_H)X_{f,x}.$$

As in the preceding filter argument, its time kernel has all absolute polynomial moments. Every piece has strictly positive spectral frequencies. Thus

$$G_x^* P = F_x^* P = 0, \quad \left(\int G_x d\mu(x) \right) P = (1 - P) D_t(g) P, \quad \left(\int F_x d\mu(x) \right) P = (1 - P) D_t(f) P. \quad (4.18)$$

At a center of latitude index m , put $\rho_x = f''(m)/(-g''(m)) > 0$. The common wavepacket decomposition gives

$$F_x = -\rho_x G_x + E_x, \quad (4.19)$$

where G_x has size $C\alpha_q(m)$ and E_x has size $C\beta_q(m)/w(m)$, at every fixed rapid order. We verify the improved error explicitly. The function $a = f + \rho_x g$ has $a''(m) = 0$. Subtract its affine Taylor polynomial, as in Equation (4.10). In a row window of width w , the third-order remainder is bounded, with all scaled derivatives and polynomially weighted tails, by

$$Cw^3 \left(\frac{f''(m)}{L+m} + \frac{\rho_x(-g''(m))}{v} \right) \leq C \frac{\beta_q(m)}{w}.$$

The last inequality uses $w^2 \leq u, v$ and $u \leq L + m$. For windows outside this central region, derivative ratios have polynomial growth and the row tails absorb those powers. The weighted row estimate applied to $C_x(a)$ proves the claimed source error; the same time filter preserves it. This proves Equation (4.19) without making any assertion about the sign of individual unfiltered row pieces.

Write $R_g = \int G_x d\mu(x)$, $R_f = \int F_x d\mu(x)$ and $T = \int \sqrt{\rho_x} G_x d\mu(x)$. Each integral is finite at fixed flux. From Equation (4.18), a reversed product with a raising operator on the left vanishes in the compression, giving $PG_x^* G_y P = P[G_x^*, G_y] P$. Expanding Equation (4.19) therefore gives the exact identity

$$\begin{aligned} P(R_g^* R_f + R_f^* R_g) P &= -2PT^*TP \\ &\quad - \iint (\sqrt{\rho_x} - \sqrt{\rho_y})^2 P[G_x^*, G_y] P d\mu(x) d\mu(y) \\ &\quad + \iint P([G_x^*, E_y] + [E_x^*, G_y]) P d\mu(x) d\mu(y). \end{aligned} \quad (4.20)$$

We give the norm bounds for the last two integrals, since their spatial summability is essential. If $D = \text{dist}(x, y)$, rapid localization and the even parity of the pieces imply, for every fixed M ,

$$\| [G_x^*, G_y] \| \leq C_M \alpha_x \alpha_y (1 + D)^{-M}, \quad \| [G_x^*, E_y] \| \leq C_M \alpha_x \frac{\beta_y}{w_y} (1 + D)^{-M}.$$

Here subscripts denote evaluation at the corresponding center. The logarithmic derivative of ρ is bounded by $C((L+m)^{-1} + v^{-1})$. Using radial resolution w and the polynomial ratio bounds of Section 3 consequently gives a fixed exponent k such that

$$|\sqrt{\rho_x} - \sqrt{\rho_y}| \leq C \sqrt{\rho_x} \frac{D}{w_x} (1 + D)^k.$$

This bound follows by integrating the logarithmic derivative along the meridian between the two latitudes when the distance is at most a small multiple of w_x ; outside that region its right side dominates the polynomial ratio bound after increasing k . Since $\rho_x \alpha_x^2 \asymp A_x$, choose M above all these polynomial exponents plus the two-dimensional volume exponent. Summing first in y bounds the second line of Equation (4.20) by $C \int A_x / w_x^2 d\mu(x)$, and its third line by $C \int A_x / w_x d\mu(x)$. Since $w \geq 1$ and $-2PT^*TP \leq 0$, Equation (4.17) is at most $C \int_0^t A_q(m)/w(m) dm$ in quadratic form. Conjugate back by $U_{q,d}(s)$ and integrate over $0 \leq s \leq 1$ to obtain Equation (4.16). $\square$

Corollary 4.6 (Cost of changing a calibration state). *Let $q^0 = 3\lfloor q/3 \rfloor$ and $r = q - q^0 \in \{0, 1, 2\}$. Consider the interaction $K_{q,3}(f)$ on $\mathcal{F}_{q-3}$ and its common-frame comparison with $K_{q^0,3}(f)$. Embed a unit vector $\xi \in Z_{n,q^0-3}$ by the common orbital labels, or instead grow it to flux $q - 3$ using the r isometries $\mathcal{U}_{Q,1}$. The following costs are each bounded by*

$$C \int_0^{q-3} \frac{A_q(m)}{v(m)} dm : \quad (4.21)$$

the change of interaction in the embedded state, and the change of its expectation between the embedded and grown states. The same bounds hold after selecting any common collection of source-center pieces, and after adding their absolute bounds. In particular the statement applies to the filled vector at flux $q^0 - 3$ and to separate northern and southern portions of the interaction.

Proof. The first assertion is the integrated interaction difference in Proposition 4.4; vacant extra modes are compressed after comparison in the common frame. For one growth step, differentiate the expectation of the interaction along $U_{Q,1}(s)$. The generator has size $\alpha_Q \leq C/v$, while the observable interaction has size A_q . The weighted commutator calculus bounds the derivative in operator norm by $C \int A_q/v$. Finite-parameter conjugation preserves this bound. There are at most two growth steps, and all size functions for their fluxes are comparable. Applying this argument to each selected source piece before summing proves the last assertion, including comparisons with the directly embedded lower-flux reference on common northern slots. $\square$

5 Reading zero modes by orthogonal branches

We now turn the pinning maps into coordinates on every zero space. At each step we either record a hole or remove an electron. The coordinates are orthogonal, although the holes themselves have not been assigned positions. The purpose of the construction is quantitative: a local observable near the north pole can change substantially only when the coordinates record a hole sufficiently far along the descent toward that pole. We prove this assertion as an operator inequality, and then use rotation averaging to obtain a uniform comparison in the one-hole spaces.

Throughout this section, all lower bounds on the flux are absorbed into one fixed integer $q_{\text{stop}} \geq 4$. We stop every descent when $q \leq q_{\text{stop}}$. Constants may depend on this stopping value, on the uniform constants of the local calculus, and on a fixed decay exponent. They never depend on q or on the particle number.

5.1 The algebraic decomposition

Let P_q be the orthogonal projection onto the Fock zero space $Z_q = \bigoplus_n Z_{n,q}$. Write $n_q = c_q^* c_q$ for occupation of the south-pole orbital. In a nonzero sector $Z_{n,q}$ set

$$h = q + 3 - 3n. \quad (5.1)$$

For $n \geq 1$, divisibility by the cube of every pair determinant gives

$$Z_{n,q} = \left\{ \prod_{i < j} (z_{i0} z_{j1} - z_{j0} z_{i1})^3 S : S \text{ is symmetric and homogeneous of degree } h \text{ in each spinor} \right\}. \quad (5.2)$$

Thus $h \geq 0$. We retain Equation (5.1) also in the vacuum sector, where $h = q + 3$.

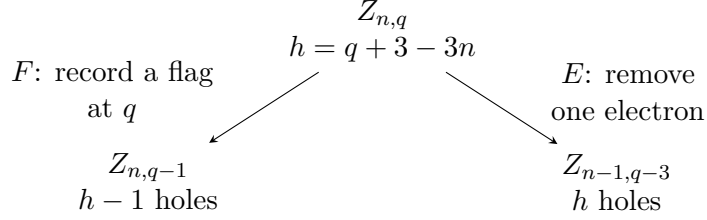


Figure 2: One step of the orthogonal decomposition. The displayed spaces are coordinate spaces for the two branches. A flag is a record of an F step; it is not a measured position of a quasihole.

Denote by $Z_{n,q}^{(d)}$ the subspace in which each electron has d zeros at the south pole. The endpoint transport from Section 4, followed by the inclusion with its last d orbitals vacant, will be denoted by

$$\mathcal{U}_{q,d} : Z_{n,q-d} \longrightarrow Z_{n,q}^{(d)}, \quad d \in \{1, 3\}.$$

It is an isometry, preserves particle number and the sum of the orbital indices, and is defined with the same choices on all number sectors.

Lemma 5.1 (Two orthogonal branches). *For $q > q_{\text{stop}}$, define*

$$\mathcal{F}_{n,q} = \ker(c_q|_{Z_{n,q}}), \quad \mathcal{E}_{n,q} = Z_{n,q} \ominus \mathcal{F}_{n,q}.$$

Then $\mathcal{F}_{n,q} = Z_{n,q}^{(1)}$. Contraction by c_q maps $\mathcal{E}_{n,q}$ bijectively onto $Z_{n-1,q}^{(3)}$. If T_q denotes the polar part of $c_q|_{Z_q}$, the maps

$$J_{q,F} = \mathcal{U}_{q,1}, \quad J_{q,E} = \mathsf{T}_q^* \mathcal{U}_{q,3}$$

are isometries onto the two orthogonal branches. Consequently,

$$Z_{n,q} \simeq Z_{n,q-1} \oplus Z_{n-1,q-3}. \quad (5.3)$$

The second summand is absent for $n = 0$. In the first summand h decreases by one; in the second it is unchanged.

Proof. The absence of orbital q is equivalent to divisibility by the north spinor coordinate in every variable, which is precisely one pin at the south pole. Evaluation of one electron at that pole leaves the three pair zeros attached to every remaining electron. This proves that the range of c_q lies in $Z_{n-1,q}^{(3)}$.

To prove surjectivity, use the affine coordinate in which the south-pole coefficient is the coefficient of the highest power. After factoring the Laughlin polynomial, the map on symmetric polynomials extracts the coefficient of z_n^h . A basis of its target consists of monomial symmetric functions indexed by partitions with at most $n - 1$ parts, all at most h . Append a part equal to h and take the corresponding monomial symmetric function in n variables, with each distinct monomial included once. Extraction returns the original monomial symmetric function. This proves surjectivity, including $h = 0$. Restriction to the orthogonal complement of the kernel is therefore bijective, and its polar part is unitary onto the range. Composing with the endpoint transports gives the stated isometries. The assertions about h follow directly from (5.1). $\square$

Figure 2 displays the changes in flux, particle number, and hole number in the two branches.

The remaining issue in this decomposition is analytic. A polar decomposition need not be local when its nonzero singular values approach zero. The next lemma rules out that problem uniformly in the flux.

5.2 A local polar map

Lemma 5.2 (Occupation gap and local representatives). *There is $\varepsilon > 0$, independent of q and of the number sector, such that*

$$P_q n_q P_q \geq \varepsilon P_{\mathcal{E}_q}, \quad \mathcal{E}_q = \bigoplus_n \mathcal{E}_{n,q}. \quad (5.4)$$

The polar map $\mathbb{T}_q$ and the projections onto the two branches, restricted to Z_q , have bounded physical representatives rapidly localized at the south pole. The representatives preserve Z_q and its orthogonal complement. They have respectively the number and index grades of c_q and of the identity.

Proof. We first prove the singular-value bound, keeping track of the restriction of pair rows. The normalized rows are

$$B_{q,b} = \sum_{\substack{0 \leq i < j \leq q \\ i+j=b}} a_{q,b}(i,j) c_j c_i, \quad a_{q,b}(i,j) = \frac{(j-i) \sqrt{\binom{q}{i} \binom{q}{j}}}{\sqrt{q \binom{2q-2}{b-1}}}, \quad 1 \leq b \leq 2q-1.$$

For completeness, their normalization follows by taking coefficients in

$$\sum_{i < j} (j-i)^2 \binom{q}{i} \binom{q}{j} x^{i+j} = (1+x)^q (x \partial_x)^2 (1+x)^q - ((x \partial_x)(1+x)^q)^2 = qx(1+x)^{2q-2}.$$

If a row contains orbital q , put $k = 2q - b$. Its removed coefficient mass is

$$p_{q,k} = \frac{k \binom{q-1}{k-1}}{\binom{2q-2}{k-1}}.$$

Here $p_{q,1} = p_{q,2} = 1$, and

$$\frac{p_{q,k+1}}{p_{q,k}} = \frac{(k+1)(q-k)}{k(2q-k-1)} \leq 1.$$

It follows that every row with a retained pair has retained mass

$$r_{q,b} = \sum_{\substack{i < j < q \\ i+j=b}} |a_{q,b}(i,j)|^2 \geq \frac{q}{4q-6} \geq \frac{1}{4}. \quad (5.5)$$

Rows with no coefficient at q have mass one. The two top rows retain no pair and are omitted.

For $d = 1$, the pinning weights obey $g_j = \frac{1}{2} \log((q-j)/q)$. Multiplication of its coefficients by $e^{-g_i - g_j}$, followed by normalization, turns a flux- $(q-1)$ row into the normalized retained flux- q row. More precisely, if $B_{q,b}^{\text{ret}}$ denotes the row with the coefficient at orbital q deleted, then $B_{q,b}^{\text{ret}} = \sqrt{r_{q,b}} B_{q,1,b}(1)$, in the notation of Equation (4.4). Let $\delta > 0$ be the uniform endpoint gap from Lemma 4.1. By (5.5), the Hamiltonian compressed to the south-empty sector has a gap at least $\delta/4$ above $\bigoplus_n \mathcal{F}_{n,q}$.

The CAR give the exact identity

$$\sum_b [B_{q,b}, n_q]^* [B_{q,b}, n_q] = n_q \sum_{k=1}^q p_{q,k} n_{q-k}.$$

The coefficient sum is

$$\sum_{k=1}^q p_{q,k} = \frac{4q-2}{q+1} < 4.$$

One way to see this is to write $p_{q,k} = k \binom{2q-k-1}{q-1} / \binom{2q-2}{q-1}$ and apply the hockey-stick identity twice. For $\psi \in Z_q$, the two top singleton rows also imply $c_{q-1}c_q\psi = c_{q-2}c_q\psi = 0$. Hence

$$\sum_b \|[B_{q,b}, n_q]\psi\|^2 \leq 2\|c_q\psi\|^2. \quad (5.6)$$

Let $Q = 1 - n_q$ and let P_F project onto $\bigoplus_n \mathcal{F}_{n,q}$. Since $B_{q,b}\psi = 0$, the restricted gap gives

$$\frac{\delta}{4}\|Q\psi - P_F\psi\|^2 \leq \sum_b \|B_{q,b}Q\psi\|^2 \leq 2\|c_q\psi\|^2.$$

The occupied and vacant components are orthogonal, so

$$\|\psi - P_F\psi\|^2 \leq (1 + 8/\delta)\|c_q\psi\|^2.$$

Thus (5.4) holds with $\varepsilon = \delta/(\delta + 8)$.

To construct local representatives, take a real smooth frequency cutoff χ equal to one near zero and supported strictly inside the uniform unperturbed gap. Apply its time filter to c_q :

$$A_q = \int_{\mathbb{R}} \hat{\chi}(t) e^{itH_q} c_q e^{-itH_q} dt,$$

with the Fourier normalization incorporated into $\hat{\chi}$. Contraction preserves zero modes. The cutoff therefore gives $A_q P_q = c_q P_q$ and $[A_q, P_q] = 0$. The local calculus makes A_q rapidly localized at the south pole, with uniform constants. Choose a smooth function η on a fixed interval containing the spectrum of $A_q^* A_q$, equal to zero near zero and to $x^{-1/2}$ for $x \geq \varepsilon$. Then

$$\tilde{T}_q = A_q \eta(A_q^* A_q)$$

is rapidly localized by smooth functional calculus and agrees with T_q on Z_q . Its products $\tilde{T}_q^* \tilde{T}_q$ and $1 - \tilde{T}_q^* \tilde{T}_q$ represent the branch projections on Z_q . The construction preserves number and index grades because H_q preserves both and $A_q^* A_q$ has grade zero. It also preserves P_q in both directions. No claim about a spectral gap of $A_q^* A_q$ outside Z_q is needed: η is smooth on that entire spectral interval. $\square$

5.3 One-step comparisons

Write

$$K_{q,d}(f) = \mathcal{U}_{q,d}^* D_q(f) \mathcal{U}_{q,d} - D_{q-d}(f)$$

on the corresponding smaller zero space. Proposition 4.4 supplies localized representatives for these operators, with the stated size and flux-comparison estimates.

Proposition 5.3 (Weighted observable at one branch). *Fix $p \in (0, 2)$, $L \geq 2$, and $f(j) = f_{L,p}(j)$. In the orthogonal coordinates (5.3),*

$$\begin{aligned} (J_{q,F} \oplus J_{q,E})^* D_q(f) (J_{q,F} \oplus J_{q,E}) &= \begin{pmatrix} D_{q-1}(f) + K_{q,1}(f) & 0 \\ 0 & D_{q-3}(f) + f(q) + K_{q,3}(f) \end{pmatrix} + R_q(f), \\ \|R_q(f)\| &\leq C_p \frac{f(q)}{(L+q)^2}. \end{aligned} \quad (5.7)$$

The notation in the left side means the unitary whose columns are the two branch isometries. The bound includes the off-diagonal blocks. For affine f the remainder is zero and $K_{q,d}(f) = 0$.

Proof. The F block is the definition of $K_{q,1}(f)$. For an affine sequence, the grades of T_q give

$$D_q(f)\mathsf{T}_q = \mathsf{T}_q D_q(f) - f(q)\mathsf{T}_q$$

on the zero spaces, and both branch projections commute with $D_q(f)$. The endpoint transports commute with the number and index operators. This proves the affine assertion, including the scalar $f(q)$ in E .

For a general f , subtract its affine Taylor polynomial at q . The remainder $a(j) = f(j) - f(q) - f'(q)(j - q)$ satisfies

$$|a(j)| \leq C_p \frac{f(q)}{(L+q)^2} (1+q-j)^{p+4}, \quad 0 \leq j \leq q.$$

The same polynomial bounds hold for the finite differences needed for localization. In the south-pole frame this says that $D_q(a)$ has one-body interaction size at most $C_p f(q)/(L+q)^2$, times a fixed polynomial in distance from the pole. Commuting it with any of the rapidly localized representatives in Lemma 5.2 thus has norm at most that size: decompose the representative into ball shells and sum its arbitrarily high inverse-power tails against this polynomial. In the E block use $\mathsf{T}_q[D_q(a), \mathsf{T}_q^*]$; for the off-diagonal blocks use the commutator with a branch projection. Compression and the endpoint isometries do not increase these norms. This proves (5.7). $\square$

For a general observable supported far to the north, the polar map contributes an even smaller error, because its representative is localized at the opposite pole. We record the precise form used below.

Lemma 5.4 (Northern comparison). *There are fixed $c \geq 0$ and $b_0 \geq 1$ with the following property. Let $O = O^*$ be neutral, of norm at most one, supported on a slot ball of radius $r \geq 2$ centered at latitude index m , and put $u = 1 + m$. If $q > Cr^{b_0}u$, identify all northern slots in the fluxes being compared. The two diagonal branch blocks of O equal*

$$O + \mathcal{D}_{q,1}(O) \quad \text{and} \quad O + \mathcal{D}_{q,3}(O)$$

up to a matrix remainder of norm $C_D q^{-D}$ for every fixed D . Each $\mathcal{D}_{q,d}(O)$ is neutral and self-adjoint. If x is its original center, then, for fixed $c, C_ < \infty$ and every $D > 0$,*

$$\begin{aligned} \|\mathcal{D}_{q,d}(O)\| &\leq Cr^c u/q^2, \\ \inf_{A_R \in \mathfrak{A}(B_R(x))} \|\mathcal{D}_{q,d}(O) - A_R\| &\leq C_D r^c u q^{-2} R^{-D} \quad (R \geq C_* r). \end{aligned} \tag{5.8}$$

For $|q - q'| \leq 2$ and fixed d , the difference between these changes, in common northern slots, obeys the same two estimates with q^{-2} replaced by q^{-3} . The exponent c and core constant C_ are independent of D .*

Proof. For F , this is Corollary 4.3. Its norm and tail estimates are Equations (4.11) and (4.12); enlarge the fixed core constant to write the outer ball radius as R instead of $r + R$. For E , commute O through the local representative of T_q before applying that corollary with $d = 3$. The support of O is a distance comparable with $\sqrt{q}$ from the south pole when b_0 is increased if necessary. Rapid locality therefore makes the polar commutator and the off-branch blocks $O(q^{-D})$ for every D . The same estimate handles the finitely many trimmed orbitals. The difference assertion for the remaining changes is the same-corollary comparison of the pinning transports at bounded flux difference. $\square$

5.4 Histories and a linear reference expectation

Iterate the two branch coordinates until the flux reaches q_{stop} . A word ω records its successive F and E choices; its terminal number and flux are denoted by n_ω and q_ω . The resulting unitary is

$$\mathcal{R}_{n,q} : Z_{n,q} \longrightarrow \bigoplus_{\omega} Z_{n_\omega, q_\omega}. \quad (5.9)$$

For a fixed readout axis and fixed filters, the maps $J_{q,F}$, $J_{q,E}$, and $\mathcal{R}_{n,q}$ are defined on physical Fock spaces independently of the auxiliary slot frames. Rebuilding frames outside the common northern slots changes only the representations used to prove locality estimates; it does not redefine these maps or their history coordinates. An F step taken at flux b records one flag with value b . At a terminal space we add $q_\omega + 3 - 3n_\omega$ flags, all with value zero. Let $\mathcal{B}(\omega)$ be this multiset. Each word has exactly h flags. The vacuum follows only F steps, with the same terminal convention. We do not resolve the bounded-dimensional terminal space further.

Fix $a \in (0, 2)$. On the direct sum in (5.9), define the positive diagonal operator

$$W_m(\omega) = \sum_{b \in \mathcal{B}(\omega)} \min \left\{ 1, \left(\frac{1+m}{1+b} \right)^a \right\}. \quad (5.10)$$

Its value is scalar on each terminal summand. In particular $W_m = 0$ on the filled zero space, and $W_m \geq (1+q)^{-a}$ whenever $h > 0$.

We next fix the scalar used to center observables. Put $q^0 = 3\lfloor q/3 \rfloor$, and let Ω_{q^0} be the normalized filled zero mode at that flux. Embed its one-particle space into U_q by sending orbital j to orbital j , and then use the slot embedding at flux q , with the orthogonal complement of the embedded one-particle space vacant. Define

$$\text{ref}_q(O) = \langle \Omega_{q^0}, O \Omega_{q^0} \rangle \quad (5.11)$$

in this padded realization. This is a positive linear functional of norm one on the slot algebra. At $q = q^0$ it is the filled expectation. For $q^0 = 0$ the convention is $U_0 = \mathbb{C}$ with its unique normalized one-electron filled vector. Outside the common northern region only linearity, the norm-one bound, and the exact filled expectation at $q = q^0$ will be used.

Lemma 5.5 (Compatibility of reference expectations). *Let two fluxes q, q' use common northern slots, and let $s \leq \min(q, q')$. Embed a fixed vector of $\mathcal{F}_s$ into either system by the same orbital labels, with the orthogonal complement of the embedded one-particle space vacant. The two states have identical expectations on the algebra of common northern slots. In particular, references at fluxes with the same nearest lower filled flux agree on that algebra.*

Proof. Expand a slot observable into normally ordered CAR monomials. Its expectation in an embedded state is a linear combination of the state's normally ordered orbital correlations. The coefficients use only the analysis-frame entries of the slots occurring in the monomial. Entries at orbital labels above s contribute zero, because those orbitals are vacant. At labels at most s , the common northern frame entries are identical. Thus every monomial has the same expectation in both realizations. This also proves the assertion after physical compression of the slot observable, which leaves these matrix elements unchanged. $\square$

The parent and its E child use filled states at q^0 and $q^0 - 3$, respectively. Their expectations are related by the filled E step. By contrast, the references at $q - 3$ and $q^0 - 3$ use the same filled state and agree on common northern slots by Lemma 5.5. For a rapidly localized operator, apply

this exact identity to a truncation supported in those slots; the remaining difference is at most twice the truncation error. The common-frame construction permits these conventions for every frame entering a comparison, with uniform estimates.

Theorem 5.6 (Local observables in history coordinates). *For every fixed $a \in (0, 2)$ there are $B, C < \infty$ such that the following holds uniformly in n, q and in the allowed slot frames. Let $O = O^*$ be a neutral slot observable of norm at most one, supported in a ball of radius $r \geq 2$ with center of latitude index m . Its compression to $Z_{n,q}$ satisfies*

$$\pm \left(\mathcal{R}_{n,q} P_{n,q} O P_{n,q} \mathcal{R}_{n,q}^* - \text{ref}_q(O) I \right) \leq C r^B W_m. \quad (5.12)$$

Here $P_{n,q}$ denotes the zero-space projection in the number- n sector. More generally, for a convergent decomposition $A = \sum_\nu A_\nu$ into neutral self-adjoint observables supported in balls about the same center, of respective radii $r_\nu \geq 2$, the right side can be replaced by $C W_m \sum_\nu r_\nu^B \|A_\nu\|$.

Proof. The proof keeps O unchanged while descending through common northern slots. Only the small changes caused by each transport are estimated inductively. This distinction prevents an increasing radius from being assigned repeatedly to the original observable.

We first describe the centered one-step identities under the northern condition of Lemma 5.4. For an E step put $t = q - 3$, so $t^0 = q^0 - 3$, and write $R_{q,E}(O)$ for its uncentered polar remainder in Lemma 5.4. The filled E step at q^0 , together with compatibility of the northern references, gives the exact scalar identity

$$\text{ref}_q(O) - \text{ref}_t(O) = \text{ref}_{t^0}(\mathcal{D}_{q^0,3}(O) + R_{q^0,E}(O)).$$

Thus the parent-to-child reference change is determined by a filled step. Define

$$\begin{aligned} R'_{q,E}(O) &= R_{q,E}(O) - \text{ref}_{t^0}(R_{q^0,E}(O))I, \\ s_q(O) &= \text{ref}_t(\mathcal{D}_{q,3}(O)) - \text{ref}_{t^0}(\mathcal{D}_{q^0,3}(O)). \end{aligned}$$

The references in the second line use the same filled state. Lemma 5.5, first on common northern slots and then on rapid tails by norm approximation, therefore gives

$$s_q(O) = \text{ref}_{t^0}(\mathcal{D}_{q,3}(O) - \mathcal{D}_{q^0,3}(O)) + O(q^{-D}).$$

The bounded-flux comparison in Lemma 5.4 supplies the additional factor q^{-1} . Substituting these scalar identities into the branch comparison yields

$$\begin{aligned} J_{q,E}^*(O - \text{ref}_q(O)) J_{q,E} &= (O - \text{ref}_t(O)) + (\mathcal{D}_{q,3}(O) - \text{ref}_t(\mathcal{D}_{q,3}(O))) \\ &\quad + s_q(O)I + R'_{q,E}(O), \\ |s_q(O)| &\leq C r^c u / q^3 + C_D q^{-D}, \quad \|R'_{q,E}(O)\| \leq C_D q^{-D}. \end{aligned} \quad (5.13)$$

Here and below the symbols denote compressions to the relevant child zero space. When $h = 0$ the comparison is its own filled calibration, so all centered contributions vanish exactly.

For an F step, $t = q - 1$. The nearest smaller filled fluxes are either identical or differ by three. In the latter case apply the filled E comparison just used. Together with (5.8), this proves

$$J_{q,F}^*(O - \text{ref}_q(O)) J_{q,F} = O - \text{ref}_t(O) + S_{q,F}(O), \quad \|S_{q,F}(O)\| \leq C r^c u / q^2 + C_D q^{-D}. \quad (5.14)$$

The off-diagonal blocks have norm $C_D q^{-D}$. An F step already records a flag, so its entire error will be charged to that flag.

Here is a uniform closure argument, including the parameter choices. Let c, b_0 be as in Lemma 5.4. Choose $b > \max\{b_0, c\}$, then choose $B > \max\{ab, c\}$ large enough to absorb the fixed polynomial costs of the local calculus. A constant M will be fixed after B . Set

$$\tau = Mr^b u.$$

If $q \leq \tau$ and $h > 0$, every flag has $b' \leq q$, so

$$W_m \geq (2Mr^b)^{-a} I.$$

The norm bound $\|O - \text{ref}_q(O)\| \leq 2$ proves (5.12) in this region with constant $2(2M)^a$. When $h = 0$ the centered compression is zero. Increasing M includes all stopping cases in this argument.

Fix a finite Q , and let K_Q be the least constant in (5.12) for fluxes at most Q , with the chosen B , all number sectors, and all allowed frames. This number is finite: for $h > 0$ use $W_m \geq (1+Q)^{-a}$, while the filled compression is exactly centered. We prove

$$K_Q \leq C_M + \frac{1}{2}K_Q \tag{5.15}$$

with C_M independent of Q .

Suppose $q > \tau$. Expand the unchanged O down the branching tree, using (5.13) and (5.14), until the current flux is at most τ . The original northern slots, center, and radius are unchanged through this part of the descent. The terminal comparisons have just been bounded. In an E child, decompose $\mathcal{D}_{q,3}(O)$ into its core and dyadic ball shells, all centered at m . Equation (4.14), with a tail order larger than $B + 2$, gives

$$\sum_{\nu} r_{\nu}^B \|\mathcal{D}_{q,3}(O)_{\nu}\| \leq C(B)r^{B+c}u/q^2.$$

Applying the definition of K_Q to each smaller-flux shell, and using linearity of the reference, bounds its centered contribution by

$$K_Q C(B)r^{B+c}u/q^2$$

times the remaining history weight in that child. Along any word, the flux strictly decreases; thus

$$\sum_{q' > \tau} \frac{u}{(q')^2} \leq C \frac{u}{\tau} = \frac{C}{Mr^b}.$$

Consequently all induction-dependent contributions are bounded by

$$K_Q \frac{C(B)}{M} r^{B+c-b} W_m.$$

Choose M so large that this is at most $\frac{1}{2}K_Q r^B W_m$. This choice is possible because $b > c$ and $r \geq 2$.

We verify that the other errors have a bound independent of K_Q . At a node with positive remaining hole number, its identity is bounded by the sum of the identities assigned to its remaining flags. A scalar electron error may therefore be charged to every such flag. For a flag at ℓ , the total charge from ancestors in (5.13) is at most

$$C \frac{r^c u}{\max\{\tau, 1 + \ell\}^2}. \tag{5.16}$$

The F error in (5.14) occurs at its own flag and obeys the same bound, up to a fixed constant. Write $w_\ell = \min\{1, (u/(1+\ell))^a\}$. Since $a < 2$ and $\tau = Mr^b u$,

$$\frac{r^c u}{\max\{\tau, 1+\ell\}^2} \leq CM^{a-2} r^{c+b(a-2)} w_\ell \leq C_M r^B w_\ell.$$

The q^{-D} remainders, including the off-diagonal blocks, obey the same estimate after choosing $D > a + 2$. For a self-adjoint off-diagonal block, its norm bounds the full block matrix above and below by that norm times the identity at its node, so the same charge applies. Nodes with $h = 0$ contribute nothing, by the exact centered filled identity. Summation of these diagonal bounds along words proves (5.15).

It follows that $K_Q \leq 2C_M$ for every Q , proving the uniform claim. Finally, the assertion for $A = \sum_\nu A_\nu$ follows by linearity of the reference and by summing the form inequalities. $\square$

The theorem estimates a compression to the zero spaces. It does not claim that ordinary local fluctuations vanish in the filled state. It also applies to coherent superpositions of histories: all charges above are diagonal operator bounds, rather than bounds conditional on a classical choice of word.

5.5 Rotation averaging in a one-hole space

The history estimate still depends on a chosen readout axis. In this subsection, a one-hole space means a space with quasihole degree $h = 1$. Representation theory then removes the axis dependence and yields an integrated estimate with no factor of the sphere area.

Corollary 5.7 (One-hole comparison). *Suppose $h = 1$, equivalently $q = 3n - 2$. Let ψ, ϕ be any two unit vectors in $Z_{n,q}$. Let $\{O_x\}$ be a measurable family of neutral self-adjoint observables of norm at most one, supported in balls of a fixed radius $r \geq 2$ about x , with center measure of bounded density. There are constants $C, B' < \infty$, independent of n, q , such that*

$$\int |\langle \psi, O_x \psi \rangle - \langle \phi, O_x \phi \rangle| dx \leq Cr^{B'}. \quad (5.17)$$

The same conclusion holds for rapidly localized families, with the right side bounded by the corresponding fixed rapid-locality seminorm. For a family whose norms are bounded by A , multiply the right side by A .

Proof. Choose $a \in (1, 2)$ in Theorem 5.6. The symmetric-polynomial factor in (5.2) has degree one in each variable. It is the irreducible spin- $n/2$ representation, so $Z_{n,q}$ is irreducible and has dimension $n + 1$.

When $h = 1$, an F branch has $h = 0$ and hence dimension one. Before that branch all steps are E , which lower the flux by three. Therefore at most one one-dimensional history has a flag at any specified nonterminal flux. The history that reaches the stopping region without an F step has bounded terminal dimension. It follows that, as an operator on $Z_{n,q}$,

$$\text{Tr}(W_0) \leq C + \sum_{b \geq 0} (1+b)^{-a} \leq C_a. \quad (5.18)$$

The trace is unchanged on conjugating back to the physical zero space. Write $\widehat{W}_0 = \mathcal{R}_{n,q}^* W_0 \mathcal{R}_{n,q}$ for that physical operator.

Let $V_q(g)$ denote the rotation representation, with normalized Haar measure dg . Schur's lemma gives

$$\int_{\text{SU}(2)} V_q(g) \widehat{W}_0 V_q(g)^* dg = \frac{\text{Tr}(\widehat{W}_0)}{n+1} I.$$

For each rotation use the readout axis whose north pole is the image of the original north pole. Precisely, write $x(g) = g \cdot \text{north}$ and $W^g = V_q(g) \widehat{W}_0 V_q(g)^*$. Use the rotated frame and rotated branch construction at $x(g)$. Theorem 5.6, applied to $O_{x(g)}$, gives

$$|\langle \psi, O_{x(g)} \psi \rangle - \langle \phi, O_{x(g)} \phi \rangle| \leq Cr^{B'} (\langle \psi, W^g \psi \rangle + \langle \phi, W^g \phi \rangle).$$

The two reference expectations cancel in this inequality. Changes between local frames cost only a fixed polynomial in r by the local calculus. Integrating rotations and multiplying by the sphere area yields at most $Cr^{B'} q \text{Tr}(W_0)/(n+1)$, which is uniformly bounded because $q = 3n - 2$. For any center measure with bounded mass in unit balls, spread the mass at each center over a ball of fixed radius. The resulting density relative to area is bounded uniformly. The operator originally assigned to x is supported in a ball of radius $r+1$ about every point in that spreading ball, so the same pointwise bound applies before integration. This reduces the general center measure to the area estimate. Finally, summing ball shells proves the assertion for rapid families. $\square$

6 Charge bounds and rotation averaging

The history estimate of Theorem 5.6 controls a local observable by weighted flags. To treat arbitrary hole number by rotation averaging, we compare these weights with a physical one-body observable, first with a slowly decreasing weight and then with a summable weight.

Increase the fixed stopping flux in the history construction, if necessary, so that it is at least eight and all pinning comparisons above it are valid, including shifts by at most three. Write $q_s = q_{\text{stop}}$ and let $\mathcal{R}_{n,q}$ denote the history unitary in (5.9). For $L \geq 2$ and $0 < p < 2$, put

$$\begin{aligned} f(x) = f_{L,p}(x) &= \left(\frac{L}{L+x} \right)^p, & C_q(f) &= \frac{q+3}{3(q+1)} \sum_{j=0}^q f(j), \\ S_{L,p}(\omega) &= \sum_{b \in \mathcal{B}(\omega)} f(b). \end{aligned} \tag{6.1}$$

When q is divisible by three, $C_q(f)$ is the filled-state expectation of $D_q(f)$: rotational invariance gives occupation $(q+3)/(3(q+1))$ in every orbital. The formula defines the scalar $C_q(f)$ also at the other fluxes. The operator $S_{L,p}$ is diagonal in histories and scalar on each terminal summand. Its sum is over the flag multiset $\mathcal{B}(\omega)$, including all terminal flags at position zero; each history has exactly $h = q+3-3n$ flags. All operator inequalities involving $S_{L,p}$ below are in history coordinates. In particular, they make no measurement assumption about flags in a zero-mode vector.

Proposition 6.1 (Pooled charge). *For each of $p = 1/8$ and $p = 3/2$ there are fixed $L \geq 2$ and $c > 0$ such that, for every flux and every nonzero zero-mode space $Z_{n,q}$,*

$$\mathcal{R}_{n,q}(C_q(f)\mathbf{1} - P_{n,q}D_q(f)P_{n,q})\mathcal{R}_{n,q}^* \geq cS_{L,p}. \tag{6.2}$$

Here compression to $Z_{n,q}$ is understood, and $P_{n,q}$ denotes its orthogonal projection. The constants are independent of q, n and of the readout axis. When $h = 0$, both sides vanish.

The obstruction to this estimate occurs at electron steps. Along a history, each F step contributes approximately $-f(q)/3$ to $D_q(f) - C_q(f)\mathbf{1}$, supplying a credit for its new flag. Each E step contributes a positive scalar correction. When at least two flags remain, divide this scalar cost between them. The scalar tail estimate below leaves the strictly positive margin

$$\frac{1}{3} - \frac{1+p}{6(1-p)} = \frac{5}{42} \quad (p = 1/8).$$

With one remaining flag, there is no such margin. We instead combine the scalar with the transported observable and compare it to a filled step; the one-hole comparison of Corollary 5.7 makes the resulting error summably small. These two cases prove the first charge estimate.

Rotation averaging then bounds integrated local-expectation differences between two zero-mode states by $C(q+1)^{7/8}$ times the sum of their hole numbers. This still grows with the area, but it is enough to prove the charge estimate again with $p = 3/2$. For a polar annulus $1+m \asymp M$, use this averaged bound while the flux is moderate relative to M ; the transport interaction has a factor $(q+1)^{-2}$ that pays for the area dependence. At larger flux, apply the original fixed-axis history estimate directly to the annulus. The two resulting sums are small when the new weight scale L is large. The summable weight then gives the area-independent local-expectation bound at the end of this section.

6.1 Scalar telescoping and summable error bounds

We estimate $D_q(f) - C_q(f)\mathbf{1}$ from above. An F step contributes the scalar $C_{q-1}(f) - C_q(f)$, and an E step contributes

$$\ell_q = C_{q-3}(f) + f(q) - C_q(f). \quad (6.3)$$

For the continuous comparison define

$$F(x) = \frac{1}{x} \int_0^x f(t) dt, \quad \ell(x) = -2F'(x) + f'(x) \quad (x > 0), \quad (6.4)$$

with the continuous extensions at zero.

Lemma 6.2 (Scalar estimates). *Fix $p \in (0, 2)$ and a lower stopping flux $q_s \geq 8$. There is a function $\varepsilon_p(L) \rightarrow 0$ as $L \rightarrow \infty$ with the following properties, uniformly for $q \geq q_s$:*

$$\left| C_{q-1}(f) - C_q(f) + \frac{1}{3}f(q) \right| \leq \varepsilon_p(L)f(q), \quad (6.5)$$

$$\sum_{q \geq q_s} \frac{|\ell_q - \ell_{3\lfloor q/3 \rfloor}|}{f(q)} \leq \varepsilon_p(L). \quad (6.6)$$

The function ℓ is nonnegative. If $0 < p < 1$, then for every collection $\mathcal{E}$ of E-step fluxes along a history, and every flag position k below these steps,

$$\sum_{\substack{q \in \mathcal{E} \\ q \geq k}} (\ell_q)_+ \leq \left(\frac{1+p}{3(1-p)} + \varepsilon_p(L) \right) f(k). \quad (6.7)$$

Changing k by a fixed bounded amount, or taking k in the terminal range, only changes $\varepsilon_p(L)$.

Proof. The exact formula

$$\ell_q = f(q) - \left(1 + \frac{2}{q+1}\right) \frac{f(q) + f(q-1) + f(q-2)}{3} + \frac{2 \sum_{j=0}^{q-3} f(j)}{(q+1)(q-2)} \quad (6.8)$$

is obtained by separating the last three summands in C_q . A useful positive form of this identity is obtained by putting $d_j = f(j) - 2f(j+1) + f(j+2)$ and $Q_q = (q+1)(q-2)$:

$$\ell_q = \frac{1}{Q_q} \sum_{j=0}^{q-3} (j+1)(j+2)d_j + \frac{2q}{3(q+1)} d_{q-2}. \quad (6.9)$$

Expanding the second differences verifies the formula coefficient by coefficient. In particular, $\ell_q \geq 0$, and constant and affine functions contribute zero.

Here are precise discrete-to-continuous error bounds:

$$|\ell_q - \ell(q)| + \sup_{x \in [q-3, q]} |\ell(x) - \ell(q)| \leq C(L+q)^{-2}, \quad (6.10)$$

$$|\ell_q - \ell_{q-r}| \leq \begin{cases} CL^{-2}, & q \leq L, \\ C(F(q) + f(q))q^{-2}, & q > L, \end{cases} \quad (6.11)$$

for $r = 0, 1, 2$. We give details because the sharper second estimate is needed for the summable weight. The identity $d_j = \int_0^1 (1 - |t - j - 1|)_+ f''(t) dt$ expresses (6.9) as $\int_0^q \kappa_q(t) f''(t) dt$. The function κ_q is linear between consecutive integers and has values

$$\kappa_q(i) = \frac{i(i+1)}{Q_q} \quad (0 \leq i \leq q-2), \quad \kappa_q(q-1) = \frac{2q}{3(q+1)}, \quad \kappa_q(q) = 0.$$

On $[0, q-2]$ its difference from t^2/q^2 is at most $C(t/q^2 + t^2/q^3)$, and on $[q-2, q]$ that difference is bounded by a constant. The error in its integral against f'' is consequently at most

$$C \left(q^{-2} \int_0^q t f''(t) dt + q^{-3} \int_0^q t^2 f''(t) dt + \int_{q-2}^q f''(t) dt \right).$$

For $q \leq L$ use $f'' \leq C/L^2$; for $q > L$ use $\int_0^q t f'' \leq 1$, $\int_0^q t^2 f'' \leq 2qF(q)$, and $f''(t) \leq Cf(t)(L+t)^{-2}$. This proves the first term of (6.10). The continuous identity $\ell(x) = x^{-2} \int_0^x t^2 f''(t) dt$ implies $\ell'(x) = f''(x) - 2\ell(x)/x$, which proves the oscillation estimate.

For (6.11), difference the positive formula directly:

$$\begin{aligned} \ell_q - \ell_{q-1} &= -\frac{2(q-1)}{Q_q Q_{q-1}} \sum_{j=0}^{q-4} (j+1)(j+2)d_j \\ &\quad + \frac{(q-1)(q-2)}{3q(q+1)} d_{q-3} + \frac{2q}{3(q+1)} d_{q-2}. \end{aligned}$$

The first sum is bounded by $CqF(q)$, as follows by its integral representation and $\int_0^q t^2 f'' \leq 2qF(q)$. This gives $CF(q)/q^2$ when $q > L$; the last two terms cost $Cf(q)/q^2$. Below L use $d_j \leq C/L^2$ in the displayed formula. A bounded number of differences proves the claimed bound.

The exact F-step identity is

$$C_{q-1}(f) - C_q(f) + \frac{1}{3}f(q) = \frac{2}{3q(q+1)} \sum_{j=0}^{q-1} (f(j) - f(q)).$$

For $q \leq L$ its ratio to $f(q)$ is $O(L^{-1})$. For $q > L$ use

$$F(q) \leq \begin{cases} f(q)/(1-p), & p < 1, \\ C_p L/q, & p > 1. \end{cases} \quad (6.12)$$

The resulting ratio is $O(L^{-1})$ also when $1 < p < 2$, since $L^{1-p}q^{p-2} \leq L^{-1}$ for $q \geq L$. The case $p = 1$ follows from the explicit logarithmic integral and is not needed below. Dividing (6.11) by $f(q)$ and summing gives $O(L^{-1})$ below L . Above L it gives $O(L^{-1})$: the summands are $O(q^{-2})$ for $p < 1$ and $O(q^{-2} + L^{1-p}q^{p-3})$ for $1 < p < 2$.

Twice integrating f'' proves

$$\ell(x) = \frac{1}{x^2} \int_0^x t^2 f''(t) dt \geq 0, \quad \int_k^\infty \ell(x) dx = 2F(k) - f(k).$$

The E-step intervals $[q-3, q]$ have disjoint interiors along a history. Consequently their continuous contribution is at most $\frac{1}{3} \int_{(k-3)_+}^\infty \ell(x) dx$. For $p < 1$, (6.12) bounds this by $\frac{1+p}{3(1-p)} f(k)$, up to $O(L^{-1})f(k)$ from the bounded endpoint shift. Finally, the tail sum of (6.10) is bounded by $C/(L+k)$, and

$$\frac{1}{L+k} \leq \frac{1}{L} f(k) \quad (0 < p < 1).$$

Thus it is $O(L^{-1})f(k)$ uniformly in k . This proves (6.7). $\square$

We next list the analytic error sums that will be used in both charge arguments. Retain the size function A_q from Proposition 4.4. With $u = 1 + m$, $v = 1 + q - m$, and $w = \sqrt{uv/(q+1)}$, it satisfies

$$\begin{aligned} A_q(m) &= \frac{u^2 f''(m)}{(q+1)^2} \asymp \frac{f(m)}{(q+1)^2} \left(\frac{u}{L+u} \right)^2, \\ B_q &= \int_0^q A_q(m) dm, \quad J_q = \int_0^q \frac{A_q(m)}{v} dm. \end{aligned} \quad (6.13)$$

The longitude integral has bounded mass and is included in constants. Proposition 4.4 gives this size for each transported change in $D(f)$.

Lemma 6.3 (Uniform error sums). *For either $p = 1/8$ or $p = 3/2$, as $L \rightarrow \infty$,*

$$\sup_{q \geq q_s} \frac{B_q}{f(q)} = O(L^{-1}), \quad \sum_{q \geq q_s} \left(\frac{J_q}{f(q)} + \frac{1}{(L+q)^2} \right) = O\left(\frac{1 + \log L}{L} \right). \quad (6.14)$$

For $p = 1/8$ there are, in addition,

$$\sum_{q \geq q_s} \frac{\sup_{0 \leq m \leq q} A_q(m)}{f(q)} = O(L^{-1}), \quad \sum_{q \geq q_s} \frac{1}{f(q)} \int_0^q \frac{A_q(m)}{w} dm = O(L^{-1/2}). \quad (6.15)$$

All implied constants are independent of L .

Proof. For $q \leq L$, integrating u^2/L^2 gives $B_q \leq C(q+1)L^{-2}$; for $q > L$, dropping the factor $(u/(L+u))^2$ gives $B_q \leq CF(q)/q$. These estimates and (6.12) prove the first bound in (6.14). On the northern half of the sphere $v \geq (q+1)/2$, whereas on the southern half $A_q(m) \leq Cf(q)(L+q)^{-2}$. Therefore

$$\frac{J_q}{f(q)} \leq \frac{CB_q}{(q+1)f(q)} + \frac{C \log(q+2)}{(L+q)^2}.$$

The first term sums to $O(L^{-1})$: above L it is bounded by Cq^{-2} for $p < 1$ and by $CL^{1-p}q^{p-3}$ for $p > 1$. The second sums to $O((1 + \log L)/L)$.

For the supremum in (6.15), use CL^{-2} when $q \leq L$ and $C(q+1)^{-2}$ otherwise. The latter divided by $f(q)$ is $O(L^{-p}q^{p-2})$, whose sum is $O(L^{-1})$ for $p < 1$. On the northern half, $w \geq c\sqrt{u}$, so the last integral is at most

$$\frac{C}{(q+1)^2} \int_0^{q/2} \frac{f(m)u^{3/2}}{(L+u)^2} dm + \frac{C\sqrt{q+1}f(q)}{(L+q)^2}.$$

After division by $f(q)$, both terms are bounded by $C\sqrt{q+1}/L^2$ for $q \leq L$ and by $Cq^{-3/2}$ for $q > L$, provided $p < 1/2$. Their sums are $O(L^{-1/2})$. $\square$

6.2 Calibration and the first charge estimate

Set $K_q := K_{q,3}(f)$ on $\mathcal{F}_{q-3}$. Its compression to $Z_{n-1,q-3}$ is the transported change in an E step from $Z_{n,q}$, as in Proposition 5.3. Its center terms have size $A_q(m)$. A direct estimate of ℓ_q would accumulate a scalar even in a filled state. We instead calibrate $K_q + \ell_q$ by a filled step at the nearest lower flux divisible by three.

Put $q^0 = 3\lfloor q/3 \rfloor$ and $r = q - q^0 \in \{0, 1, 2\}$. Start with the filled state at flux $q^0 - 3$ and increase the flux r times by F transport at the same south pole. The resulting unit vector χ_q lies at flux $q - 3$ and has the minimum possible hole number r . If $r = 0$, it is the filled state itself.

Lemma 6.4 (Calibrated electron step). *For q above the fixed stopping range,*

$$|\ell_q + \langle K_q \rangle_{\chi_q}| \leq C \left(J_q + |\ell_q - \ell_{q^0}| + \frac{f(q)}{(L+q)^2} \right). \quad (6.16)$$

For any collection of northern center terms of K_q , their expectation in χ_q may instead be replaced by the linear reference functional of Theorem 5.6 at child flux $q - 3$; the additional error is at most CJ_q . This remains true when the choice between the two centerings is made separately on disjoint annuli.

Proof. In the filled E step at q^0 , both parent and child compressions of $D(f) - C(f)\mathbf{1}$ vanish. Proposition 5.3 therefore gives $|\ell_{q^0} + \langle K_{q^0} \rangle| \leq Cf(q^0)(L+q^0)^{-2}$. Corollary 4.6 compares this filled-step expectation first with the directly embedded lower-flux state and then with its at most two F-growth transports. Its bound is CJ_q for each comparison, proving (6.16) after adding $|\ell_q - \ell_{q^0}|$.

For a northern source center, the linear reference functional of Theorem 5.6, applied at child flux $q - 3$, is the expectation in the same directly embedded filled state at $q^0 - 3$, using the common slots. The last assertion of Corollary 4.6 applies to any selected collection of source pieces and bounds the sum of absolute comparison errors by CJ_q . It therefore also permits the annulus-dependent choice of centering in the statement. $\square$

For completeness, the elementary rule for summing history bounds is as follows. At a node of flux q with positive hole number, every descendant flag is at an index at most q , and hence

$$S_{L,p} \geq hf(q)\mathbf{1} \geq f(q)\mathbf{1} \quad (6.17)$$

on that node's full descendant space. A node error bounded by $\eta_q f(q)\mathbf{1}$ can consequently be replaced by $\eta_q S_{L,p}$ there. If $\sum_q \eta_q \leq \eta$, the sum of all these node errors is at most $\eta S_{L,p}$: on each

history, each flag appears only at ancestor fluxes, each visited at most once. This is a comparison of diagonal operators after the node form bounds have been applied. It also controls an off-diagonal remainder by its operator norm. At a node with no holes, use the exact vanishing of the centered compression instead of this rule.

Proof of Proposition 6.1 for $p = 1/8$. Apply Proposition 5.3 recursively to $D_q(f) - C_q(f)\mathbf{1}$. The norm remainder at a node is at most $Cf(q)(L+q)^{-2}$, and its total cost is $o_L(1)S_{L,p}$ by (6.17) and Lemma 6.3. F transport has norm at most CB_q . Together with (6.5), its contribution is

$$-(\frac{1}{3} - o_L(1))f(q)$$

for the flag created at that step.

At an E step with one remaining hole, $q \equiv 1 \pmod{3}$, so χ_q also has one hole. Compare the child vector to χ_q using Corollary 5.7. The difference of the K_q expectations is at most $C \sup_m A_q(m)$. This is a two-sided form bound since it holds for every unit vector in the child space. The calibration error and all one-hole errors have total cost $o_L(1)S_{L,p}$ by Lemmas 6.2 and 6.3.

At an E step with at least two remaining holes, Proposition 4.5 bounds the increase from K_q by $C \int A_q(m)/w dm$. Its total cost is again $o_L(1)S_{L,p}$. It remains to pay the positive part of ℓ_q . Divide that scalar equally among the remaining flags. A fixed flag then pays at most one half of each $(\ell_q)_+$ at such an ancestor. The E-step intervals are disjoint, so Lemma 6.2 bounds the total payment for a flag at k by

$$\left(\frac{1+p}{6(1-p)} + o_L(1) \right) f(k) = (\frac{3}{14} + o_L(1))f(k).$$

At terminal fluxes there are only finitely many spaces. Uniformly on them, $f(j) = 1 + O(L^{-1})$, and hence

$$D_q(f) - C_q(f)\mathbf{1} = -\frac{h}{3}\mathbf{1} + O(L^{-1}).$$

For $h = 0$ the expression is exactly zero by rotational invariance; for $h \geq 1$ the error is $O(L^{-1})h$, which is also $O(L^{-1})$ times the terminal flag weight. Terminal flags therefore receive the same credit $1/3 - o_L(1)$. Combining all bounds yields

$$\mathcal{R}_{n,q}(D_q(f) - C_q(f)\mathbf{1})\mathcal{R}_{n,q}^* \leq -(\frac{1}{3} - \frac{3}{14} - o_L(1))S_{L,p}.$$

The strict margin is $5/42$. Choose $L = L_0$ so that the total error is less than $5/84$ and take $c = 5/84$. $\square$

We record what this first estimate gives after rotation averaging. It will be applied to annular portions of the next transport interaction, with their center size factored out.

Corollary 6.5 (First averaged comparison). *Let W be a neutral Hermitian interaction with bounded strength per unit area and fixed rapid locality bounds. For unit vectors $\psi \in Z_{n,q}$ and $\chi \in Z_{n',q}$ with hole numbers h, h' ,*

$$|\langle W \rangle_\psi - \langle W \rangle_\chi| \leq C_W(q+1)^{7/8}(h+h'). \quad (6.18)$$

For radius- r interactions of unit local norm one can take $C_W \leq C(1+r)^B$ for a fixed B .

Proof. To specify the rotation average on a physical zero space, let $V_{n,q}(g)$ denote its rotation representation and define

$$S_{L,p}^g = V_{n,q}(g) \mathcal{R}_{n,q}^* S_{L,p} \mathcal{R}_{n,q} V_{n,q}(g)^*.$$

Thus $S_{L,p}^g$ acts on $Z_{n,q}$, whereas $S_{L,p}$ itself acts in history coordinates. This is the history construction obtained by rotating a fixed readout frame by g ; its estimates are uniform in g . Use this convention separately in each number sector.

For a local term centered at $x(g) = g \cdot \text{north}$, use Theorem 5.6 in the frame rotated by g , with exponent $a = 7/4$. The local term itself need not belong to a rotation-covariant family; the estimate is pointwise in g . Since $(1+b)^{-a} \leq C f_{L_0,1/8}(b)$, the local comparison is bounded by the corresponding rotated $S_{L_0,1/8}$. Its reference scalar cancels between the two vectors. By irreducibility of U_q , normalized rotation averaging gives the one-particle identity

$$\int_{SU(2)} D_q(f)^g dg = \frac{\sum_{j=0}^q f(j)}{q+1} \mathcal{N}, \quad (6.19)$$

where Haar measure has mass one. Proposition 6.1 for $p = 1/8$ consequently gives

$$\int_{SU(2)} \langle S_{L_0,1/8}^g \rangle_\psi dg \leq \frac{h}{3c(q+1)} \sum_{j=0}^q f_{L_0,1/8}(j).$$

The physical area is $2\pi q$, and the center density is bounded. Multiplying by that area and using $\sum_{j=0}^q f_{L_0,1/8}(j) \leq C(q+1)^{7/8}$ proves the result. The same argument applies to any center measure with bounded mass in unit balls: spread each center uniformly over a ball of fixed radius and regard its observable as centered at the new point. The resulting measure has bounded area density, and the support radius increases by only a fixed constant. Thus the preceding integral estimate applies to both discrete and continuous center measures. Finally, a rapid shell expansion multiplies the radius- r estimate by a summable series of shell norms times $(1+r)^B$. $\square$

6.3 The summable weight

We now fix $p = 3/2$, keeping the already chosen L_0 and the constant in Corollary 6.5 fixed. The new scale L will be chosen at the end. Calibrate every E step with positive hole number by Lemma 6.4. Its scalar error, the branch remainders, and all F-step errors have total cost $o_L(1)S_{L,3/2}$ by the estimates already proved. It remains to bound the centered E interactions.

Divide centers into annuli $M \leq 1 + m < 2M$, where $M = 1, 2, 4, \dots$. Write

$$a_M = \left(\frac{M}{L+M} \right)^2, \quad f_M = f(M), \quad \delta = \frac{1}{24}.$$

On this annulus the local interaction size is at most $C(q+1)^{-2} f_M a_M$. Choose a fixed sufficiently large C_0 and separate fluxes at $T_M = C_0 M^{1+\delta}$. An annulus is present only if $q+1 \geq M$.

Proof of Proposition 6.1 for $p = 3/2$. First consider $q \leq T_M$. Center the annular interaction at χ_q . If the child has $h > 0$ holes, the calibration state has $r \leq h$ holes, so Corollary 6.5, applied at child flux $q-3$, bounds the centered interaction in absolute quadratic form by

$$C(q+1)^{-2} f_M a_M (q+1)^{7/8} h \mathbf{1}. \quad (6.20)$$

The child flag weight is at least $hf(q)$. For $M \leq q+1 \leq T_M+1$, $f_M/f(q) \leq CM^{p\delta}$. Thus summing the relative cost (6.20) over these fluxes gives

$$Ca_M M^{p\delta} \sum_{q+1 \geq M} (q+1)^{-1-1/8} \leq Ca_M M^{-(1/8-p\delta)} = Ca_M M^{-1/16}. \quad (6.21)$$

The finitely many fluxes near the stopping range obey the same estimate after increasing its constant.

Now let $q > T_M$. With C_0 sufficiently large, the annulus lies in the common northern region. Center it by the linear reference functional in Theorem 5.6. Lemma 6.4 accounts for this change of centering with the already budgeted error CJ_q . For a center in the annulus, each descendant history ω satisfies

$$\begin{aligned} W_m(\omega) &\leq C \sum_{k \in \mathcal{B}(\omega)} \min \left(1, \left(\frac{M}{1+k} \right)^a \right) \\ &\leq \frac{C}{f_M} \sum_{k \in \mathcal{B}(\omega)} f(k) = \frac{C}{f_M} S_{L,3/2}(\omega). \end{aligned} \quad (6.22)$$

This is also a diagonal operator inequality in the descendant history space. Indeed, for $k \leq M$, $f(k) \geq f(M)$; for $k > M$, use $a = 7/4 > p$ and $M/(1+k) \leq C(L+M)/(L+k)$. The annulus has area $O(M)$. The local readout estimate therefore bounds its centered interaction by

$$C(q+1)^{-2} f_M a_M \frac{M}{f_M} S_{L,3/2}.$$

For rapid interactions this formula follows by applying the radius- r bound to each shell and summing its norm times $(1+r)^B$. The shell center stays in the same annulus even when its support extends beyond it; shells reaching the south are covered by the large-radius part of the readout estimate. Summing over $q > T_M$ gives

$$Ca_M M \sum_{q > T_M} (q+1)^{-2} S_{L,3/2} \leq Ca_M M^{-\delta} S_{L,3/2}. \quad (6.23)$$

Both resulting sums over dyadic M tend to zero with L . Explicitly, for every $0 < \eta < 2$,

$$\sum_{M \in \{1,2,4,\dots\}} \left(\frac{M}{L+M} \right)^2 M^{-\eta} \leq C_\eta L^{-\eta}. \quad (6.24)$$

For $M \leq L$ this follows by summing $L^{-2} M^{2-\eta}$; for $M > L$ it follows by summing $M^{-\eta}$. Apply this with $\eta = 1/16$ and $\eta = 1/24$ to (6.21) and (6.23). The node summation rule after (6.17) now gives an error $o_L(1)S_{L,3/2}$ for the entire centered E contribution.

There is no uncompensated electron scalar in this argument. Every F flag and every terminal flag contributes $-(1/3 - o_L(1))f(k)$, while all E contributions and all remainders cost $o_L(1)S_{L,3/2}$. Choose a fixed $L = L_1$, after L_0 and the constant in Corollary 6.5, so that the combined error is at most $1/6$. This proves (6.2) with $c = 1/6$. $\square$

6.4 Rotation averaging and local interactions

The summability of $f_{L,3/2}$ removes the power of the area in Corollary 6.5. We state the resulting estimate in the form needed for the energy argument.

Theorem 6.6 (Local expectations are linear in the number of holes). *Let $q = 3(N - 1)$ and let W be a neutral Hermitian interaction whose center density and rapid locality bounds are uniform in q . For every unit vector $\psi \in Z_{n,q}$,*

$$|\langle W \rangle_\psi - \langle W \rangle_{\text{Laughlin}, N}| \leq C_W(N - n). \quad (6.25)$$

For interactions supported in radius r with local norm at most one, C_W is bounded by $C(1 + r)^B$ for a fixed exponent B . The same conclusion holds for zero-mode density matrices by linearity, with the expected value of $N - \mathcal{N}$ on the right.

Proof. Use the local readout estimate with $a = 7/4$, now comparing W_0 to $S_{L_1, 3/2}$. Since the flux is divisible by three, the reference is the filled Laughlin expectation. The rotation average (6.19) and the second part of Proposition 6.1 give

$$\int_{SU(2)} \langle S_{L_1, 3/2}^g \rangle_\psi dg \leq \frac{h}{3c(q+1)} \sum_{j=0}^q f_{L_1, 3/2}(j) \leq \frac{Ch}{q+1},$$

because L_1 is fixed and $\sum_{j \geq 0} f_{L_1, 3/2}(j) < \infty$. Integration over centers costs area $O(q+1)$, exactly as in the proof of Corollary 6.5. The result is $C_W h = 3C_W(N - n)$; absorb the factor three into C_W . Polynomial radius costs and rapid shell summability are preserved by that argument. Decomposing a density matrix into number sectors and then into pure states proves the last assertion. $\square$

7 Local removal of interaction energy

The estimate on zero modes controls the cost of a missing electron. We now connect an arbitrary state to the zero space by a local process that removes particles. Its interaction energy decays exponentially, and its expected particle loss is bounded by its initial interaction energy. In particular, adding electrons above Laughlin filling has a uniform energy cost.

Write $B_p = B(v_p)$, $0 \leq p \leq 2q - 2$, for the normalized pair annihilators of (2.5), and put $d_a = 2q - 1$. For a rotation $g \in SU(2)$, $A(g)$ denotes the conjugate of an operator A by the Fock representation. Haar measure dg has mass one. The rotation average of any normalized pair row resolves the interaction:

$$d_a \int B_p(g)^* B_p(g) dg = H_q. \quad (7.1)$$

This is Schur's lemma on the spin- $(q - 1)$ pair space V_q .

Theorem 7.1 (Local energy descent). *There are constants $c > 0$, $L < \infty$, and q_0 such that, for every $q \geq q_0$, there is a measurable family of rapidly local physical operators J_y on $\mathcal{F}_q$ satisfying*

$$\int J_y^* J_y d\nu_q(y) = H_q, \quad \int J_y^* H_q J_y d\nu_q(y) \leq H_q^2 - cH_q. \quad (7.2)$$

Every J_y has a definite particle loss $r_y \in \{2, \dots, L\}$, so that $[\mathcal{N}, J_y] = -r_y J_y$. The center measure has bounded mass per unit ball, and all rapid-locality bounds are independent of q and of the particle sector.

The first operation in the construction is $B_0(g)$, which removes a pair near the point $g \cdot \text{north}$. We then partially empty a fixed neighborhood of that point. This second operation suppresses the nearby pair correlations left behind by the first removal. We first identify the part of H_q^2 that pays for the surviving correlations.

7.1 The four-body term available for descent

The maps $\mathcal{L}_k$ and W_4 are defined in (2.4) and (2.7). Thus $\mathcal{L}_4(|u\rangle\langle v|) = B(u)^*B(v)$ and $W_4(v \otimes w) = v \wedge w$. Let $P_{q,\ell}$ be the projection in $V_q \otimes V_q$ onto total spin $2(q-1) - \ell$, for $0 \leq \ell \leq 2q-2$, and set

$$R_q^{\text{hi}} = \sum_{\ell=23}^{2q-2} P_{q,\ell}, \quad R_q^{\text{lo}} = I - R_q^{\text{hi}}. \quad (7.3)$$

For large q , R_q^{lo} contains precisely the small deficits $\ell = 0, \dots, 22$. Proposition A.1, proved in the appendix, gives

$$H_q^2 \geq c_0 H_q + \mathcal{L}_4(W_4 R_q^{\text{hi}} W_4^*) \quad (0 < c_0 < 1/25) \quad (7.4)$$

for all sufficiently large q , uniformly on $\mathcal{F}_q$.

This statement contains more information than the unperturbed spectral gap. The normal-ordering identity for H_q^2 contains the full four-body term $\mathcal{L}_4(W_4 W_4^*)$. The finite comparison of [16, Sections 2–6] uses only the blocks $\ell \leq 22$. The appendix includes the finite certificate and its transfer to finite q , and keeps the other blocks with coefficient one. The loss construction will use precisely this part of the square.

7.2 Emptying a neighborhood without increasing distant pair energy

Choose a smooth nondecreasing $T : \mathbb{R} \rightarrow [0, 1]$ such that $T = 0$ on $(-\infty, 1]$ and $T = 1$ on $[3, \infty)$. We choose it flat at the two endpoints and, in addition, require

$$|T^{(r)}(x)| \leq C_r \sqrt{1 - T(x)} \quad (r \geq 1). \quad (7.5)$$

For example, with $\eta(x) = e^{-1/x}$ for $x > 0$ and $\eta(x) = 0$ for $x \leq 0$, one may take $T(x) = \eta(x-1)/(\eta(x-1) + \eta(3-x))$. The estimate follows by differentiating near $x = 3$; an exponential $e^{-1/(3-x)}$ times any fixed negative power of $3-x$ is bounded by a constant times $e^{-1/(2(3-x))}$. Away from that endpoint the estimate is immediate from smoothness.

For an integer $M \geq 2$, put $t_j = T(j/M)$ and $n_j = c_j^* c_j$. The two operators

$$K_{j,0} = I - n_j + t_j n_j, \quad K_{j,1} = \sqrt{1 - t_j^2} c_j \quad (7.6)$$

define a trace-preserving channel, since $K_{j,0}^* K_{j,0} + K_{j,1}^* K_{j,1} = I$. Compose these channels in increasing order of j , for $0 \leq j < 3M$. Write K_α for the resulting products of Kraus operators, where $\alpha \in \{0, 1\}^{3M}$, and write $\mathcal{E}_M$ for the channel. Its dual on observables is $\mathcal{E}_M^*(O) = \sum_\alpha K_\alpha^* O K_\alpha$.

The canonical anticommutation relations imply a useful exact formula. On an even normal-ordered monomial, the dual of the j th channel multiplies by t_j for each occurrence of c_j or c_j^* . Indeed, if neither occurs, the even monomial commutes with both operators of mode j and completeness gives the claim. If exactly one occurs, the loss contribution vanishes and the no-loss contribution gives t_j . If both occur, writing them as an occupation factor gives t_j^2 . Consequently

$$\mathcal{E}_M^*(H_q) = \sum_{b=1}^{2q-1} \tilde{B}_{b-1}^* \tilde{B}_{b-1}, \quad \tilde{B}_{b-1} = \sum_{\substack{i < j \\ i+j=b}} t_i t_j \overline{v_{b-1}(i, j)} c_j c_i. \quad (7.7)$$

Here and below terms with an orbital index outside $\{0, \dots, q\}$ are absent.

Lemma 7.2 (Pair energy after local loss). *There is a sequence $\varepsilon_M \rightarrow 0$ such that, for every $M \geq 2$ and every $q \geq 30M$,*

$$\sum_{b=1}^{2q-1} \tilde{B}_{b-1}^* \tilde{B}_{b-1} \leq \sum_{b>M} B_{b-1}^* B_{b-1} + \varepsilon_M I \quad \text{on } \mathcal{F}_q. \quad (7.8)$$

Proof. The following two elementary estimates specialize the general pair localization bounds to the cutoff used here. In particular, the exponential estimate on distant rows will let us preserve coefficient one on their energy. Both follow directly from the coefficients in (2.5). For $b \leq q/2$ and every fixed integer $r \geq 0$,

$$\sum_{\substack{i < j \\ i+j=b}} |v_{b-1}(i, j)| |i - b/2|^r \leq C_r b^{r/2+1/4}. \quad (7.9)$$

For $b > 10M$ and $q \geq 30M$,

$$\sum_{\substack{i < j, i+j=b \\ i < 3M}} |v_{b-1}(i, j)| \leq C b^{3/2} e^{-c_1 b}. \quad (7.10)$$

Constants may change after decreasing $c_1 > 0$. Here is a direct verification. Set

$$p_{q,b}(i) = \frac{\binom{q}{i} \binom{q}{b-i}}{\binom{2q}{b}}.$$

The exact pair formula gives

$$|v_{b-1}(i, b-i)|^2 = \frac{2(2q-1)(2i-b)^2}{b(2q-b)} p_{q,b}(i), \quad \frac{p_{q,b}(i+1)}{p_{q,b}(i)} = \frac{(q-i)(b-i)}{(i+1)(q-b+i+1)}. \quad (7.11)$$

When $b \leq q/2$, these ratios, their inverses, and symmetry about $b/2$ show

$$p_{q,b}(i) \leq C b^{-1/2} \exp\{-c_2(i - b/2)^2/b\}.$$

For completeness, on the upper half $i \geq (b-1)/2$, the first factor in the ratio is at most one and the second is at most $\exp\{-(2i+1-b)/(b+1)\}$. Summing logarithms from a middle index gives the Gaussian upper bound relative to the middle value. In the range $|i - b/2| \leq \sqrt{b}$, the absolute value of that logarithm is at most $C(1 + |i - b/2|)/b$. Summing these lower bounds and using $\sum_i p_{q,b}(i) = 1$ bounds the middle value by $C/\sqrt{b}$. Insertion in (7.11), followed by summing Gaussian moments, proves (7.9).

To prove (7.10), there is nothing to check if $b > q + 3M$, because the sum is empty. Otherwise $b \leq q + 3M \leq 11q/10$, so $2(2q-1)/[b(2q-b)] \leq C/b$ in (7.11). For every allowed $i \leq 2b/5$ and $b \geq 20$, the ratio in (7.11) is at least $4/3$. Starting with $i < 3M < 3b/10$ and taking at least $b/10 - O(1)$ steps toward $2b/5$ gives $p_{q,b}(i) \leq C e^{-cb}$, since each probability is at most one. The square root of (7.11) is therefore at most $C\sqrt{b} e^{-c_1 b}$, and there are at most b terms. This proves (7.10).

The modified row vanishes when $b \leq M$. For $M < b \leq 10M$, put $x = b/(2M)$ and $t = T(x)$. The smooth even function $z \mapsto T(x+z)T(x-z)$ has zero first derivative at zero. By (7.5), every derivative of positive order there is bounded by $C_r \sqrt{1-t}$. Taylor expansion through degree 41, with a bounded remainder of order 42, and (7.9) yield

$$\tilde{B}_{b-1} = t^2 B_{b-1} + E_b, \quad \|E_b\| \leq C M^{-3/4} \sqrt{1-t} + C M^{-20}. \quad (7.12)$$

We used $\|c_j c_i\| \leq 1$. More explicitly, the degree- r contribution is bounded by

$$C_r M^{-r} b^{r/2+1/4} \sqrt{1-t}, \quad 2 \leq r \leq 41,$$

and the remainder is at most $CM^{-42} b^{21+1/4}$.

Let $\delta = M^{-8}$ and $s = 1 - t + \delta$. For any vector ϕ , Cauchy's inequality gives

$$\|(t^2 B_{b-1} + E_b)\phi\|^2 \leq (t^4 + s)\|B_{b-1}\phi\|^2 + (1 + t^4/s)\|E_b\phi\|^2.$$

Since $t^4 + s \leq 1 + \delta$, (7.12) bounds the second term by $CM^{-3/2}\|\phi\|^2$. Summing over the $O(M)$ central rows costs $CM^{-1/2}\|\phi\|^2$. The additional coefficient δ also disappears in this error: (7.9) with $r = 0$ implies $\sum_{M < b \leq 10M} \|B_{b-1}\|^2 \leq CM^{3/2}$.

Finally, for $b > 10M$ the coefficients of $\tilde{B}_{b-1} - B_{b-1}$ are supported where $i < 3M$ or $j < 3M$. The increasing ordering reduces this to $i < 3M$. Equation (7.10) bounds this difference in operator norm by $Cb^{3/2}e^{-c_1 b}$, while normalization of the pair vector gives $\|B_{b-1}\| \leq \sqrt{b}$. The norm of the difference of the two row squares is therefore summable over $b > 10M$, with sum at most $Ce^{-c_3 M}$. Combining the three ranges proves the assertion, with $\varepsilon_M \leq C(M^{-1/2} + M^{-13/2} + e^{-c_3 M})$. $\square$

7.3 Rotation averaging suppresses the small-deficit blocks

Retaining only the rows whose index sum exceeds M gives a positive operator on the two pair factors:

$$A_{q,M} = d_a \int |v_0(g)\rangle\langle v_0(g)| \otimes \left(\sum_{k+1 > M} |v_k(g)\rangle\langle v_k(g)| \right) dg. \quad (7.13)$$

If the sum includes all k , the operator equals I by Schur's lemma. Consequently $0 \leq A_{q,M} \leq I$. This bound handles every retained block with the same coefficient one as in (7.4). It remains to make the coefficients on the finitely many unretained blocks small and bound their lifts by H_q .

Lemma 7.3 (Vanishing weight on fixed deficits). *For $A_{q,M}$ in (7.13), there are coefficients $0 \leq a_{q,M,\ell} \leq 1$ such that*

$$A_{q,M} = \sum_{\ell=0}^{2q-2} a_{q,M,\ell} P_{q,\ell}.$$

For every fixed $\ell \geq 0$,

$$\lim_{M \rightarrow \infty} \limsup_{q \rightarrow \infty} a_{q,M,\ell} = 0. \quad (7.14)$$

Moreover, for each fixed ℓ there is $C_\ell < \infty$ such that

$$\mathcal{L}_4(W_4 P_{q,\ell} W_4^*) \leq C_\ell H_q \quad (7.15)$$

for all sufficiently large q .

Proof. The tensor product of two spin- $(q-1)$ representations is multiplicity free. Hence rotation invariance and $0 \leq A_{q,M} \leq I$ prove the first assertion. Put $a = q-1$ and $d_{q,\ell} = 4q-3-2\ell$. If $C_q(\ell, k)$ is the coupling coefficient of $v_0 \otimes v_k$ in total spin $2a-\ell$ at total lowering degree k , then

$$a_{q,M,\ell} = \frac{2q-1}{d_{q,\ell}} \sum_{k+1 > M} |C_q(\ell, k)|^2. \quad (7.16)$$

The unrestricted sum, including all k , has coefficient exactly one.

We give the fixed- k limit, including its normalization. At a fixed total degree, the two spin lowering operators, divided by $\sqrt{2a}$, converge to multiplication by two independent oscillator variables X, Y . The highest vector of deficit ℓ converges to $(X - Y)^\ell / \sqrt{2^\ell \ell!}$: applying the raising operator gives successive coefficient ratio $-\sqrt{(\ell - p)/(p + 1)}$ in the limit, which fixes this normalized vector. Normalized subsequent lowering multiplies by $(X + Y)/\sqrt{2(k - \ell)}$ at degree k . Thus the degree- k coupled vector converges, in the orthonormal monomial basis, to

$$\frac{(X - Y)^\ell (X + Y)^{k-\ell}}{2^{k/2} \sqrt{\ell! (k - \ell)!}}.$$

Its coefficient of $Y^k / \sqrt{k!}$ has squared modulus $2^{-k} \binom{k}{\ell}$, and hence

$$|C_q(\ell, k)|^2 \longrightarrow 2^{-k} \binom{k}{\ell}.$$

There are only finitely many degrees below a fixed M . Subtracting these from the exact unrestricted coefficient one gives

$$\lim_{q \rightarrow \infty} a_{q,M,\ell} = 1 - \frac{1}{2} \sum_{\ell \leq k \leq M-1} 2^{-k} \binom{k}{\ell}.$$

The identity $\sum_{k=\ell}^{\infty} 2^{-k} \binom{k}{\ell} = 2$ proves (7.14). This finite-head argument requires no uniform convergence over an unbounded set of coupling coefficients.

For (7.15), choose a normalized highest vector $z_{q,\ell} = \sum_{p=0}^{\ell} s_p v_p \otimes v_{\ell-p}$, with $\sum_p |s_p|^2 = 1$. Its wedge annihilator is $\sum_p \bar{s}_p B_{\ell-p} B_p$. Every $B_{\ell-p}$ uses only the first $\ell + 2$ orbitals and has norm at most $\sqrt{\ell + 2}$. Therefore

$$\|B(W_4 z_{q,\ell}) \psi\|^2 \leq (\ell + 2) \sum_{p=0}^{\ell} \|B_p \psi\|^2.$$

Resolve $P_{q,\ell}$ by the orbit of its highest vector and use (7.1). This yields

$$\mathcal{L}_4(W_4 P_{q,\ell} W_4^*) \leq \frac{d_{q,\ell}}{2q - 1} (\ell + 2)(\ell + 1) H_q,$$

whose coefficient is bounded for fixed ℓ . □

Proof of Theorem 7.1. For the moment fix M , and define

$$J_{g,\alpha} = K_\alpha(g) B_0(g), \quad d\nu_q(g, \alpha) = (2q - 1) dg$$

with counting measure in α . Completeness of the loss channel and (7.1) give $\int J_y^* J_y d\nu_q = H_q$. Lemma 7.2, applied to $B_0(g)\psi$ and then averaged, gives

$$\begin{aligned} \int J_y^* H_q J_y d\nu_q(y) &= d_a \int B_0(g)^* \mathcal{E}_{M,g}^*(H_q) B_0(g) dg \\ &\leq \mathcal{L}_4(W_4 A_{q,M} W_4^*) + \varepsilon_M H_q. \end{aligned} \tag{7.17}$$

The equality between the averaged pair products and the lifted four-body operator uses ordered tensor factors. There is no factor of two: $B(v \wedge w) = B(w)B(v)$ and the lift sums each ordered pair once.

By Lemma 7.3 and positivity of the lift,

$$\mathcal{L}_4(W_4 A_{q,M} W_4^*) \leq \mathcal{L}_4(W_4 R_q^{\text{hi}} W_4^*) + \left(\sum_{\ell=0}^{22} C_\ell a_{q,M,\ell} \right) H_q.$$

Fix $c_0 = 1/50$ in (7.4). Choose M large enough, and then q large enough, that $\varepsilon_M + \sum_{\ell=0}^{22} C_\ell a_{q,M,\ell} \leq c_0/2$. Equations (7.4) and (7.17) prove (7.2) with $c = c_0/2$.

Each K_α is a product of $3M$ operators of particle loss zero or one, so $J_{g,\alpha}$ has loss between 2 and $L = 3M + 2$. All modes involved in $J_{g,\alpha}$ have index below $3M$, apart from identity factors. To see operator-norm locality directly, use the frame isometry $S_q : U_q \rightarrow \ell^2(\Lambda_q)$ and write $c(f) = B(f)$ on the slot Fock space. If P_R cuts off slots outside the radius- R ball around $g \cdot \text{north}$, the canonical anticommutation relations give

$$\|c(S_q e_j(g)) - c(P_R S_q e_j(g))\| = \|(I - P_R) S_q e_j(g)\|_2 \leq C_{D,M} R^{-D}.$$

The last estimate follows from Lemma 3.1 and the slot volume bound, with a larger decay order before summing squares. The same estimate for a product follows by telescoping its finitely many factors, or by Lemma 3.4. Thus the jumps are rapidly local around $g \cdot \text{north}$, uniformly in q . There are only 2^{3M} auxiliary labels, with M now fixed. The pushforward of $(2q - 1)dg$ to the physical sphere has area density $(2q - 1)/(2\pi q)$; the remaining rotation angle has mass one. Including all auxiliary labels therefore gives total center density $2^{3M}(2q - 1)/(2\pi q)$, uniformly bounded in q for this fixed M . These observations prove all locality and density assertions. $\square$

7.4 Particle loss, excess charge, and a coercive Fock Hamiltonian

Let $q = 3(N - 1)$ and let ρ_t evolve by the trace-preserving equation whose dual generator is

$$\mathcal{D}_q(O) = \int J_y^* O J_y d\nu_q(y) - \frac{1}{2} \{H_q, O\}. \quad (7.18)$$

This is a finite-dimensional Lindblad generator [10]. Positivity and trace preservation also follow directly by iterating variation of constants, with the completely positive gain map $\rho \mapsto \int J_y \rho J_y^* d\nu_q(y)$ and the no-jump map $\rho \mapsto e^{-tH_q/2} \rho e^{-tH_q/2}$. We write $\langle O \rangle_t = \text{Tr}(\rho_t O)$.

Proposition 7.4 (Control of particle number). *There are constants $C_{\text{num}} < \infty$, $\mu > 0$, $a > 0$, and $C_G < \infty$, independent of sufficiently large $q = 3(N - 1)$, such that*

$$\mathcal{N} - N \leq C_{\text{num}} H_q. \quad (7.19)$$

The operator

$$G_q = H_q + \mu(N - \mathcal{N}) \quad (7.20)$$

satisfies

$$G_q \geq a(H_q + |N - \mathcal{N}|) \geq g(I - P_{L,N}) \quad (7.21)$$

for a constant $g > 0$, with $P_{L,N}$ extended by zero on the other Fock sectors. For every initial density matrix, the evolution (7.18) obeys

$$\langle H_q \rangle_t \leq e^{-ct} \langle H_q \rangle_0, \quad (7.22)$$

$$0 \leq \langle \mathcal{N} \rangle_0 - \langle \mathcal{N} \rangle_t \leq C_{\text{num}} \langle H_q \rangle_0, \quad (7.23)$$

$$\langle G_q \rangle_t \leq C_G \langle G_q \rangle_0. \quad (7.24)$$

Every limit point ρ_∞ is supported on $\ker H_q$, and

$$0 \leq \text{Tr} \rho_\infty (N - \mathcal{N}) \leq C_G \mu^{-1} \langle G_q \rangle_0. \quad (7.25)$$

Proof. Theorem 7.1 gives $\mathcal{D}_q(H_q) \leq -cH_q$, proving (7.22). Since $[\mathcal{N}, J_y] = -r_y J_y$,

$$-\mathcal{D}_q(\mathcal{N}) = \int r_y J_y^* J_y d\nu_q(y), \quad 2H_q \leq -\mathcal{D}_q(\mathcal{N}) \leq LH_q.$$

Integration proves (7.23) with $C_{\text{num}} = L/c$. Compactness of the density matrices in each fixed finite-dimensional Fock space gives limit points as $t \rightarrow \infty$. By energy decay each is supported on $\ker H_q$. Lemma 2.1 shows that this kernel contains no sector of particle number exceeding N . Applying (7.23) to a state of definite particle number n and passing to a limit point gives $n - N \leq C_{\text{num}} \langle H_q \rangle_0$. This proves (7.19) on each sector, and hence on Fock space.

Choose $0 < \mu \leq \min(1, (2C_{\text{num}})^{-1})$. In sectors $n \leq N$, $G_q = H_q + \mu(N - n)$ dominates $\mu(H_q + N - n)$. In sectors $n > N$, (7.19) gives $G_q \geq H_q/2$ and $H_q + n - N \leq (1 + C_{\text{num}})H_q$. This proves the first inequality in (7.21) with $a = \min(\mu, [2(1 + C_{\text{num}})]^{-1})$. For $n \neq N$, $|N - n| \geq 1$; for $n = N$, the uniform gap and uniqueness in Theorem 2.2 give $H_{N,q} \geq (1/25)(I - P_{L,N})$. Thus the second inequality holds with $g = a/25$.

Finally, energy decay and number control give

$$\langle G_q \rangle_t \leq \langle G_q \rangle_0 + \mu C_{\text{num}} \langle H_q \rangle_0 \leq (1 + \mu C_{\text{num}}/a) \langle G_q \rangle_0.$$

Take $C_G = 1 + \mu C_{\text{num}}/a$. Passing to a limit point, where $H_q = 0$, proves (7.25). $\square$

8 Relative bounds and stability

The preceding sections provide two estimates with different roles. The zero-mode bound controls a local observable after particles have been removed. The loss evolution connects an arbitrary state to that zero space at a cost measured by its initial interaction energy. We first combine them into a relative form bound. Spectral transport will then put the disorder perturbation in the class covered by that bound.

Throughout this section, $q = 3(N - 1)$ is sufficiently large, $\Omega \in Z_{N,q}$ is the unit Laughlin vector, and $H = H_q$. Choose the fixed $\mu > 0$ from Proposition 7.4 and write

$$G = H + \mu(N - \mathcal{N}). \tag{8.1}$$

Thus G is positive, has kernel $\mathbb{C}\Omega$, and has a uniform gap. Moreover it controls both H and $|N - \mathcal{N}|$.

8.1 A relative bound for terms that annihilate the ground state

We use the local norm $\|A\|_{x,\ell}$ of (3.10) at one fixed order ℓ . Partition the sphere into cells of uniformly bounded diameter whose centers satisfy the two-dimensional volume bound of Section 3. For a continuous interaction, collect all terms centered in a cell and assign their integral to that cell's center. Moving a center a bounded distance changes the local norm by a fixed factor. We normalize the total localization norm in each cell to at most one.

Physical operators act trivially on the complementary slot modes; the inequalities below are on physical Fock space. Averaging ball approximants over total-slot-number rotations preserves support and approximation error, so approximants of a neutral operator may be chosen neutral. Taking Hermitian parts gives Hermitian approximants.

Choose ℓ larger than the polynomial support-radius exponent in Theorem 6.6, and larger than any fixed geometric exponents needed below. A dyadic decomposition into ball approximants

shows that the zero-mode estimate extends to interactions with bounded total $\|\cdot\|_{x,\ell}$ per cell. Explicitly, the shell of radius 2^k has norm $O(2^{-k\ell})$, while its zero-mode cost is $O(2^{kB})$ for a fixed $B < \ell$; the series converges.

Proposition 8.1 (Relative form bound). *For this fixed sufficiently large ℓ , there is a constant C independent of q such that*

$$-CG \leq \sum_x A_x \leq CG \quad (8.2)$$

whenever the A_x are physical, neutral, Hermitian operators satisfying $A_x\Omega = 0$ and $\|A_x\|_{x,\ell} \leq 1$. If the sum or integral of localization norms in each cell is at most S , the conclusion becomes $-CSG \leq \sum_x A_x \leq CSG$.

Proof. Fix q for the moment and let $K = K_q$ be the least constant in (8.2) for the stated class. It is finite: every admissible sum annihilates Ω on both sides, its norm is bounded by the finite number of cells, and G has a positive gap on $\Omega^\perp$. This observation gives no useful uniform bound; we obtain one from the loss evolution.

Let J_y be the jumps of Theorem 7.1, including their bounded auxiliary indices in the measure dy . They obey

$$\int J_y^* J_y dy = H, \quad J_y \Omega = 0 \quad \text{for almost every } y.$$

The second assertion follows by taking the expectation of the first in Ω . Put $C_{xy} = [A_x, J_y]$ and $D_{xy} = 1 + d(x, y)$. Since A_x is even, the ordinary commutator vanishes for disjointly supported approximants, including when J_y is odd. Lemma 3.4(iii) gives

$$\|C_{xy}\| \leq CD_{xy}^{-\ell}, \quad \|C_{xy}\|_{x,\ell} \leq C, \quad \|C_{xy}^* C_{xy}\|_{x,\ell} \leq CD_{xy}^{-\ell}. \quad (8.3)$$

The last estimate retains the same order ℓ required of A_x . Thus, after integration in y , the positive operators $C_{xy}^* C_{xy}$ remain in the class whose best relative-bound constant is K . Fixing one polynomial norm makes this closure possible.

Every C_{xy} annihilates Ω , because both A_x and J_y do. Therefore $C_{xy}^* C_{xy}$ is a physical, positive, neutral, centered operator to which the definition of K applies. Choose

$$2 < b < \ell - 2. \quad (8.4)$$

The two-dimensional volume bound and (8.3) give

$$\sum_x \int D_{xy}^b C_{xy}^* C_{xy} dy \leq CKG. \quad (8.5)$$

Indeed for each fixed x the integrated operator has localization norm at most $C \int D_{xy}^{b-\ell} dy \leq C$, and kills Ω . Collect it as a single admissible term at cell x . This also explains the continuous-family extension in the statement.

Let ρ_t be the state evolution dual to the loss generator, starting at an arbitrary density matrix ρ_0 . All observables in the desired inequality preserve number, so replacing ρ_0 by its number-diagonal part changes none of the relevant expectations. For $A = \sum_x A_x$, the Lindblad identity reads

$$\frac{d}{dt} \text{Tr}(\rho_t A) = \text{Re} \sum_x \int \text{Tr}(\rho_t J_y^* C_{xy}) dy.$$

Apply Cauchy–Schwarz to this sum with weights D_{xy}^{-b} and D_{xy}^b . Since $\sum_x D_{xy}^{-b} \leq C$ uniformly in y , (8.5) yields

$$\left| \frac{d}{dt} \text{Tr}(\rho_t A) \right| \leq C (\text{Tr}(\rho_t H))^{1/2} (K \text{Tr}(\rho_t G))^{1/2}. \quad (8.6)$$

The energy-descent estimates give constants $c, C > 0$ such that

$$\text{Tr}(\rho_t H) \leq e^{-ct} \text{Tr}(\rho_0 H), \quad \text{Tr}(\rho_t G) \leq C \text{Tr}(\rho_0 G), \quad H \leq CG.$$

Integrating (8.6) therefore costs at most $C\sqrt{K} \text{Tr}(\rho_0 G)$.

Along a sequence $t_j \rightarrow \infty$, finite dimensionality gives a limit state ρ_∞ supported on Z_q . Its sectors have $n \leq N$ by Lemma 2.1. The zero-mode bound and $\langle \Omega, A\Omega \rangle = 0$ show that

$$|\text{Tr}(\rho_\infty A)| \leq C \text{Tr}(\rho_\infty (N - \mathcal{N})) = C\mu^{-1} \text{Tr}(\rho_\infty G) \leq C \text{Tr}(\rho_0 G).$$

Combining the two bounds, and then taking the supremum over all admissible interactions and states, gives

$$K \leq C + C\sqrt{K}.$$

Solving this quadratic inequality for $\sqrt{K}$ proves a bound independent of q , as required. $\square$

8.2 Transporting the ground state

The perturbation is not termwise centered in the sense of Proposition 8.1. Spectral transport produces that property. We give the argument explicitly because it must follow the perturbed ground line rather than keep the original line fixed. The smooth frequency filter implements the quasiadiabatic idea of [6] in the exact spectral-flow form of [2]; the locality estimates needed here were established in Section 3.

Let φ be real and measurable, $\|\varphi\|_\infty \leq 1$. On Fock space set

$$V = d\Gamma(T_q(\varphi)), \quad H_s = H + sV, \quad -1 \leq s \leq 1.$$

The coherent resolution (2.3) writes V as a bounded-density integral of neutral Hermitian terms

$$V_x = \kappa_q \varphi(x) c(k_x)^* c(k_x).$$

They are rapidly local with uniform bounds by Lemma 3.1. The same holds for H , using the rotations of $v_0 = e_0 \wedge e_1$:

$$H = (2q - 1) \int_{SU(2)} B(v_0(g))^* B(v_0(g)) dg.$$

The measure has uniformly bounded density on the physical sphere. In particular every H_s has uniform interaction bounds for $|s| \leq 1$, even though $\|V\|$ can grow with N .

Fix $g_0 = 1/25$ and work initially on a connected parameter interval containing zero where the N -sector ground state is simple and the gap above it exceeds $g_0/2$. Such an interval exists at every fixed flux by finite-dimensional perturbation theory. Choose once and for all a smooth odd real function F satisfying

$$F(\omega) = -\frac{1}{\omega} \quad (|\omega| \geq g_0/2), \quad F(\omega) = 0 \quad (|\omega| \leq g_0/4). \quad (8.7)$$

It can be chosen with an inverse Fourier kernel having every absolute polynomial moment. The $1/\omega$ tail is square-integrable and has integrable derivatives of all positive orders; near time zero

Plancherel gives local integrability, and repeated integration by parts gives arbitrary decay for large time.

For each term of V , define $Y_x(s)$ by applying this frequency filter with respect to H_s . In an eigenbasis of H_s ,

$$\langle a|Y_x(s)|b\rangle = F(E_a - E_b)\langle a|V_x|b\rangle. \quad (8.8)$$

Oddness of F makes Y_x anti-Hermitian. Every Y_x is physical, neutral, and rapidly local with uniform bounds; put $Y(s) = \int Y_x(s) dA(x)$ and solve

$$U'(s) = Y(s)U(s), \quad U(0) = I.$$

This unitary preserves number. If P_s is the N -sector ground projection, the usual spectral derivative formula and (8.7) give $P'_s = [Y(s), P_s]$. Consequently $U(s)P_0U(s)^* = P_s$. Neutrality is important here: the matrix elements crossing the ground projection stay in the N -particle sector. No assertion about a perturbed Fock-space gap is needed.

For each x separately, define

$$\begin{aligned} Q_x(s) &= U(s)^*(V_x + [H_s, Y_x(s)])U(s), \\ a_x(s) &= \langle \Omega, Q_x(s)\Omega \rangle, \quad A_x(s) = Q_x(s) - a_x(s)I. \end{aligned} \quad (8.9)$$

These are Hermitian neutral physical operators. For an excited N -sector eigenvector a and the ground vector 0 , their unconjugated off-diagonal matrix element is

$$(1 + (E_a - E_0)F(E_a - E_0))\langle a|V_x|0\rangle = 0.$$

Hermiticity gives the reverse vanishing. Thus each $Q_x(s)$ preserves $\mathbb{C}\Omega$, and $A_x(s)\Omega = 0$ exactly. The commutator with H_s and the finite-parameter conjugation by $U(s)$ preserve rapid locality, uniformly on the bootstrap interval. The scalar satisfies $|a_x(s)| \leq \|Q_x(s)\|$; it can be assigned to the same center. Hence the interaction $A_x(s)$ satisfies Proposition 8.1 with a uniform size bound.

Let $E_0(s)$ be the N -sector ground energy. Differentiating the pulled-back Hamiltonian and taking its expectation in Ω gives

$$\frac{d}{ds}(U(s)^*H_sU(s) - E_0(s)I) = \int A_x(s) dA(x), \quad -CG \leq \int A_x(s) dA(x) \leq CG. \quad (8.10)$$

All differentiation is finite-dimensional; the integral notation may equally be collected into cells. On the N -sector, $G = H$. Integrating in either parameter direction yields

$$U(s)^*H_sU(s) - E_0(s)I \geq (1 - C|s|)H \quad \text{on } \bigwedge^N U_q. \quad (8.11)$$

The kernel on this interval is $\mathbb{C}\Omega$, so the min-max principle gives a gap of at least $(1 - C|s|)g_0$.

Choose $\lambda_* > 0$ with $\lambda_* \leq 1$ and $C\lambda_* \leq 1/4$. The improved lower bound $3g_0/4$ prevents the gap from reaching the bootstrap boundary $g_0/2$ anywhere in $[-\lambda_*, \lambda_*]$. Indeed a first boundary point would contradict continuity of the ordered eigenvalues and (8.11). The construction therefore extends over this entire interval at every sufficiently large flux. Taking $\Delta_* = g_0/2$ proves Theorem 1.1.

A Retaining the unused four-body blocks

The local loss construction requires a stronger operator inequality than the unperturbed gap: most of the normal-ordered four-body term must remain on the right-hand side. We give the full finite-dimensional calculation that establishes this strengthening. The seven auxiliary squares, their rational certificate, and the finite-flux transfer are adapted from [16, Sections 2–6 and 8]. The proof below retains the four-body blocks that are discarded when only the gap is sought.

We use the lift and wedge maps of (2.4) and (2.7), with increasing orthonormal wedges of norm one. Thus $B(v \wedge w) = B(w)B(v)$, and the domain of W_4 uses ordered pair factors. Write $a = q - 1$ and $H = H_q$; the conventions give

$$\mathcal{L}_k(|v\rangle\langle w|) = B(v)^\dagger B(w), \quad (\text{A.1})$$

$$B_p = B(v_p), \quad H = \mathcal{L}_2(\Pi_{V_q}) = \sum_{p=0}^{2q-2} B_p^\dagger B_p. \quad (\text{A.2})$$

The dagger in this appendix denotes the same adjoint as the star in the main text.

The tensor product $V_q \otimes V_q$ contains one copy of each spin $2q - 2 - l$, $0 \leq l \leq 2q - 2$. Let $P_{q,l}$ be its orthogonal spin projection and define

$$R_q^{\text{lo}} = \sum_{l=0}^{22} P_{q,l}, \quad R_q^{\text{hi}} = I - R_q^{\text{lo}} \quad (q \geq 24). \quad (\text{A.3})$$

The spin-block label used in the calculation below is $D = l + 1$; thus R_q^{lo} consists exactly of $1 \leq D \leq 23$.

Proposition A.1 (Retained four-body blocks). *For every fixed $0 < c_0 < 1/25$ there exists q_0 such that, for every integer $q \geq q_0$, the following inequality holds on the entire Fock space $\mathcal{F}_q$:*

$$H_q^2 \geq c_0 H_q + \mathcal{L}_4(W_4 R_q^{\text{hi}} W_4^\dagger). \quad (\text{A.4})$$

The threshold is independent of particle number.

We prove the proposition by comparing a positive rotational average of seven explicit squares with the normal-ordered terms of H_q^2 . The comparison uses the target four-body blocks $1 \leq D \leq 23$ only. All other target blocks consequently remain with coefficient one.

Lemma A.2 (Normal-ordered square). *On $\mathcal{F}_q$,*

$$H^2 = H + \mathcal{L}_3(W_3(W_3^\dagger W_3 - I)W_3^\dagger) + \mathcal{L}_4(W_4 W_4^\dagger). \quad (\text{A.5})$$

Proof. In normal ordering $B_p^\dagger(B_p B_r^\dagger)B_r$, only the two inner annihilators cross the two inner creators. Two, one, or zero contractions leave two-, three-, or four-body terms, respectively. On two particles, $H = \Pi_{V_q}$ is a projection, so the two-body coefficient is Π_{V_q} . The term with no contractions is

$$\sum_{p,r} B(v_p \wedge v_r)^\dagger B(v_p \wedge v_r) = \mathcal{L}_4(W_4 W_4^\dagger).$$

There is no extra factor of two: this sum, and the defining basis of $V_q \otimes V_q$, both use ordered pairs. On three particles, $H = W_3 W_3^\dagger$, since W_3 restricted to $v_p \otimes U_q$ is the map $B_p^\dagger$ from one to three particles. Subtracting its two-body contribution from H^2 therefore determines the remaining coefficient as

$$(W_3 W_3^\dagger)^2 - W_3 W_3^\dagger = W_3(W_3^\dagger W_3 - I)W_3^\dagger.$$

Normal-ordered coefficients are determined successively by these restrictions, proving the identity on Fock space. $\square$

A.1 Spin coefficients and their limits

The three-body comparison will use the exact spin spectrum of the wedge map. The four-body comparison will use only finitely many coupling coefficients and their limits.

For this auxiliary calculation we simultaneously change the signs of all pair rows from (2.5), so that $v_0 = -e_0 \wedge e_1$, and use the normalized lowering bases e_p and v_p . This common phase change leaves H_q and the intrinsic wedge maps and spin projections unchanged. For spin j , lowering from label p to $p+1$ has coefficient $\sqrt{(p+1)(2j-p)}$. Write $(s)_p = s(s-1) \cdots (s-p+1)$, with $(s)_0 = 1$.

Lemma A.3 (Pair coefficients). *The coefficient of $e_i \wedge e_j$ in v_p , for $i < j$, vanishes unless $i+j = p+1$, and otherwise equals*

$$v_p(i, j) = (i-j) \sqrt{\frac{(q)_i (q)_j p!}{q(2q-2)_p i! j!}}. \quad (\text{A.6})$$

For fixed p, i, j , its limit is

$$v_p^*(i, j) = (i-j) \sqrt{\frac{p!}{2^p i! j!}} \mathbf{1}_{i+j=p+1}. \quad (\text{A.7})$$

Proof. Apply simultaneous lowering p times to $-e_0 \wedge e_1$ and divide by $\sqrt{(2q-2)_p p!}$. Expand the lowering operator binomially. Combining the two contributions to each increasing pair gives (A.6). Taking the limit gives (A.7). The normalization can also be checked directly: the square of the coefficient in (A.6) is

$$(i-j)^2 \frac{\binom{q}{i} \binom{q}{j}}{q \binom{2q-2}{p}},$$

and differentiating $(1+x)^q$ gives

$$\sum_{i,j=0}^q (i-j)^2 \binom{q}{i} \binom{q}{j} x^{i+j} = 2qx(1+x)^{2q-2}.$$

Take the coefficient of x^{p+1} and halve the sum to restrict to $i < j$. □

The elementary tensor product rule decomposes spins j_1, j_2 into one copy of each spin $j_1 + j_2 - z$, for $0 \leq z \leq \min(2j_1, 2j_2)$. Let $R_{3,z}$ be the corresponding projection in $V_q \otimes U_q$, and put

$$d_a = 2q - 1, \quad d_{3,z} = 3q - 1 - 2z.$$

For $q \geq 2$, the allowed indices are $0 \leq z \leq q$.

Lemma A.4 (Three-body Gram eigenvalues).

$$W_3^\dagger W_3 - I = \sum_{z=0}^q q_z(q) R_{3,z}, \quad q_z(q) = (-1)^z \frac{(q)_z}{(2q-2)_z} \left(3z - 1 - \frac{z(z+1)}{q} \right). \quad (\text{A.8})$$

In particular $W_3 R_{3,z} = 0$ for $z = 0, 1, 3$ whenever these indices are allowed. For fixed z ,

$$q_z(q) \longrightarrow (3z-1)(-1/2)^z. \quad (\text{A.9})$$

Proof. Identify $V_q \otimes U_q$ with tensors antisymmetric in their first two positions. Then W_3 is $\sqrt{3}$ times full antisymmetrization. Its Gram operator is the compression of $I - P_{13} - P_{23}$, where P_{ij} swaps tensor positions. The two swap compressions coincide, since conjugation by P_{12} interchanges them and $P_{12} = -I$ on the domain. By equivariance and multiplicity-freeness they act by the same scalar b_z on each spin block.

A highest vector of that block has coefficients s_p on $v_p \otimes e_{z-p}$, $0 \leq p \leq z$, satisfying

$$\frac{s_{p+1}}{s_p} = -\sqrt{\frac{(z-p)(q-z+p+1)}{(p+1)(2q-2-p)}}. \quad (\text{A.10})$$

This follows by applying the raising operator. Using (A.6), the coefficient of the compressed P_{23} image on $v_0 \otimes e_z$ is

$$s_0 b_z = \frac{1}{2} \left[s_{z-1} \sqrt{\frac{z(q)_z}{q(2q-2)_{z-1}}} - s_z(z-1) \sqrt{\frac{(q)_z}{(2q-2)_z}} \right],$$

with the first term absent for $z = 0$. Here the ordered tensor coefficients of a normalized wedge include a factor $1/\sqrt{2}$. Iteration of (A.10) gives

$$\frac{s_z}{s_0} = (-1)^z \sqrt{\frac{(q)_z}{(2q-2)_z}}, \quad \frac{s_{z-1}}{s_0} = (-1)^{z-1} \sqrt{\frac{z(q-1)_{z-1}}{(2q-2)_{z-1}}} \quad (z \geq 1).$$

Substitution gives $q_z = -2b_z$ in the form (A.8). The three stated null sectors have $q_z = -1$, and (A.9) follows directly. $\square$

We will repeatedly use a fixed-deficit limit of this same calculation. In a product of spins j_1, j_2 , total lowering deficit means $T = j_1 + j_2 - m$, where m is the total magnetic quantum number. Coefficients below are with respect to normalized monomials $X^p Y^{T-p} / \sqrt{p!(T-p)!}$, and are zero outside the stated index ranges.

Lemma A.5 (Fixed-deficit coupling limit). *Suppose $j_1, j_2 \rightarrow \infty$ and $j_2/(j_1 + j_2) \rightarrow u^2$, where $0 < u < 1$. Put $v = \sqrt{1-u^2}$. For fixed integers $0 \leq z \leq T$, the normalized coupled vector of spin $j_1 + j_2 - z$ at total lowering deficit T can be chosen with real coefficients converging to the coefficients $s_{z,T,p}^{(u)}$ of*

$$\frac{(uX - vY)^z (vX + uY)^{T-z}}{\sqrt{z!(T-z)!}}. \quad (\text{A.11})$$

Descendants in every copy are obtained by normalized lowering, so the matching of bases between equal-spin copies is intertwining.

Proof. At $T = z$, the raising equation gives the adjacent ratio

$$-\sqrt{\frac{(z-p)(2j_2-z+p+1)}{(p+1)(2j_1-p)}}.$$

With sign $(-1)^z$ at $p = 0$, normalization yields the coefficients of $(uX - vY)^z / \sqrt{z!}$ in the limit. The change of variables $(X, Y) \mapsto (uX - vY, vX + uY)$ is orthogonal, so these polynomials have norm one in the normalized-monomial inner product. Normalized lowering from T to $T+1$ divides the sum of the two lowering operators by

$$\sqrt{(T-z+1)\{2(j_1+j_2-z)-(T-z)\}}.$$

In the limit this acts by multiplication by $(vX + uY)/\sqrt{T-z+1}$. Induction proves the statement for fixed descendants. $\square$

A.2 Rotational averages of explicit squares

We construct a positive rotational average and compare its three- and four-body terms with those of H_q^2 . This section specifies the finite inequalities needed; Subsection A.5 proves them by exact rational arithmetic.

Haar measure dg on $SU(2)$ is normalized to have total mass one. For an operator A , write $A(g) = U(g)AU(g)^\dagger$, with the appropriate Fock representation. Schur averaging sends a rank-one operator on an irreducible spin- j space to its trace times $I/(2j+1)$. For several equivalent copies, it acts in the same way on the spin factor and retains the matrix between copies.

For each of the seven rows in Subsection A.5, the integer t_ρ lies in $\{0, \dots, 7\}$. Set

$$\mathcal{S}_\rho = \{(p, j) \in \mathbb{Z}^2 : 0 \leq p \leq 7, 0 \leq j \leq 8, 0 \leq p + j - t_\rho \leq 8\}.$$

The row specifies real coefficients λ_ρ and α_{pj}^ρ for $(p, j) \in \mathcal{S}_\rho$; unlisted entries are zero. Every coefficient sum for row ρ below is over $\mathcal{S}_\rho$. Our comparison operator is

$$K = \sum_{\rho=1}^7 F_\rho^\dagger F_\rho, \quad F_\rho = \lambda_\rho B_{t_\rho} + \sum_{(p,j) \in \mathcal{S}_\rho} \alpha_{pj}^\rho c_{p+j-t_\rho}^\dagger c_j B_p. \quad (\text{A.12})$$

Thus only modes $0, \dots, 8$ occur in these auxiliary squares.

For a single row, omit ρ and put $i = p + j - t$, $k = p' + l - t$. Define

$$O_{p,j,k} = B(v_p \wedge e_j \wedge e_k) = c_k c_j B_p.$$

Reordering $c_i c_k^\dagger = \delta_{ik} - c_k^\dagger c_i$ gives the two-, three-, and four-body terms of $F^\dagger F$:

$$\lambda^2 B_t^\dagger B_t, \quad (\text{A.13})$$

$$\sum_{p,j} \lambda \alpha_{pj} \left[(c_i B_t)^\dagger c_j B_p + (c_j B_p)^\dagger c_i B_t \right] + \sum_{p,j;p',l} \alpha_{pj} \alpha_{p'l} \delta_{ik} (c_j B_p)^\dagger c_l B_{p'}, \quad (\text{A.14})$$

$$- \sum_{p,j;p',l} \alpha_{pj} \alpha_{p'l} O_{p,j,k}^\dagger O_{p',l,i}. \quad (\text{A.15})$$

Let A_2, A_3, A_4 denote these terms, summed over rows and averaged with factor d_a . Thus

$$\begin{aligned} \mathcal{K}_q &:= d_a \int K(g) dg = A_2 + A_3 + A_4, \quad A_2 = \eta H, \\ \eta &= \sum_\rho \lambda_\rho^2 = \frac{93527408868499}{10^{14}}. \end{aligned} \quad (\text{A.16})$$

A.2.1 The three-body comparison

Before wedging, (A.14) is a matrix on $V_q \otimes U_q$. Its nonzero entries preserve $T = p + j$, and $T \leq 15$. Averaging therefore gives

$$A_3 = \mathcal{L}_3 \left(W_3 \sum_{z=0}^{15} \frac{d_a}{d_{3,z}} e_z^q R_{3,z} W_3^\dagger \right), \quad (\text{A.17})$$

for all sufficiently large q , where e_z^q is the trace of the unaveraged matrix in that spin block. By Lemma A.5, with $u = 1/\sqrt{3}$, these traces converge to

$$e_z = \sum_\rho \sum_{T=\max(z, t_\rho)}^{15} \left[2\lambda_\rho s_{z,T,t_\rho} \sum_{p+j=T} \alpha_{pj}^\rho s_{z,T,p} + \left(\sum_{p+j=T} \alpha_{pj}^\rho s_{z,T,p} \right)^2 \right], \quad (\text{A.18})$$

where $s = s^{(1/\sqrt{3})}$ and empty inner sums are zero. The condition $i = k$ in (A.14) is exactly the equality of the two values of T .

The exact calculation in Subsection A.5 proves

$$e_z < \frac{3}{2}(3z-1)(-1/2)^z, \quad z \in \{2, 4, 5, \dots, 15\}. \quad (\text{A.19})$$

There are finitely many strict inequalities, $d_{3,z}/d_a \rightarrow 3/2$, and the omitted blocks $z = 0, 1, 3$ are annihilated by W_3 . Consequently, for all sufficiently large q ,

$$A_3 \leq \mathcal{L}_3 \left(W_3 \sum_{z=0}^{15} q_z(q) R_{3,z} W_3^\dagger \right). \quad (\text{A.20})$$

A.2.2 The four-body comparison before taking limits

Let

$$\widehat{W} : V_q \otimes \bigwedge^2 U_q \longrightarrow \bigwedge^4 U_q$$

be the wedge map. The second tensor factor decomposes into spins $q - r$, with r odd, $1 \leq r \leq q$. In its lowering bases use the sign of Lemma A.5; at $r = 1$ this is precisely the basis v_p .

A copy of spin $2q - 1 - D$ is obtained by coupling $a = q - 1$ with $q - r$ at deficit $D - r$. Denote its vector at total deficit T by $|D, T; r\rangle_q$. Its physical orbital-index sum is $1 + T$, and it is a highest vector when $T = D$. For fixed D, T and sufficiently large q , the allowed copy labels are

$$\mathcal{R}_D = \{1, 3, 5, \dots\} \cap [1, D], \quad D \leq T.$$

Define the real finite-flux coefficients by

$$|D, T; r\rangle_q = \sum_{\substack{p+j+k=T \\ j < k}} S_r^q(D, T; p, j, k) v_p \otimes (e_j \wedge e_k),$$

and extend them antisymmetrically in j, k . By two applications of Lemma A.5, they converge to

$$S_r(D, T; p, j, k) = \sqrt{2} s_{r, j+k, j}^{(1/\sqrt{2})} s_{D-r, T-r, p}^{(1/\sqrt{2})}, \quad (\text{A.21})$$

interpreted as zero if $r > j + k$. The formula holds for either ordering of j, k , by antisymmetry. It follows by applying Lemma A.5 first to the two single-particle spins and then to the two pair spins.

After averaging (A.15), its spin- $2q - 1 - D$ block before wedging is a copy matrix E_D^q times the spin identity, multiplied by d_a/d_D , where

$$d_D = 4q - 1 - 2D.$$

Only $1 \leq D \leq 23$ contribute: every unaveraged term has

$$T = p + j + k = p' + l + i \leq 23.$$

For these finitely many D , entrywise convergence gives $E_D^q \rightarrow E_D$, with

$$(E_D)_{rs} = - \sum_{\rho} \sum_{\substack{p, j; p', l \\ T \geq D}} \alpha_{pj}^{\rho} \alpha_{p'l}^{\rho} S_r(D, T; p, j, k) S_s(D, T; p', l, i), \quad (\text{A.22})$$

where $i = p + j - t_\rho$, $k = p' + l - t_\rho$, and $T = p + j + k$. The exact finite-flux matrix E_D^q is given by the same formula with each S_r replaced by S_r^q .

The target term $\mathcal{L}_4(W_4 W_4^\dagger)$ is obtained from the $r = 1$ summand, since that summand is $V_q \otimes V_q$. Thus its copy matrix, in the units just used, is $(d_D/d_a)\chi$, where

$$\chi_{rs} = \mathbf{1}_{r=s=1}, \quad d_D/d_a \longrightarrow 2.$$

The summands of the target with $1 \leq D \leq 23$ equal $\mathcal{L}_4(W_4 R_q^{\text{lo}} W_4^\dagger)$. Since A_4 has no other spin blocks, the comparison with A_4 uses only this part of the target. We keep $\mathcal{L}_4(W_4 R_q^{\text{hi}} W_4^\dagger)$ separately.

Set

$$w_{Dr}^q = \widehat{W}|D, D; r\rangle_q, \quad (\text{A.23})$$

$$w_{Dr}^* = \sum_{\substack{p+j+k=D \\ j < k}} S_r(D, D; p, j, k) v_p^* \wedge e_j \wedge e_k, \quad (G_D)_{rs} = \langle w_{Dr}^*, w_{Ds}^* \rangle. \quad (\text{A.24})$$

The star is a limit label, not an adjoint. The exact certificate is

$$G_D(2\chi - E_D + \varepsilon I)G_D \geq 0, \quad 1 \leq D \leq 23, \quad \varepsilon = 3 \cdot 10^{-6}. \quad (\text{A.25})$$

We next show how these finite inequalities give the stated bound uniformly in particle number. Subsection A.5 supplies their integer data and exact rational verification.

A.3 A relative error bound for finite spheres

The limiting Gram matrices have kernels, so their positivity cannot be transferred by assuming that their ranges vary continuously. Instead, we compare the annihilators themselves. An exact diagonal change of variables reduces every error to a fixed finite family of pair-generated annihilators. This gives an error relative to H_q , uniformly on the full Fock space.

For the local argument, let $\mathcal{F}_{\text{loc}}$ be the exterior algebra on the 25 orthonormal modes $e_0, \dots, e_{24}$. For $q \geq 24$ we identify it with the Fock space of the first 25 orbital modes in U_q . By (A.7), each limiting pair vector v_p^* with $p \leq 23$ lies in $\bigwedge^2 \mathbb{C}^{25}$. Contraction by these vectors is defined on $\mathcal{F}_{\text{loc}}$ by the same exterior-algebra convention as before.

Lemma A.6 (Compression and annihilator bounds). *Let W be a linear map between finite-dimensional Hilbert spaces, $G = W^\dagger W$, and let C be self-adjoint on the domain of W . Then*

$$GCG \geq 0 \iff WCW^\dagger \geq 0.$$

On $\mathcal{F}_{\text{loc}}$, for any fixed finite collection $A_b = c_k c_j B(v_p^*)$ with $0 \leq p \leq 23$ and $0 \leq j, k \leq 24$,

$$\sum_b \|A_b \psi\|^2 \leq C_0 \langle \psi, h_* \psi \rangle, \quad h_* := \sum_{p=0}^{23} B(v_p^*)^\dagger B(v_p^*), \quad (\text{A.26})$$

where C_0 depends only on the collection.

Proof. The two positivity statements both assert that the quadratic form of C is nonnegative on $\text{ran } G = \text{ran } W^\dagger$. For the second assertion, use $\|c_k c_j B(v_p^*) \psi\| \leq \|B(v_p^*) \psi\|$ and sum; one may take C_0 to be the largest multiplicity of a pair label p . $\square$

Lemma A.7 (Uniform transfer). *For $q \geq 24$ and $1 \leq D \leq 23$, put*

$$C_D^q = (d_D/d_a)\chi - E_D^q + \varepsilon I, \quad h_q = \sum_{p=0}^{23} B_p^\dagger B_p.$$

Assuming (A.25), there is a scalar $\rho_q \geq 0$, independent of D and tending to zero as $q \rightarrow \infty$, such that

$$\sum_{r,s \in \mathcal{R}_D} (C_D^q)_{rs} B(w_{Dr}^q)^\dagger B(w_{Ds}^q) \geq -\rho_q h_q \quad (\text{A.27})$$

on the entire Fock space $\mathcal{F}_q$.

Proof. An exact change of variables. Take $q \geq 24$. All annihilators in (A.27), including those in h_q , involve only the modes $0, \dots, 24$. First work on $\mathcal{F}_{\text{loc}}$. Define the positive diagonal operator

$$\Delta_q |A\rangle = \left(\prod_{x \in A} d_q(x) \right) |A\rangle, \quad d_q(x) = \sqrt{(q)_x / q^x},$$

on each basis wedge $|A\rangle$. This local Fock space is fixed as q varies. Thus Δ_q and Δ_q^{-1} converge in norm to I ; also $0 < \Delta_q \leq I$.

The pair coefficient formula (A.6) gives

$$v_p(x, y) = r_p(q) d_q(x) d_q(y) v_p^*(x, y), \quad r_p(q) = \sqrt{(2q)^p / (2q - 2)_p} \geq 1.$$

Checking the factors attached to the removed orbital labels on a basis wedge gives the exact identities

$$B_p = r_p(q) \Delta_q^{-1} B(v_p^*) \Delta_q, \quad (\text{A.28})$$

$$B(v_p \wedge e_j \wedge e_k) = \frac{r_p(q)}{d_q(j) d_q(k)} \Delta_q^{-1} B(v_p^* \wedge e_j \wedge e_k) \Delta_q. \quad (\text{A.29})$$

These are identities on the local Fock space, with the inverse diagonal acting on the output particle sector. Repeated removed labels make both sides zero. In particular, solving (A.28) for $B(v_p^*) \Delta_q$ and using $r_p(q) \geq 1$ and $\|\Delta_q\| \leq 1$ gives

$$\Delta_q h_* \Delta_q \leq h_q. \quad (\text{A.30})$$

The error is controlled by pair energy. Fix D . Index a finite family by $b = (p, j, k)$ with $p + j + k = D$ and $j < k$, and put

$$A_b = B(v_p^* \wedge e_j \wedge e_k), \quad \beta_{rb}^q = S_r^q(D, D; p, j, k) \frac{r_p(q)}{d_q(j) d_q(k)}.$$

Equations (A.23) and (A.29) yield

$$B(w_{Dr}^q) = \Delta_q^{-1} \left(\sum_b \beta_{rb}^q A_b \right) \Delta_q.$$

The coefficients are real, and Lemma A.5 gives $\beta_{rb}^q \rightarrow \beta_{rb}^* := S_r(D, D; p, j, k)$. Hence the finite real symmetric matrices

$$M_q = (\beta^q)^T C_D^q \beta^q \quad \text{converge to} \quad M_* = (\beta^*)^T (2\chi - E_D + \varepsilon I) \beta^*.$$

The quadratic expression in (A.27) is consequently

$$\Delta_q \left(\sum_{b,c} (M_q)_{bc} A_b^\dagger \Delta_q^{-2} A_c \right) \Delta_q.$$

Its limiting expression before the outer diagonals is

$$\sum_{b,c} (M_*)_{bc} A_b^\dagger A_c = \sum_{r,s} (2\chi - E_D + \varepsilon I)_{rs} B(w_{Dr}^*)^\dagger B(w_{Ds}^*) \geq 0. \quad (\text{A.31})$$

Indeed, the certificate (A.25) and Lemma A.6, applied to the map with columns w_{Dr}^* , give positivity before lifting; the lift $\mathcal{L}_4$ preserves positivity.

For clarity, the perturbation estimate does not require $M_* \geq 0$. Let R_q be the block matrix with entries

$$(R_q)_{bc} = (M_q)_{bc} \Delta_q^{-2} - (M_*)_{bc} I.$$

Its norm tends to zero, since both its index set and the local Fock space are fixed. For every vector ϕ ,

$$\left| \sum_{b,c} \langle A_b \phi, (R_q)_{bc} A_c \phi \rangle \right| \leq \|R_q\| \sum_b \|A_b \phi\|^2 \leq C_0 \|R_q\| \langle \phi, h_* \phi \rangle.$$

Combine this estimate with (A.31), take $\phi = \Delta_q \psi$, and apply (A.30). This proves (A.27) on the local Fock space with a coefficient tending to zero. Taking a maximum over the fixed set $1 \leq D \leq 23$ gives a common ρ_q .

Additional particles are spectators. Order the 25 local modes before all other modes in the Fock factorization. Every operator in (A.27) acts on the local factor alone. Tensoring the local inequality with the identity proves it on $\mathcal{F}_q$, with the same ρ_q , regardless of the number of occupied modes outside this factor. $\square$

The allowance εI in the certificate must also be paid for in pair energy. Here it is useful to sum over the orthonormal highest vectors *before* estimating their wedges. Bessel's inequality then avoids an extra factor equal to the number of spin copies.

Proposition A.8 (Four-body estimate). *For the coefficients in Subsection A.5,*

$$A_4 \leq \mathcal{L}_4(W_4 R_q^{\text{lo}} W_4^\dagger) + (1222\varepsilon + o(1))H, \quad q \rightarrow \infty, \quad (\text{A.32})$$

uniformly on $\mathcal{F}_q$.

Proof. Haar averaging the matrix unit $|D, D; r\rangle_q \langle D, D; s|_q$ gives the corresponding copy matrix unit tensored with the spin identity, divided by d_D . Therefore, on the blocks $1 \leq D \leq 23$, the low-block target minus A_4 , with the εI allowance added, is

$$d_a \sum_{D=1}^{23} \int \sum_{r,s} (C_D^q)_{rs} B(w_{Dr}^q(g))^\dagger B(w_{Ds}^q(g)) dg.$$

Lemma A.7, conjugated by each rotation, bounds this below by $-o(1)H$, because Schur averaging gives

$$d_a \int h_q(g) dg = 24H.$$

The number of blocks in the comparison is fixed, so this error remains uniform in particle number. To bound the allowance, fix D and write

$$|D, D; r\rangle_q = \sum_{\substack{p+j+k=D \\ j < k}} S_{rb}^q v_p \otimes (e_j \wedge e_k), \quad b = (p, j, k).$$

The uncoupled vectors on the right are orthonormal. So are the highest vectors on the left: distinct r belong to orthogonal spin summands of the second pair factor. Thus $S^q(S^q)^T = I$. Apply the contraction $S^q \otimes I$ to the Hilbert-space-valued vector $(c_k c_j B_p \psi)_b$. It follows that

$$\sum_{r \in \mathcal{R}_D} \|B(w_{Dr}^q) \psi\|^2 \leq \sum_{\substack{p+j+k=D \\ j < k}} \|c_k c_j B_p \psi\|^2 \leq \sum_{\substack{p+j+k=D \\ j < k}} \|B_p \psi\|^2.$$

This uses orthogonality before wedging; the vectors w_{Dr}^q themselves need not be orthogonal. The number of triples on the right is

$$n_D = \sum_{n=1}^D \left\lceil \frac{n}{2} \right\rceil = \left\lfloor \frac{(D+1)^2}{4} \right\rfloor.$$

Conjugate by rotations and integrate. Each pair norm contributes $\langle \psi, H \psi \rangle / d_a$, so the total allowance is bounded by

$$\varepsilon \sum_{D=1}^{23} n_D H = \varepsilon \left(\sum_{m=1}^{12} m^2 + \sum_{m=1}^{11} m(m+1) \right) H = 1222\varepsilon H.$$

Removing the allowance proves (A.32). No block of R_q^{hi} has entered this estimate. $\square$

A.4 The three-body tail and the retained-block inequality

The preceding comparison has used only the low four-body target. It remains to control the three-body blocks above $z = 15$ by pair energy and combine the estimates.

Lemma A.9 (Three-body tail). *For sufficiently large q ,*

$$\mathcal{L}_3 \left(W_3 \sum_{z=16}^q q_z(q) R_{3,z} W_3^\dagger \right) \geq -\delta_q H, \quad (\text{A.33})$$

where

$$\delta_q = \sum_{\substack{17 \leq z \leq q \\ z \text{ odd}}} \frac{d_{3,z}}{d_a} |q_z(q)| (z+1), \quad \delta_q \longrightarrow \frac{61}{4096}. \quad (\text{A.34})$$

Proof. For $16 \leq z \leq q$, the bracket in (A.8) is positive, since it is at least $2z - 2$. Thus only odd z contribute negatively. A normalized highest vector of $R_{3,z}$ is $\sum_{p=0}^z s_p v_p \otimes e_{z-p}$. Its orbit resolves that spin projector with factor $d_{3,z}$. After wedging and lifting, its quadratic form is

$$\left\| \sum_{p=0}^z s_p c_{z-p}(g) B_p(g) \psi \right\|^2 \leq \sum_{p=0}^z \|B_p(g) \psi\|^2.$$

Averaging each summand proves (A.33) with (A.34).

For $q \geq 4$, the ratios in $(q)_z/(2q-2)_z$ are nonincreasing, so

$$\frac{(q)_z}{(2q-2)_z} \leq \left(\frac{q}{2q-2}\right)^z \leq (2/3)^z, \quad \frac{d_{3,z}}{d_a} \leq 2.$$

The summands, extended by zero for $z > q$, are therefore dominated by $6z(z+1)(2/3)^z$. Dominated convergence and (A.9) give

$$\lim_{q \rightarrow \infty} \delta_q = \frac{3}{2} \sum_{z=17,19,\dots} (3z-1)(z+1)2^{-z} = \frac{61}{4096}.$$

The last identity follows by writing $z = 17 + 2m$ and using the geometric-series identities for $\sum t^m$, $\sum mt^m$, and $\sum m^2 t^m$ at $t = 1/4$. $\square$

Proof of Proposition A.1. Write $T_{\text{lo}} = \mathcal{L}_4(W_4 R_q^{\text{lo}} W_4^\dagger)$ and $T_{\text{hi}} = \mathcal{L}_4(W_4 R_q^{\text{hi}} W_4^\dagger)$. The normal-square identity, the three-body comparison (A.20), and the tail estimate give

$$H^2 \geq (1 - \delta_q)H + A_3 + T_{\text{lo}} + T_{\text{hi}}.$$

By (A.32), $T_{\text{lo}} \geq A_4 - (1222\varepsilon + o(1))H$. Using $\mathcal{K}_q = \eta H + A_3 + A_4 \geq 0$ from (A.16), we obtain

$$H^2 \geq \mathcal{K}_q + (1 - \eta - \delta_q - 1222\varepsilon - o(1))H + T_{\text{hi}}.$$

The coefficient of H tends to

$$1 - \frac{93527408868499}{10^{14}} - \frac{61}{4096} - 1222 \frac{3}{10^6} = \frac{4616733319001}{10^{14}} > \frac{1}{25}. \quad (\text{A.35})$$

For fixed $0 < c_0 < 1/25$, choose q large enough that this coefficient exceeds c_0 and discard the positive operator $\mathcal{K}_q$. This is (A.4). The matrix comparisons involved finitely many fixed orbital modes, the transfer estimates held on full Fock space, and the tail estimate was uniform in particle number. Therefore the same threshold applies simultaneously to every sector. $\square$

A.5 The finite rational certificate

We finish by verifying the two finite inequalities (A.19) and (A.25) used above. The seven integer rows and all rational formulas are given here. Starting from the integer rows in Subsection A.5.1, evaluate the rational three-body traces in Subsection A.5.2 and the rational four-body matrices in Subsection A.5.3. The margin and coefficient-sign tables in those sections record the resulting checks. The formulas below determine every value using only integers and rational numbers.

A.5.1 Row data

Let $P = 10^7$. The first two numbers of each row are t and $\ell = P\lambda$. An entry $pj : a$ specifies the integer a_{pj} , with p and j parsed as separate single-digit labels; omitted entries are zero. Continuation lines belong to the preceding row. Define

$$\alpha_{pj} = \frac{a_{pj}}{P} \sqrt{\frac{2^p \binom{p+j}{p}}{2^t \binom{p+j}{t}}} = \frac{a_{pj}}{P} \sqrt{\frac{2^{p-t} t! (p+j-t)!}{p! j!}}. \quad (\text{A.36})$$

Every nonzero entry has $0 \leq p + j - t \leq 8$.

```

0 7181518 05:-849651 07:-2958744 15:751659 16:-43084 17:-146216 32:720949 35:-127514
42:222437 43:-262822 53:-117700 61:-55024 71:-166049
1 -112991 05:142772 06:-142342 07:527055 08:50742 16:197917 18:-18343 24:5764 27:-21183
32:75832 35:-6115 45:2660 53:-9455 61:-27091 63:-3494 71:479 72:-15331
1 -4247311 05:-12207250 06:-14064191 07:-14063663 08:-15662679 16:483831 18:3987214
24:-814558 27:888887 32:966462 35:145978 45:-171722 53:34350 61:389319 63:97603
71:412447 72:-53786
2 3438746 14:6689266 15:6620205 16:6625065 17:4969272 18:4878238 27:-1387380 36:-586188
37:344087 42:-1120634 45:-318530 46:559169 55:157455 61:1207103 63:211617 64:33240
71:577001 73:-142578
5 -98645 05:-2719776 08:-8519913 14:-949523 17:-1952105 18:-1562606 28:-8919966 32:248826
35:-1017623 37:3656005 38:-2081871 42:-33163 43:-305905 46:35677 47:-1320749
48:-1040378 53:502094 55:1060633 58:-998003 61:-113617 63:184919 64:209239 65:-68198
67:-266297 72:-606696 73:-563620 74:437345 76:-432189
5 -3160276 05:-1698129 08:10575875 14:-252576 17:-8002616 18:22592844 28:7370640
32:289294 35:1400647 37:1987740 38:4030699 42:-1731401 43:-1034062 46:-1106371
47:-1257051 48:1461735 53:1408149 55:1136847 58:1878501 61:-2838516 63:-734099
64:1062416 65:11745 67:611312 72:-480776 73:20459 74:316925 76:37092
7 1441784 07:3441286 08:-6097155 37:-1002399 47:-251321 58:1143225 66:-17397 71:-1007464
73:-662286 76:-344621 78:-1115284

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In particular $P^2\eta = \sum_{\rho} \ell_{\rho}^2 = 93527408868499$.

A.5.2 Three-body arithmetic

For $c \in \{1, 2\}$, an integer $T \geq 0$, and integers z, p , define

$$U(c, z, T, p) = \sum_{h=\max(0, p+z-T)}^{\min(p, z)} (-1)^{z-h} \binom{z}{h} \binom{T-z}{p-h} c^{p-h}, \quad (\text{A.37})$$

and set it to zero if z or p lies outside $[0, T]$. Expanding (A.11) gives

$$s_{z, T, p}^{(1/\sqrt{3})} = U(2, z, T, p) \sqrt{\frac{2^z \binom{T}{z}}{3^T 2^p \binom{T}{p}}}.$$

Substitution in (A.18) and (A.36) cancels all radicals:

$$P^2 e_z = \sum_{\rho} \sum_{T=\max(z, t_{\rho})}^{15} \frac{2^z \binom{T}{z}}{3^T 2^{t_{\rho}} \binom{T}{t_{\rho}}} X_{\rho, T} (X_{\rho, T} + 2\ell_{\rho} U(2, z, T, t_{\rho})), \quad (\text{A.38})$$

where

$$X_{\rho, T} = \sum_{p+j=T} a_{pj}^{\rho} U(2, z, T, p),$$

with value zero on empty levels. For each $z \in \{2, 4, 5, \dots, 15\}$, evaluate (A.38) and form the exact rational margin $m_z = \frac{3}{2}(3z-1)(-1/2)^z - e_z$. Write $10^6 m_z = n_z/d_z$ with integers n_z, d_z and $d_z > 0$. Integer division gives the floors $\lfloor n_z/d_z \rfloor$ listed below:

z	2	4	5	6	7	8	9	10	11	12	13	14	15
floor	1298313	211317	249	250	249	250	250	250	249	250	250	6470	250

All are positive, proving (A.19).

A.5.3 Four-body arithmetic

Fix $1 \leq D \leq 23$; all copy indices below lie in $\mathcal{R}_D$. Put

$$u_r = \frac{1}{r!(D-r)!}, \quad (E_D)_{rs} = \sqrt{u_r u_s} Y_{rs}, \quad (G_D)_{rs} = \sqrt{u_r u_s} Z_{rs}.$$

The matrices Y, Z are rational. Define

$$\mathbf{V}_r(D, T; p, j, k) = \mathbf{U}(1, r, j+k, j) \mathbf{U}(1, D-r, T-r, p),$$

with value zero for $r > j+k$. Two applications of (A.11) give

$$S_r(D, T; p, j, k) = \mathbf{V}_r(D, T; p, j, k) \sqrt{2^{1-(j+k)-T+r} \frac{j!k!p!u_r}{(T-D)!}}. \quad (\text{A.39})$$

Therefore

$$\begin{aligned} P^2 Y_{rs} = & - \sum_{\rho} \sum_{\substack{p, j; p', l \\ T \geq D}} a_{pj}^{\rho} a_{p'l}^{\rho} \mathbf{V}_r(D, T; p, j, k) \mathbf{V}_s(D, T; p', l, i) \\ & \times 2^{1-2T+p+p'-t_{\rho}+(r+s)/2} \frac{i!k!t_{\rho}!}{(T-D)!}, \end{aligned} \quad (\text{A.40})$$

where $i = p + j - t_{\rho}$, $k = p' + l - t_{\rho}$, and $T = p + j + k$. Both entry sums are ordered. To verify the cancellation, multiply the two factors in (A.39) by the two factors (A.36); their remaining square root is exactly the factorial and power-of-two factor displayed in (A.40).

For an increasing quadruple A of nonnegative indices with sum $D+1$, put $f(A) = \prod_{b \in A} b!$. If (x, y, j, k) lists its elements, let $\sigma(x, y, j, k)$ be the sign of sorting this list into increasing order. In each summand below put $p = x + y - 1$, and define

$$L_r(A) = 2^{(1-2D+r)/2} \sum_{\substack{x < y \text{ in } A \\ \{j, k\} = A \setminus \{x, y\}, j < k}} \sigma(x, y, j, k) (x-y) p! j! k! \mathbf{V}_r(D, D; p, j, k). \quad (\text{A.41})$$

Combining (A.7) and (A.39), the coefficient of $w_{D_r}^*$ on the Slater wedge A is $\sqrt{u_r/f(A)} L_r(A)$. Thus

$$Z_{rs} = \sum_{\substack{A \text{ increasing quadruple} \\ \sum_{b \in A} b = D+1}} \frac{L_r(A) L_s(A)}{f(A)}. \quad (\text{A.42})$$

All exponents of two in these formulas are integers, since r, s are odd.

Let $D_u = \text{diag}(u_r)$ and $D_{\sqrt{u}} = \text{diag}(\sqrt{u_r})$. The matrix in (A.25) equals $D_{\sqrt{u}} M D_{\sqrt{u}}$, where

$$M = Z B Z, \quad B = \text{diag}((2\mathbf{1}_{r=1} + \varepsilon) u_r) - D_u Y D_u. \quad (\text{A.43})$$

Thus it suffices to check $M \geq 0$ using rational arithmetic.

To check positivity for each $D = 1, \dots, 23$, form Y and Z from (A.40) and (A.42), then form B and M from (A.43), using $\varepsilon = 3/10^6$. All sums range over the printed rows and the finite index sets specified above. Compute with exact fractions throughout. If $d = |\mathcal{R}_D|$, initialize $J = I_d$. For $b = 1, \dots, d$, compute in order

$$X = M J, \quad c_b = \frac{\text{tr } X}{b}, \quad J = c_b I_d - X. \quad (\text{A.44})$$

Then $\det(xI + M) = x^d + c_1x^{d-1} + \dots + c_d$. For example, this recurrence follows by multiplying the polynomial adjugate of $xI + M$ by $xI + M$ and comparing coefficients, together with the derivative-of-determinant trace identity. Exact evaluation of (A.37), (A.40), (A.41), (A.42), and (A.44) gives:

D	signs	D	signs	D	signs
1	0	9	++000	17	++++++000
2	0	10	+0000	18	+++++0000
3	00	11	+++000	19	++++++000
4	00	12	++0000	20	++++++0000
5	+00	13	++++000	21	+++++++000
6	000	14	+++0000	22	+++++++0000
7	+000	15	+++++000	23	+++++++000
8	+000	16	++++0000		

The signs list the coefficients after the leading 1, in descending degree. For each computed coefficient, choose a positive denominator and inspect its integer numerator: + denotes a positive numerator and 0 denotes zero. Every coefficient is nonnegative. Hence

$$\det(xI + M) > 0 \quad (x > 0).$$

Since M is real symmetric, a negative eigenvalue would give a positive root of this polynomial. Therefore $M \geq 0$, proving (A.25).

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